SERVICE MANUAL



SKYACTIV-D 1.5 ENGINE

SERVICE - REPAIR - MAINTENANCE

HOW TO USE THIS MANUAL

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Range of Topics

• This manual contains procedures for performing all required service operations. The procedures are divided into the following basic operations:

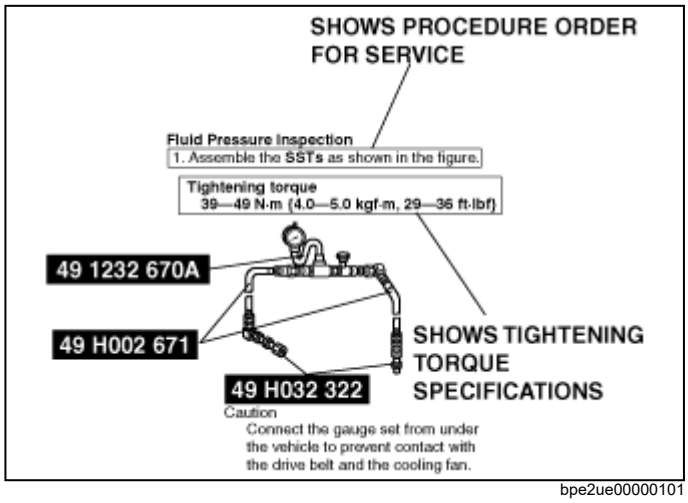
— Removal/Installation
— Disassembly/Assembly
— Replacement
— Inspection
— Adjustment

• Simple operations which can be performed easily just by looking at the actual unit (i.e., removal/installation of parts, cleaning of parts, and visual inspection) have been omitted.

Service Procedure

Inspection, adjustment

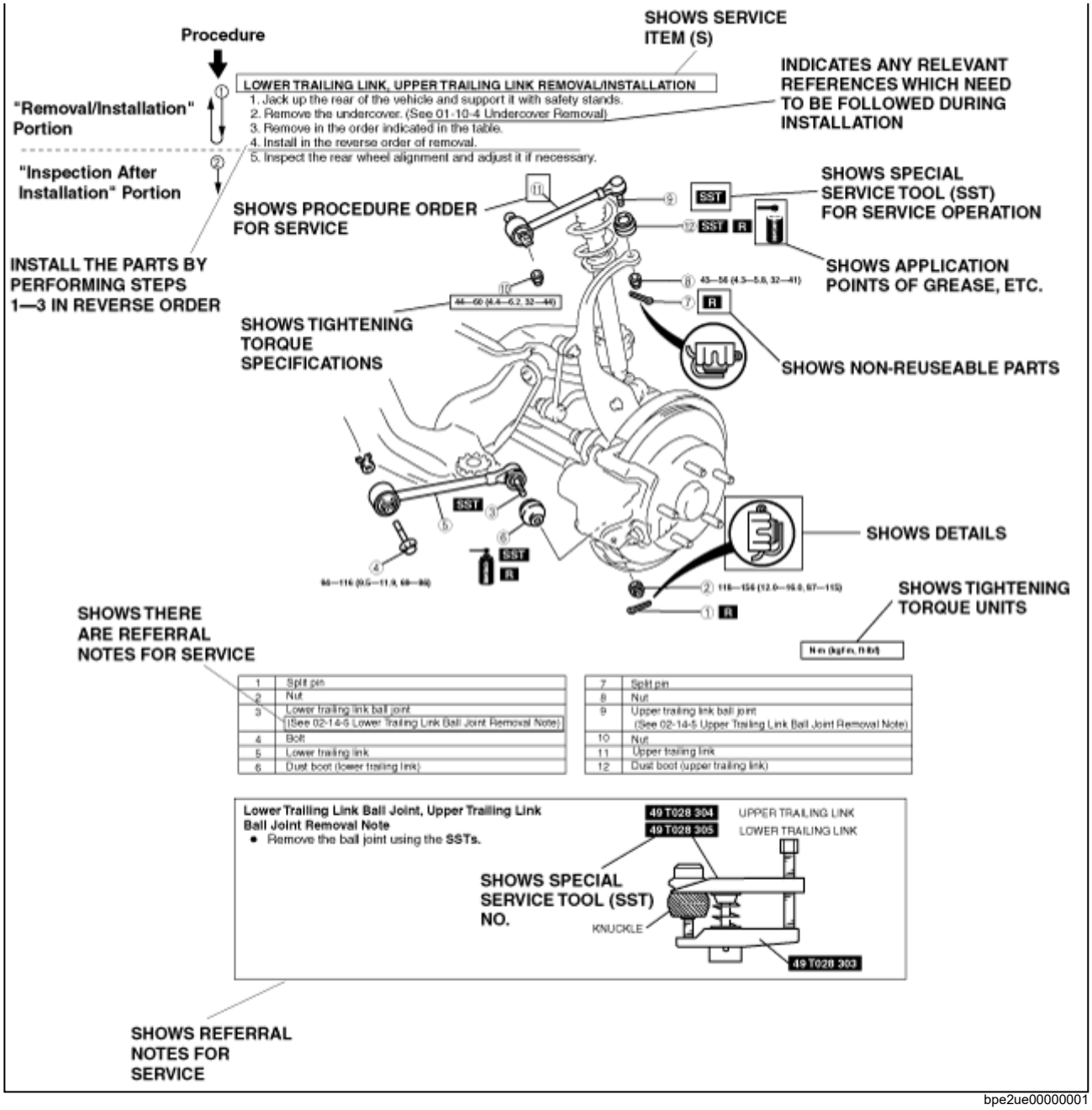
• Inspection and adjustment procedures are divided into steps. Important points regarding the location and contents of the procedures are explained in detail and shown in the illustrations.



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Repair procedure

1. Most repair operations begin with an overview illustration. It identifies the components, shows how the parts fit together, and describes visual part inspection. However, only removal/installation procedures that need to be performed methodically have written instructions.
2. Expendable parts, tightening torques, and symbols for oil, grease, and sealant are shown in the overview illustration. In addition, symbols indicating parts requiring the use of special service tools or equivalent are also shown.
3. Procedure steps are numbered and the part that is the main point of that procedure is shown in the illustration with the corresponding number. Occasionally, there are important points or additional information concerning a procedure. Refer to this information when servicing the related part.



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Symbols

• There are eight symbols indicating oil, grease, fluids, sealant, and the use of SST or equivalent. These symbols show application points or use of these materials during service.

Symbol	Meaning	Kind
OIL	Apply oil	New appropriate engine oil or gear oil
BRAKE FLUID	Apply brake fluid	New appropriate brake fluid
ATF	Apply automatic transaxle/ transmission fluid	New appropriate automatic transaxle/ transmission fluid
GREASE	Apply grease	Appropriate grease
SEALANT	Apply sealant	Appropriate sealant
P	Apply petroleum jelly	Appropriate petroleum jelly
R	Replace part	O-ring, gasket, etc.
SST	Use SST or equivalent	Appropriate tools

Advisory Messages

- You will find several Warnings, Cautions, Notes, Specifications and Upper and Lower Limits in this manual.

Warning

- A Warning indicates a situation in which serious injury or death could result if the warning is ignored.

Caution

- A Caution indicates a situation in which damage to the vehicle or parts could result if the caution is ignored.

Note

- A Note provides added information that will help you to complete a particular procedure.

Specification

- The values indicate the allowable range when performing inspections or adjustments.

Upper and lower limits

- The values indicate the upper and lower limits that must not be exceeded when performing inspections or adjustments.

UNITS

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Electric current	A (ampere)
Electric power	W (watt)
Electric resistance	ohm
Electric voltage	V (volt)
Length	mm (millimeter)
	in (inch)
Negative pressure	kPa (kilo pascal)
	mmHg (millimeters of mercury)
	inHg (inches of mercury)
Positive pressure	kPa (kilo pascal)
	kgf/cm^2 (kilogram force per square centimeter)
	psi (pounds per square inch)
Number of revolutions	rpm (revolutions per minute)
Torque	N·m (Newton meter)
	kgf·m (kilogram force meter)
	kgf·cm (kilogram force centimeter)
	ft·lbf (foot pound force)
	in·lbf (inch pound force)
Volume	L (liter)
	US qt (U.S. quart)
	Imp qt (Imperial quart)
	ml (milliliter)
	cc (cubic centimeter)
	cu in (cubic inch)
	fl oz (fluid ounce)
Weight	g (gram)
	oz (ounce)

Conversion From SI Units (Système International d'Unités)

- All numerical values in this manual are based on SI units. Numbers shown in conventional units are converted from these values.

Number Of Digits For Converted Values

- The number digits for converted values is the same as the number of significant figures*1 of the SI unit.
- For the torque value, the number of significant figures is, in principle, is 2 digits, in consideration of market practicalities. However, if the number of decimal places at the upper and lower limits of the converted value differs, the one with least number of decimal places is used. In addition, if the integer part is 3 digits or more, the integer part becomes the significant number of figures.

*1 :The number of significant figures is the number of digits from the left-most non-zero digit to the right-most digit including 0. (Example: 0.12 is 2 digits, 41.0 is 3 digits)

Converted Value Rounding Off And Rounding Up/down

- If there is no tolerance in the SI unit value, after conversion, rounding off is to within the number of significant digits.
- If there is tolerance in the SI unit value and the figure after conversion indicates the upper limit, the number of digits is rounded down to within the number of significant figures. If it indicates the lower limit, they are rounded up to within the number of significant figures.
- Even if the SI unit value is the same, the converted value may differ based on whether that value is the upper or lower limit.

ABBREVIATIONS

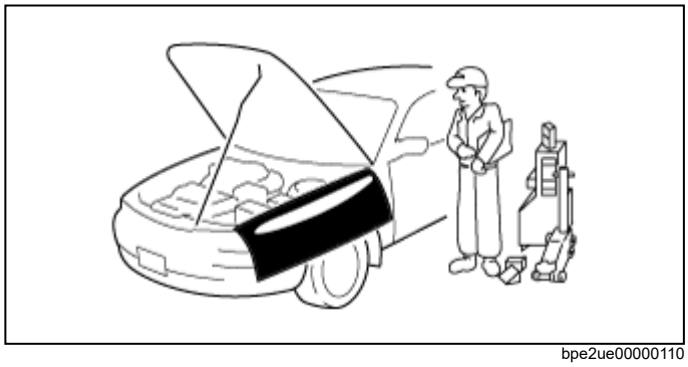
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ATX	Automatic Transaxle
EX	Exhaust
HLA	Hydraulic Lash Adjuster
IN	Intake
MTX	Manual Transaxle
OCV	Oil Control Valve
TDC	Top Dead Center
SST	Special Service Tool

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Preparation of Tools and Measuring Equipment

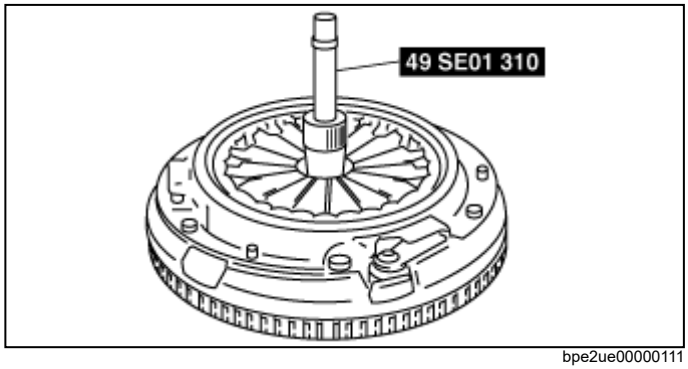
• Be sure that all necessary tools and measuring equipment are available before starting any work.



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Special Service Tools

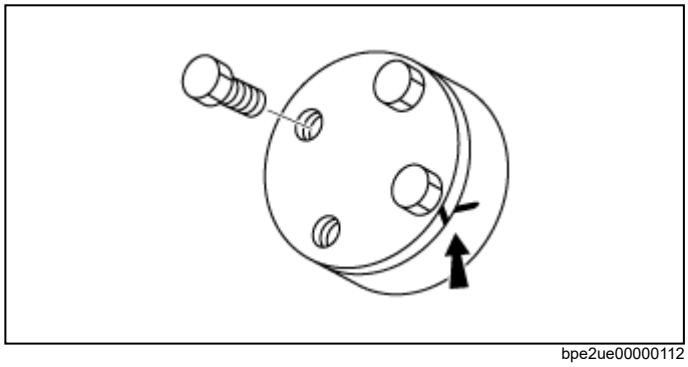
• Use special service tools or equivalent when they are required.



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Disassembly

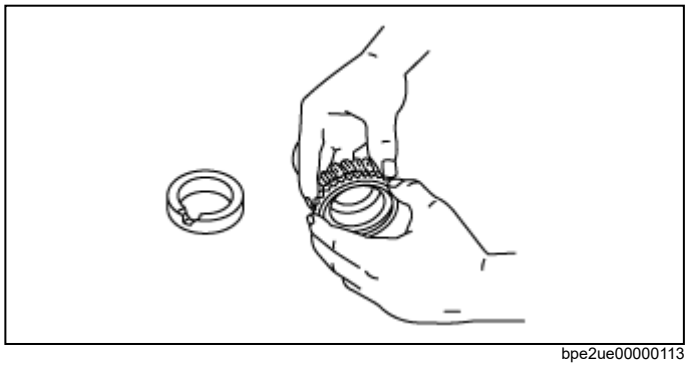
• If the disassembly procedure is complex, requiring many parts to be disassembled, all parts should be marked in a place that will not affect their performance or external appearance and identified so that reassembly can be performed easily and efficiently.



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Inspection During Removal, Disassembly

• When removed, each part should be carefully inspected for malfunction, deformation, damage and other problems.



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Arrangement of Parts

• All disassembled parts should be carefully arranged for reassembly.

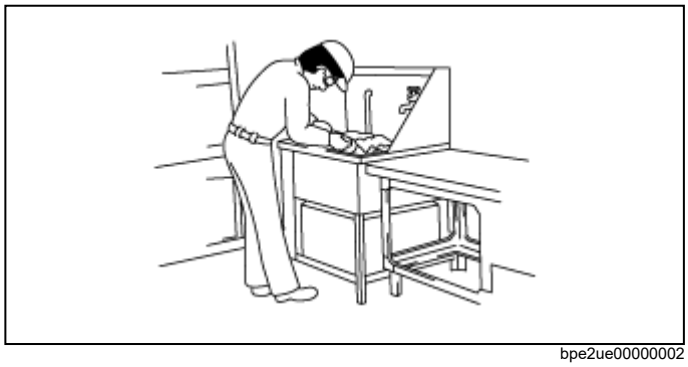


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- Be sure to separate or otherwise identify the parts to be replaced from those that will be reused.

Cleaning of Parts

- All parts to be reused should be carefully and thoroughly cleaned in the appropriate method.



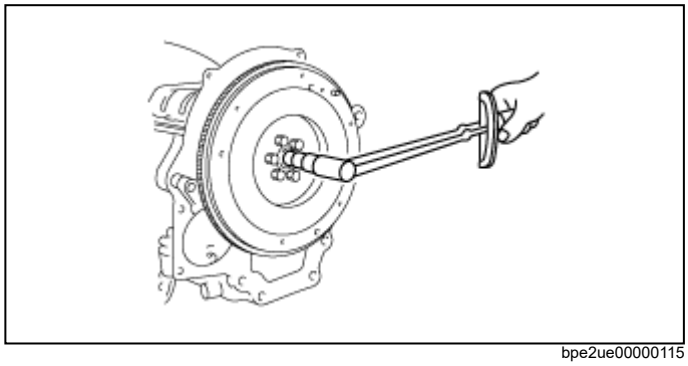
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Warning

- Using compressed air can cause dirt and other particles to fly out causing injury to the eyes. Wear protective eye wear whenever using compressed air.

Reassembly

- Standard values, such as torques and certain adjustments, must be strictly observed in the reassembly of all parts.

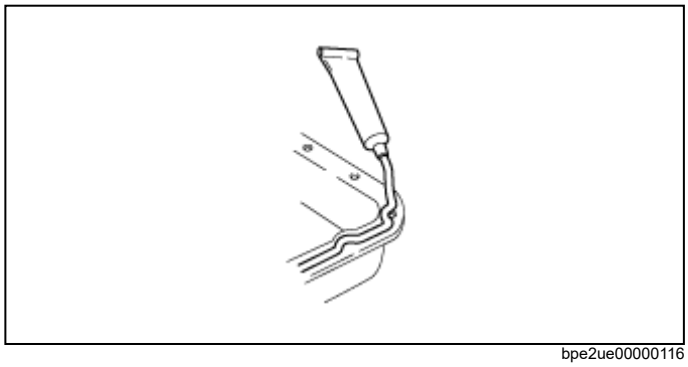


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- If removed, the following parts should be replaced with new ones:

  — Oil seals
  — Gaskets
  — O-rings
  — Lockwashers
  — Cotter pins
  — Nylon nuts

- Depending on location:

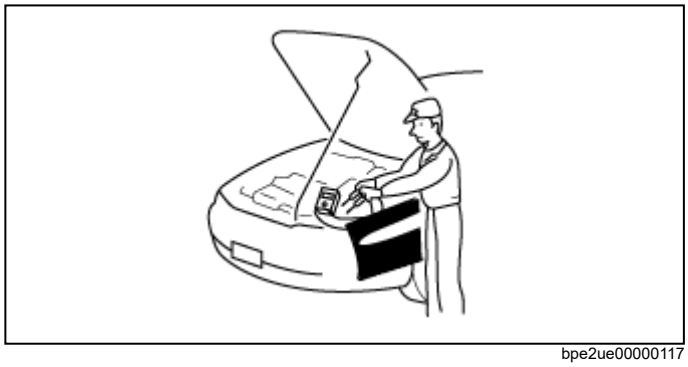


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  — Sealant and gaskets, or both, should be applied to specified locations. When sealant is applied, parts should be installed before sealant hardens to prevent leakage.
  — Oil should be applied to the moving components of parts.
  — Specified oil or grease should be applied at the prescribed locations (such as oil seals) before reassembly.

Adjustment

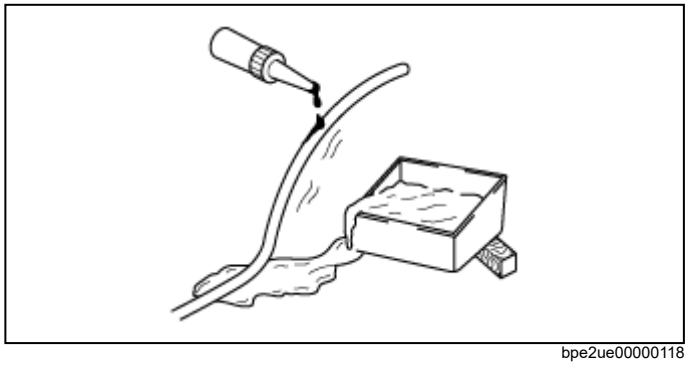
• Use suitable gauges and testers when making adjustments.



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Rubber Parts and Tubing

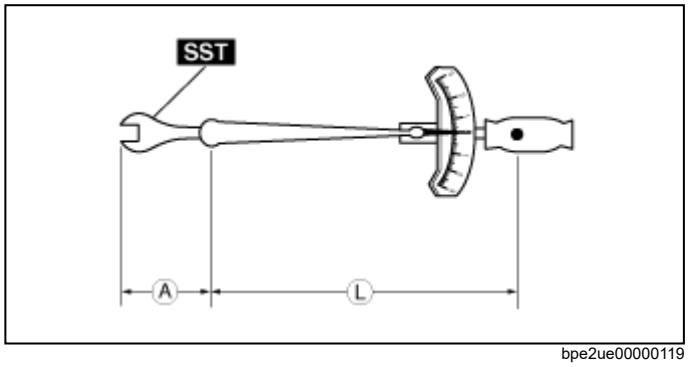
• Prevent gasoline or oil from getting on rubber parts or tubing.



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Torque Formulas

• When using a torque wrench-SST or equivalent combination, the written torque must be recalculated due to the extra length that the SST or equivalent adds to the torque wrench. Recalculate the torque by using the following formulas. Choose the formula that applies to you.



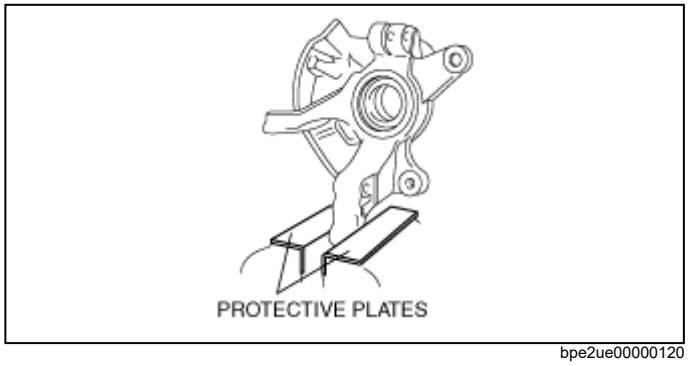
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Torque Unit	Formula
N·m	N·m ×[L/(L+A)]
kgf·m	kgf·m ×[L/(L+A)]
kgf·cm	kgf·cm ×[L/(L+A)]
ft·lbf	ft·lbf ×[L/(L+A)]
in·lbf	in·lbf ×[L/(L+A)]

A :The length of the SST past the torque wrench drive.
L :The length of the torque wrench.

Vise

• When using a vise, put protective plates in the jaws of the vise to prevent damage to parts.

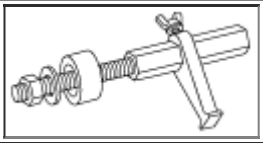


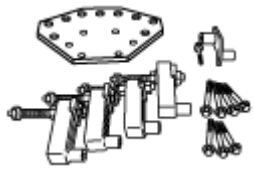
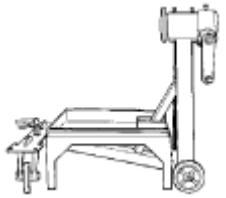
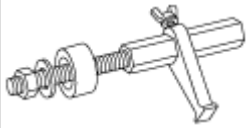
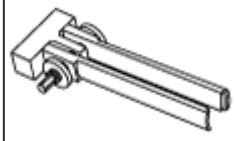
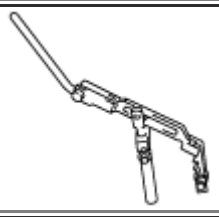
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ENGINE SST

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1: Mazda SST number
2: Global SST number
Example

1:49 E011 1A0 2: – Ring gear brake set	

1:49 L010 1A0 2: – Engine hanger set		1:49 0107 680A 2: – Engine stand		1:49 E011 1A0 2: – Ring gear brake set	
1:49 0636 100B 2: – Valve spring lifter arm		1:49 B012 0A2A 2: – Pivot		1: 49 JE02 0A2 2: — Universal valve tool (Europe only)	
1:49 1285 071 2: – Bearing puller		1:49 S033 101 2: – Dust cover installer		1:49 B012 015 2: – Valve guide installer	
1:49 B012 016 2: – Attachment		1:49 G028 205 2: – Oil seal installer		–	

ENGINE OVERHAUL SERVICE WARNING

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Warning

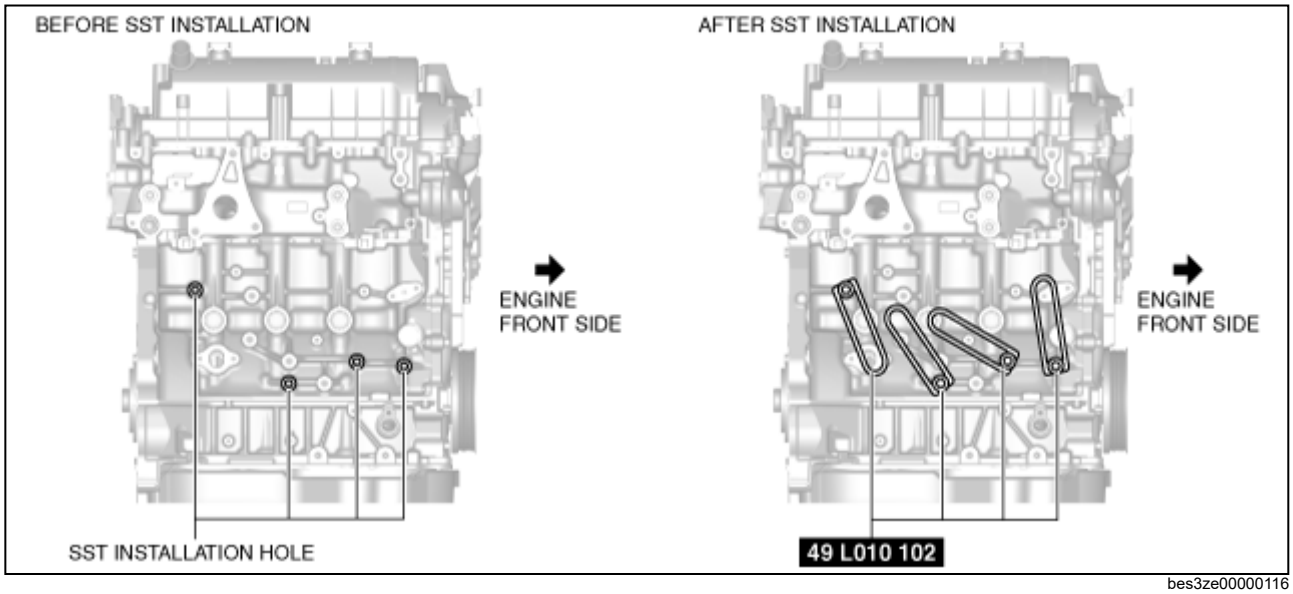
- Continuous exposure to USED engine oil has been shown to cause skin cancer in laboratory mice. Protect your skin by washing with soap and water immediately after performing work.

ENGINE MOUNTING/DISMOUNTING

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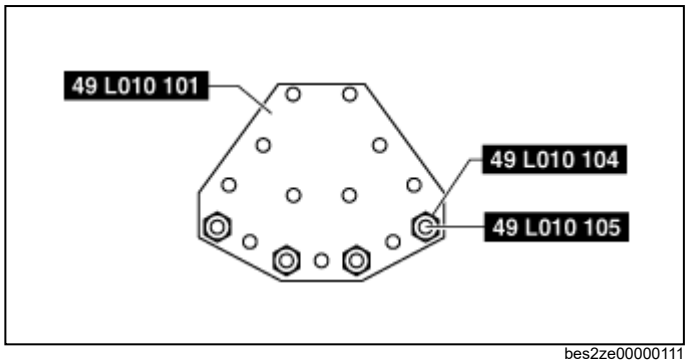
Mounting

1. Install the SST (arms) to the four positions as shown in the figure and temporarily tighten the bolts (Part No. 9YA20-1003 or M10 × 1.5 length 90 mm {3.55 in}).



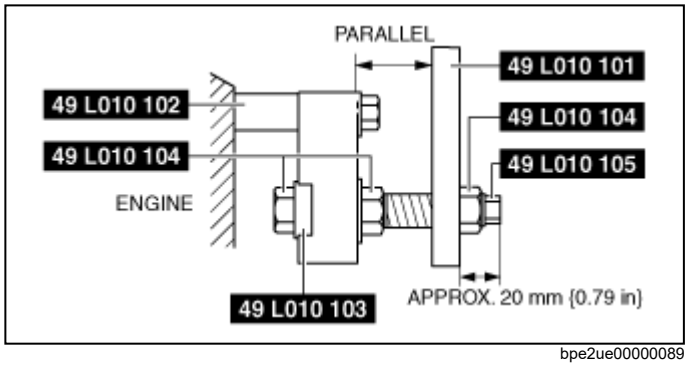
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2. Install the SSTs (bolts and nuts) to the four positions of the SST (plate) as shown in the figure.



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3. Install the SST (bolts, nuts and plate) set in Step 2 to the SST (arms) set in Step 1 using the SSTs (hook and nuts).



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4. Adjust the bolts so that approx. 20 mm {0.79 in}of thread is exposed from the side of the plate.
5. Adjust the bolts and nuts so that the plate and arms are parallel.
6. Tighten the SSTs (bolts and nuts) to affix the SST firmly.
7. Install the engine to the SST (engine stand).
8. Remove the oil drain plug and drain the engine oil.
9. Replace the gasket with a new one and install the oil pan drain plug.

Tightening torque

30—41 N·m {3.1—4.1 kgf·m, 23—30 ft·lbf}

Dismounting

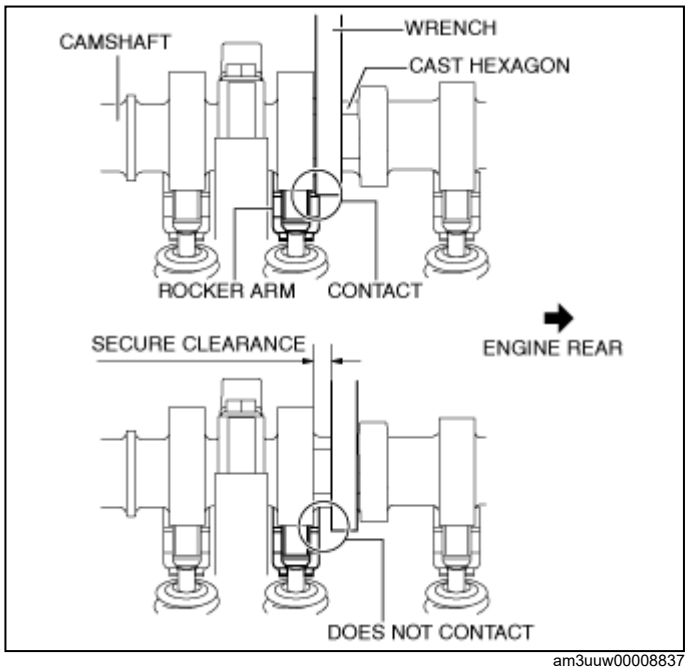
1. Dismount in the reverse order of mounting.

TIMING CHAIN DISASSEMBLY

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Caution

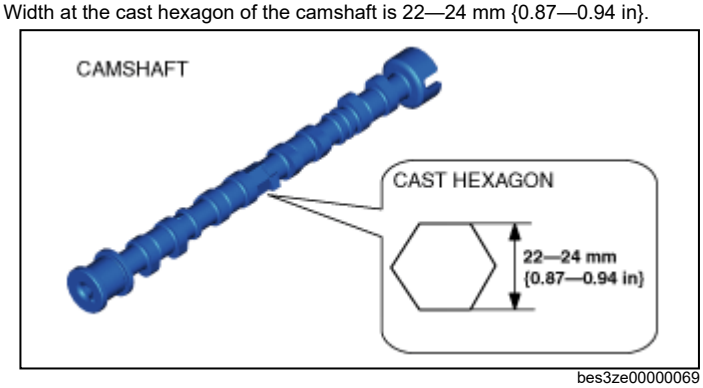
• When rotating the camshaft using a wrench on the cast hexagon, the wrench may contact the rocker arm and damage the rocker arm. To prevent damage to the rocker arm when holding the camshaft on the cast hexagon, use the wrench at engine rear side as shown in the figure to secure a clearance between the cam.



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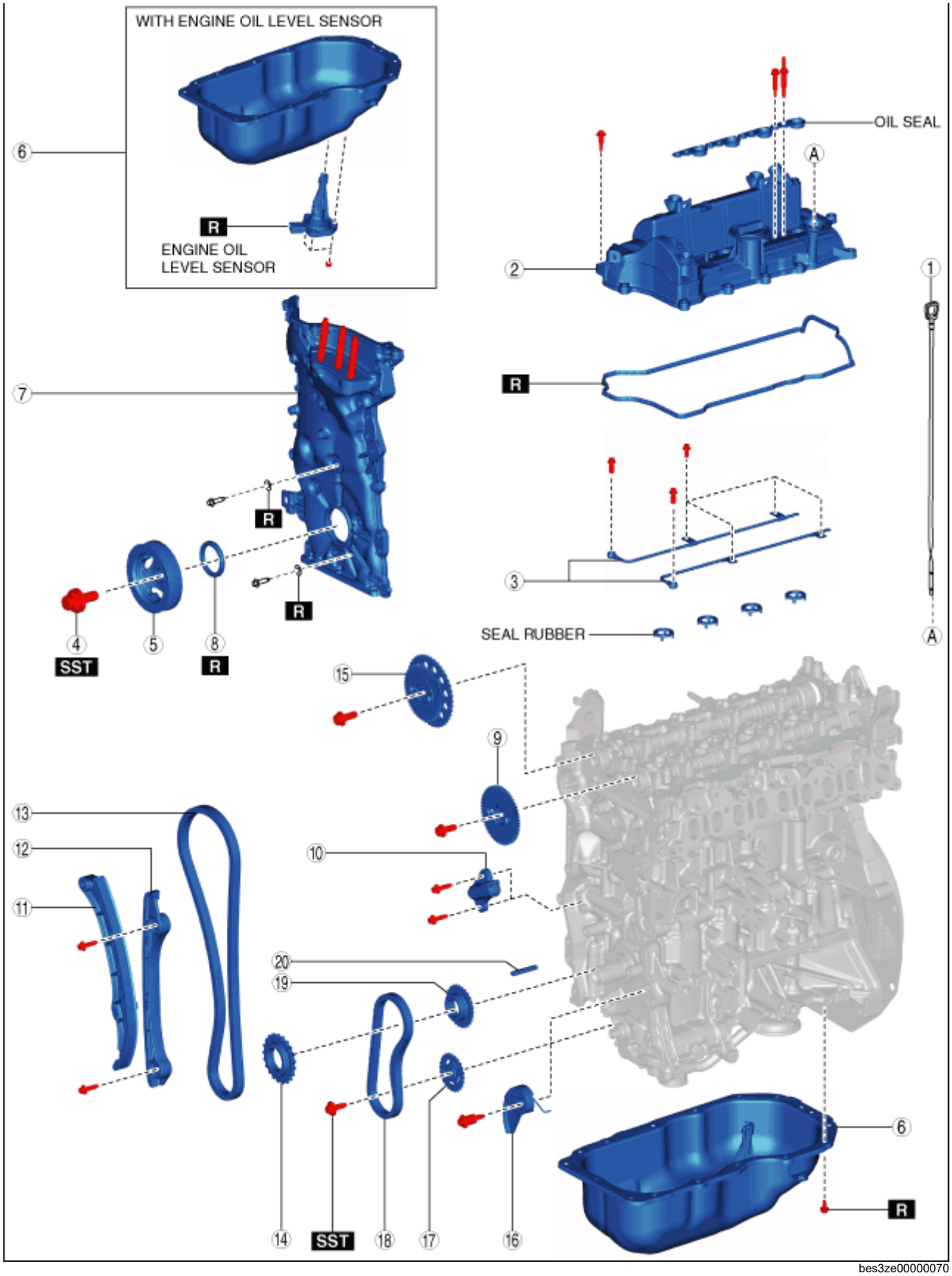
Note

• Width at the cast hexagon of the camshaft is 22—24 mm {0.87—0.94 in}.



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1. Disassemble in the order indicated in the table.



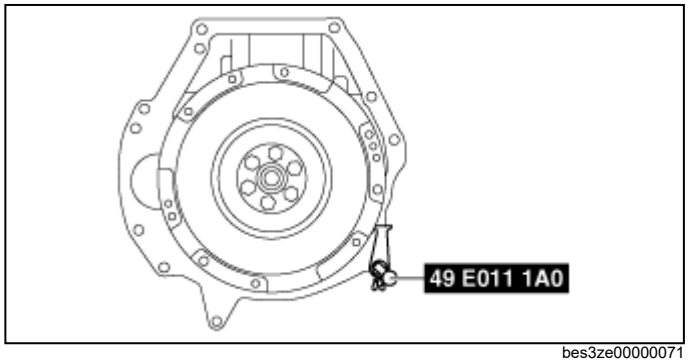
1	Dipstick
2	Cylinder head cover
3	Oil shower pipe
4	Crankshaft pulley lock bolt (See Crankshaft Pulley Lock Bolt Disassembly Note.)
5	Crankshaft pulley
6	Oil pan component (See Oil Pan Component Disassembly Note.)
7	Engine front cover (See Engine Front Cover Disassembly Note.)
8	Front oil seal (See Front Oil Seal Disassembly Note.)

9	Intake camshaft sprocket (See Intake Camshaft Sprocket Disassembly Note.)
10	Timing chain tensioner (See Timing Chain Disassembly Note.)
11	Timing chain tensioner arm (See Timing Chain Disassembly Note.)
12	Timing chain guide (See Timing Chain Disassembly Note.)
13	Timing chain (See Timing Chain Disassembly Note.)
14	Crankshaft sprocket (See Timing Chain Disassembly Note.)
15	Exhaust camshaft sprocket (See Exhaust Camshaft Sprocket Disassembly Note.)
16	Oil pump chain tensioner
17	Oil pump driven sprocket (See Oil Pump Driven sprocket Disassembly Note.)
18	Oil pump chain
19	Oil pump drive sprocket
20	Key

Crankshaft Pulley Lock Bolt Disassembly Note

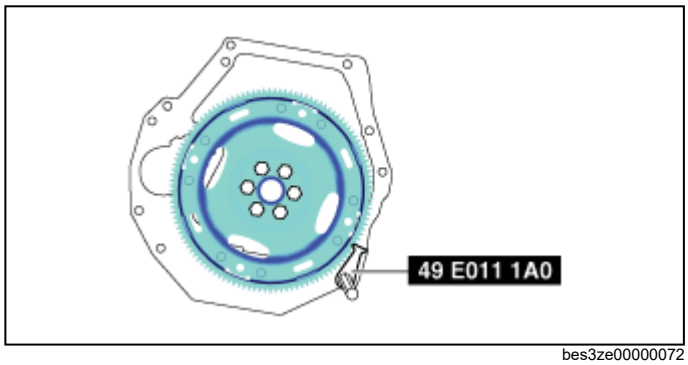
1. Hold the crankshaft using the SST.

MTX



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ATX



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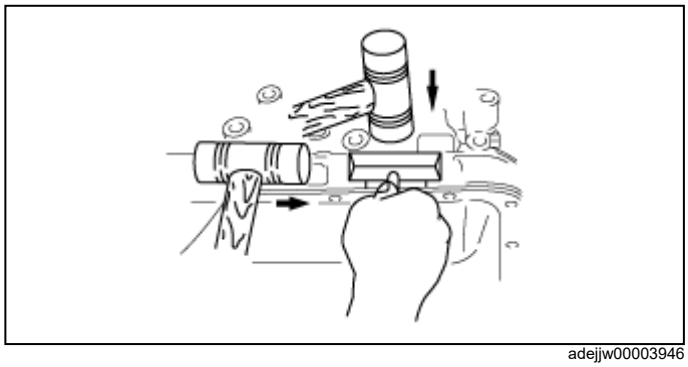
2. Remove the crankshaft pulley lock bolt.

Oil Pan Component Disassembly Note

Caution

- If the engine oil level sensor is removed, it is necessary to replace it with a new one. (with engine oil level sensor)

1. Remove the oil pan component using a separator tool.



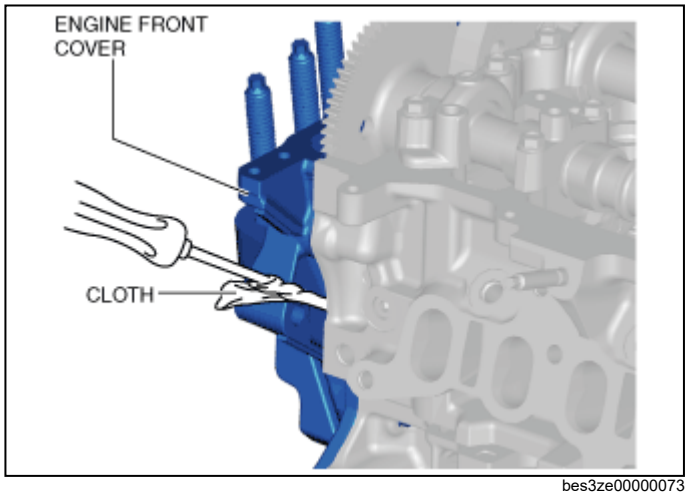
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Engine Front Cover Disassembly Note

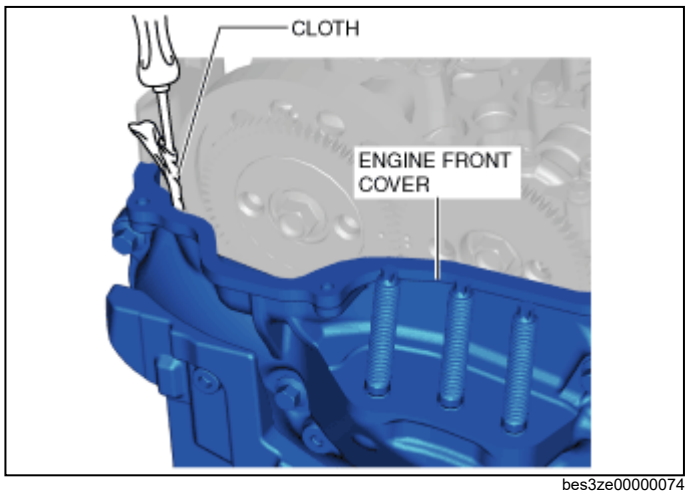
1. Remove the engine front cover installation bolts.
2. Using a screwdriver wrapped in a cloth, peel the sealant away a little at a time, and remove the engine front cover.

Caution

- Do not apply excessive force to the screwdriver. Otherwise, the engine front cover could be damaged.
- Be careful not to scratch or damage the seal surface. Otherwise, it could cause oil leakage.



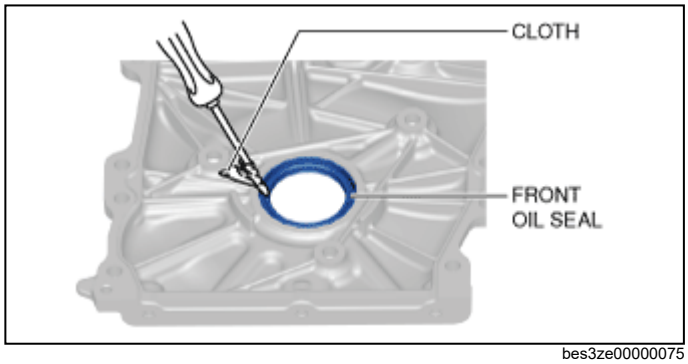
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Front Oil Seal Disassembly Note

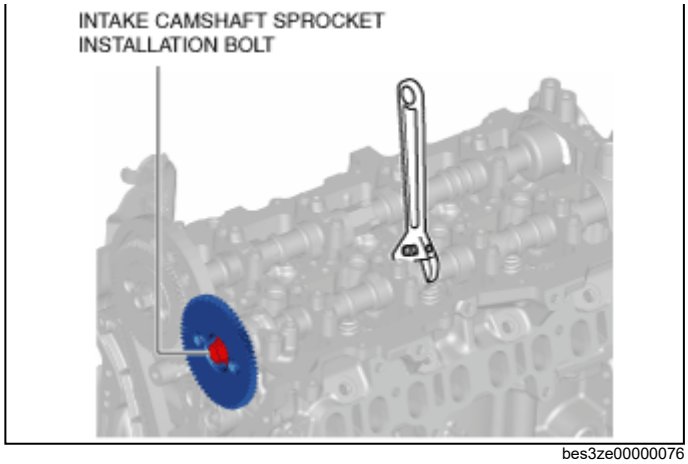
1. Remove the oil seal using a flathead screwdriver with the tip protected by a clean cloth.



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Intake Camshaft Sprocket Disassembly Note

1. Hold the intake camshaft using a wrench on the cast hexagon, and remove the intake camshaft sprocket installation bolt.



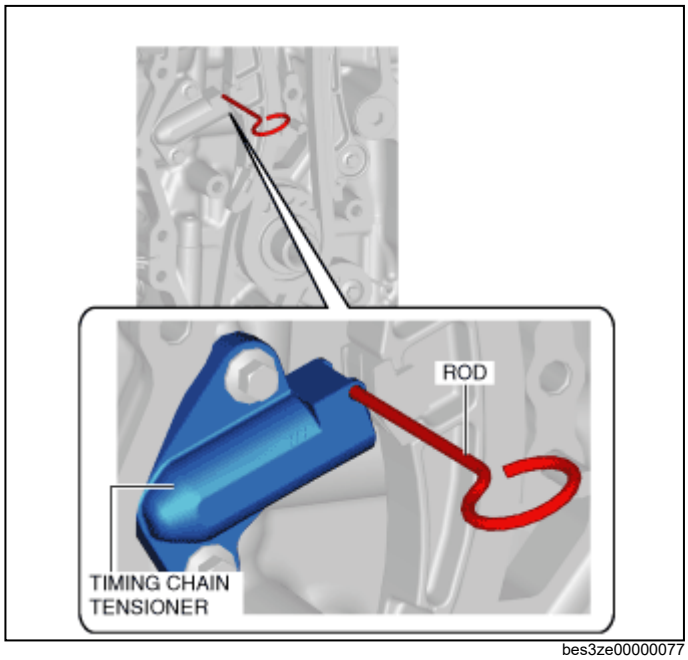
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2. Remove the intake camshaft sprocket.

Timing Chain Disassembly Note

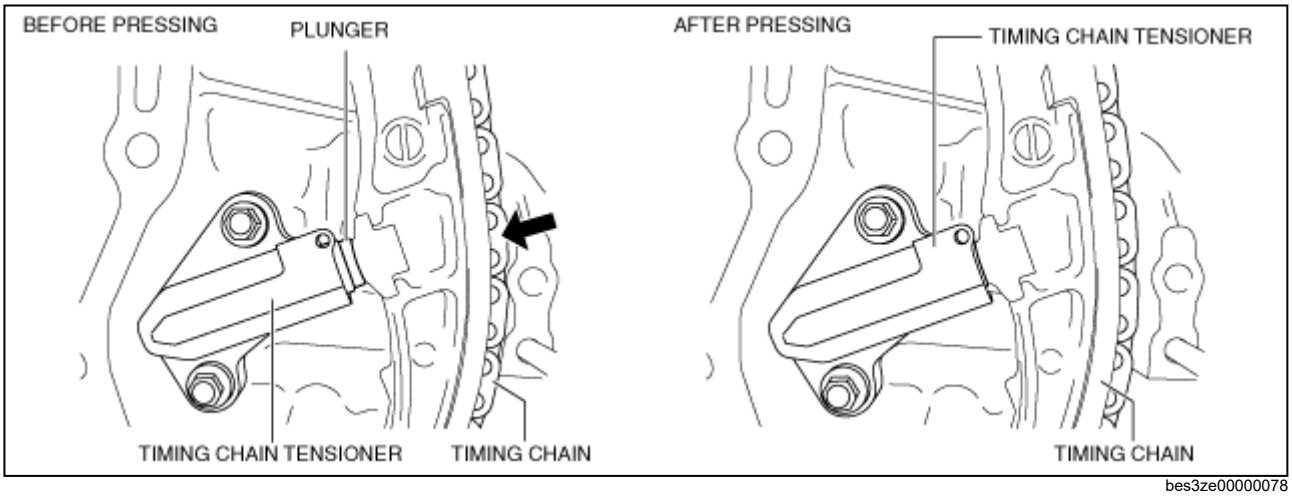
1. Loosen the timing chain tensioner using the following procedure:

(1) Insert a metal rod (diameter approx. 3.2 mm {0.13 in}, length approx. 60—70 mm {2.4—2.7 in}) into the hole in the body of the timing chain tensioner.



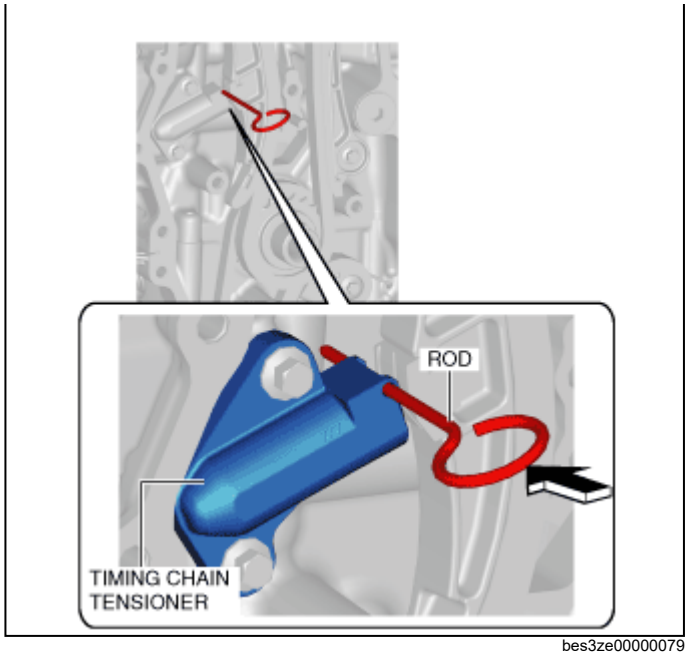
bes3ze00000077

(2) Press the timing chain in the direction of the arrow and press on the plunger of the timing chain tensioner.



bes3ze00000078

(3) With the plunger pressed in, further insert the metal rod set in (1) above.



bes3ze00000079

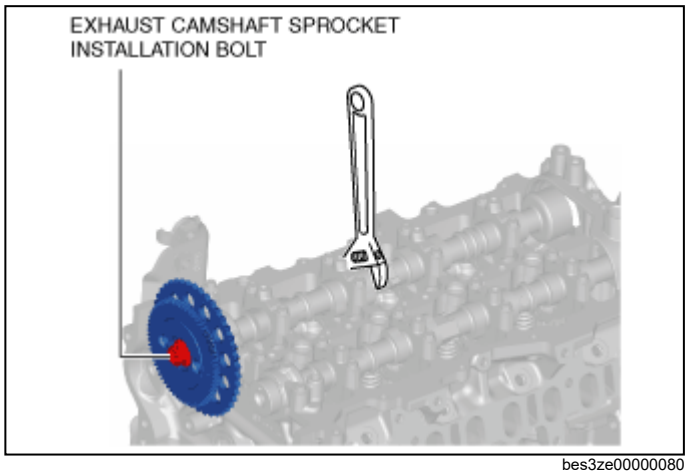
Note
- The rod secures the plunger and releases the tension.

(4) Loosen the power of the hand pressing the plunger and verify that the pressed-in rod does not move.

2. Remove the timing chain tensioner and timing chain tensioner arm.
3. Remove the timing chain guide.
4. Remove the timing chain and crankshaft sprocket as a single unit.

Exhaust Camshaft Sprocket Disassembly Note

1. Hold the exhaust camshaft using a wrench on the cast hexagon, and remove the exhaust camshaft sprocket installation bolt.



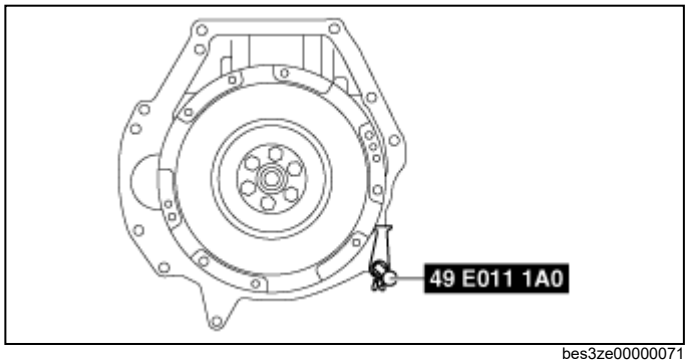
bes3ze00000080

2. Remove the exhaust camshaft sprocket.

Oil Pump Driven sprocket Disassembly Note

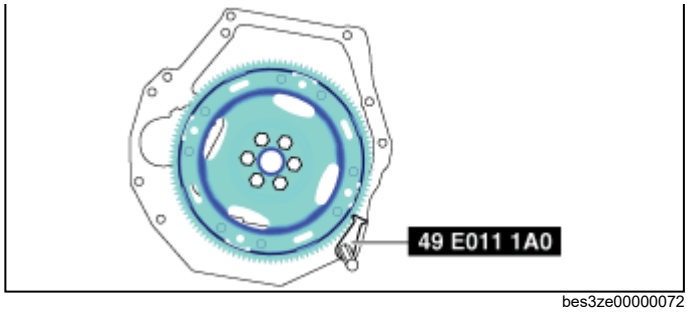
1. Hold the crankshaft using the SST.

MTX



bes3ze00000071

ATX



bes3ze00000072

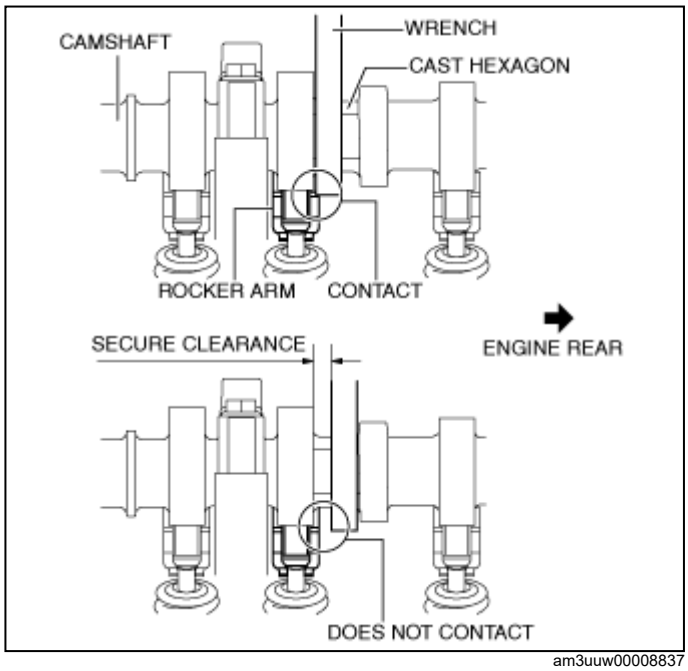
2. Remove the oil pump driven sprocket.

CYLINDER HEAD DISASSEMBLY (I)

id011000500400

Caution

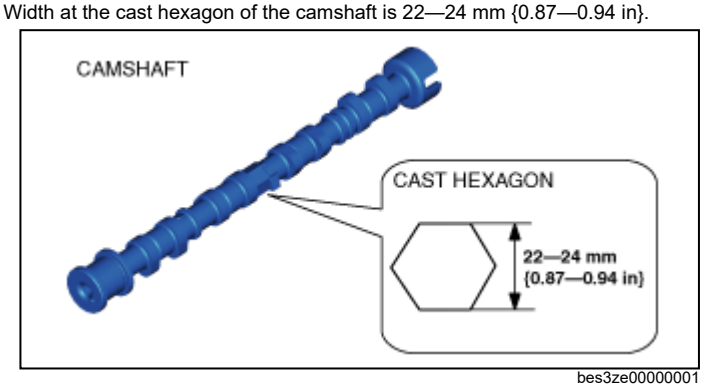
• When rotating the camshaft using a wrench on the cast hexagon, the wrench may contact the rocker arm and damage the rocker arm. To prevent damage to the rocker arm when holding the camshaft on the cast hexagon, use the wrench at engine rear side as shown in the figure to secure a clearance between the cam.



am3uuw00008837

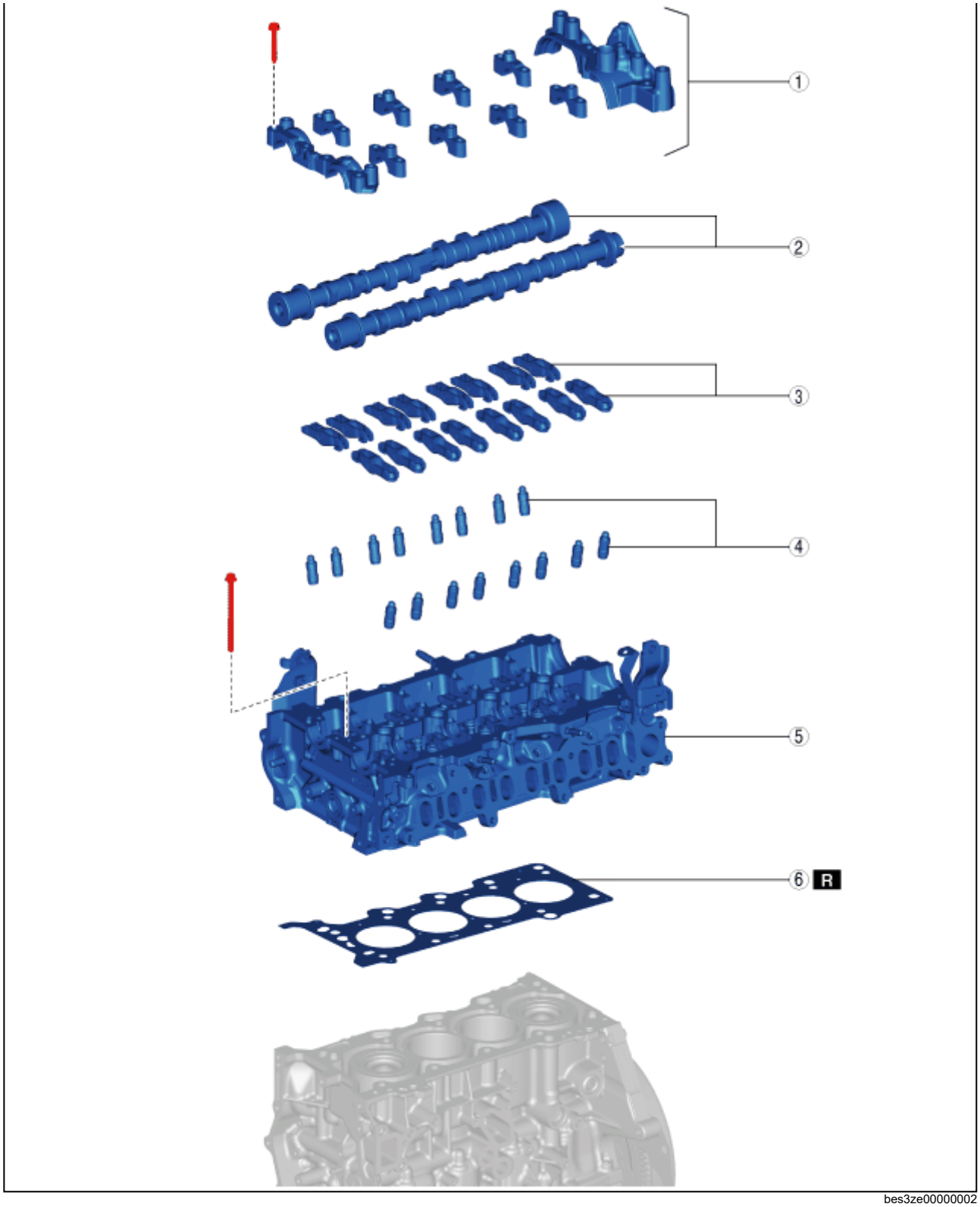
Note

• Width at the cast hexagon of the camshaft is 22—24 mm {0.87—0.94 in}.



bes3ze00000001

1. Disassemble in the order indicated in the table.

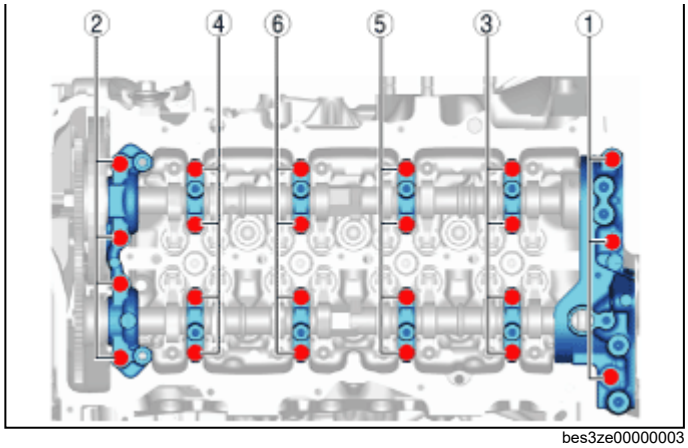


bes3ze00000002

1	Camshaft cap (See Camshaft Cap Disassembly Note.)
2	Camshaft
3	Rocker arm (See Rocker Arm Disassembly Note.)
4	HLA (See HLA Disassembly Note.)
5	Cylinder head (See Cylinder Head Disassembly Note.)
6	Cylinder head gasket

Camshaft Cap Disassembly Note

1. Before removing the camshaft cap, inspect the camshaft end play. (See CAMSHAFT INSPECTION.)
2. Loosen the camshaft cap installation bolts in two or three passes in the order shown in the figure and remove the camshaft caps.



bes3ze00000003

Rocker Arm Disassembly Note

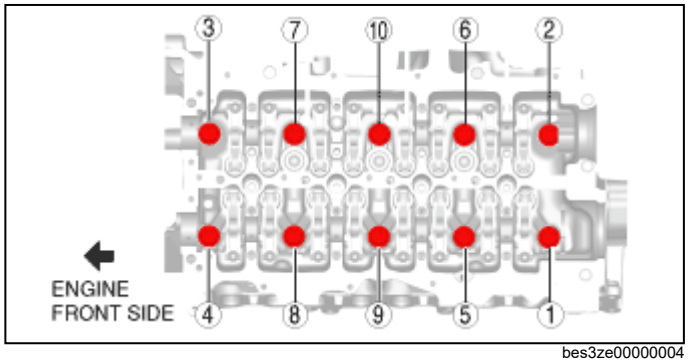
1. Keep the rocker arms in the order of removal to enable reassembly in their original positions.

HLA Disassembly Note

1. Keep the HLAs in the order of removal to enable reassembly in their original positions.

Cylinder Head Disassembly Note

1. Loosen the cylinder head installation bolts in two or three passes in the order shown in the figure and remove them.



bes3ze00000004

Caution

- The end of the glow plug protrudes from the cylinder head (tightened surface of cylinder block). When removing/installing the cylinder head and placing the cylinder head on a workbench, be careful not to allow the end of the glow plug to contact the cylinder block or workbench. Otherwise, the end of the glow plug could be damaged.

2. Remove the cylinder head.

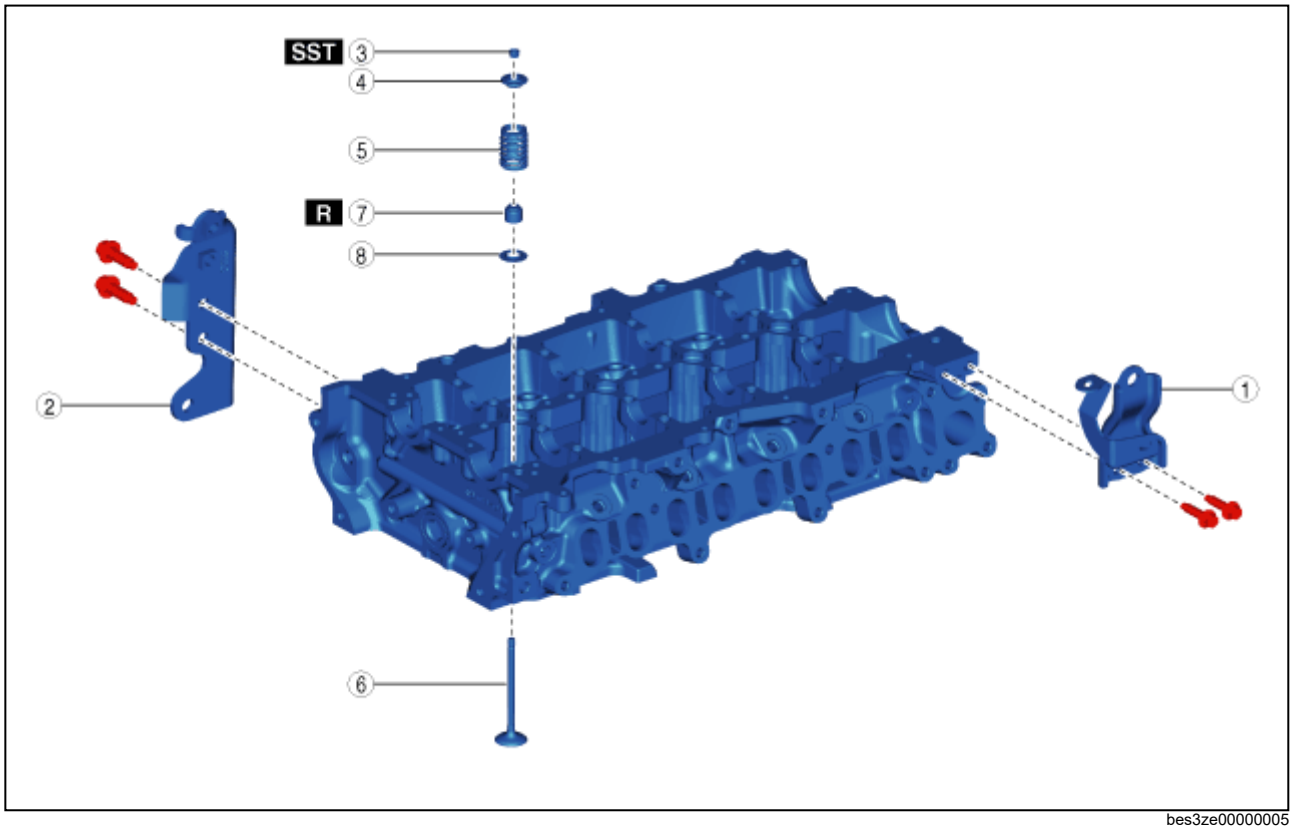
CYLINDER HEAD DISASSEMBLY (II)

id011000500500

Caution

• The end of the glow plug protrudes from the cylinder head (tightened surface of cylinder block). When removing/installing the cylinder head and placing the cylinder head on a workbench, be careful not to allow the end of the glow plug to contact the cylinder block or workbench. Otherwise, the end of the glow plug could be damaged.

1. Disassemble in the order indicated in the table.



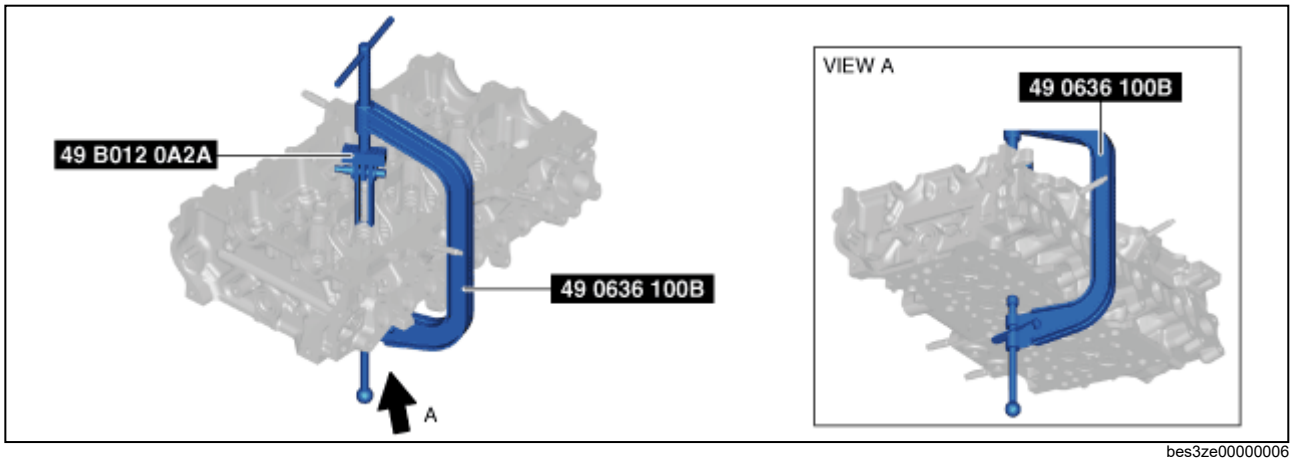
bes3ze00000005

1	Seal cover bracket
2	Hanger bracket
3	Valve keeper (See Valve Keeper Disassembly Note.)
4	Upper valve spring seat
5	Valve spring
6	Valve
7	Valve seal (See Valve Seal Disassembly Note.)
8	Lower valve spring seat

Valve Keeper Disassembly Note

When using the SSTs (49 0636 100B, 49 B012 0A2A)

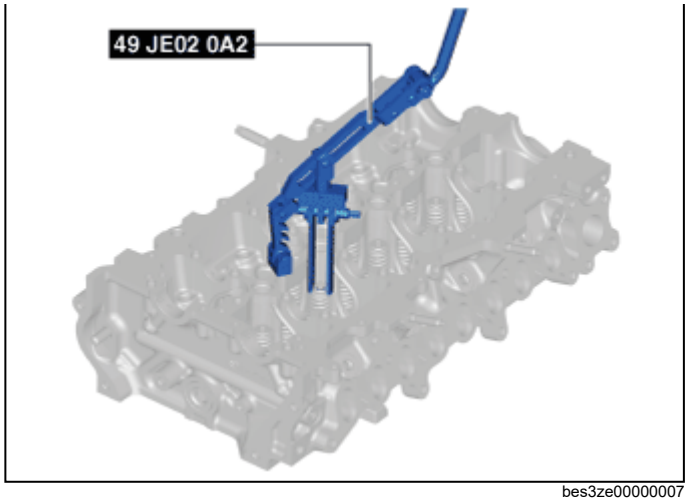
1. Remove the valve keepers using the SSTs.



bes3ze00000006

When using the SST (49 JE02 0A2) [Europe only]

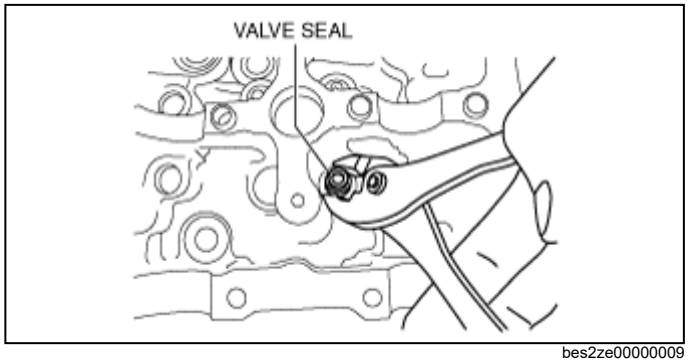
1. Remove the valve keepers using the SST.



bes3ze00000007

Valve Seal Disassembly Note

1. Remove the valve seals using the pliers.



bes2ze00000009

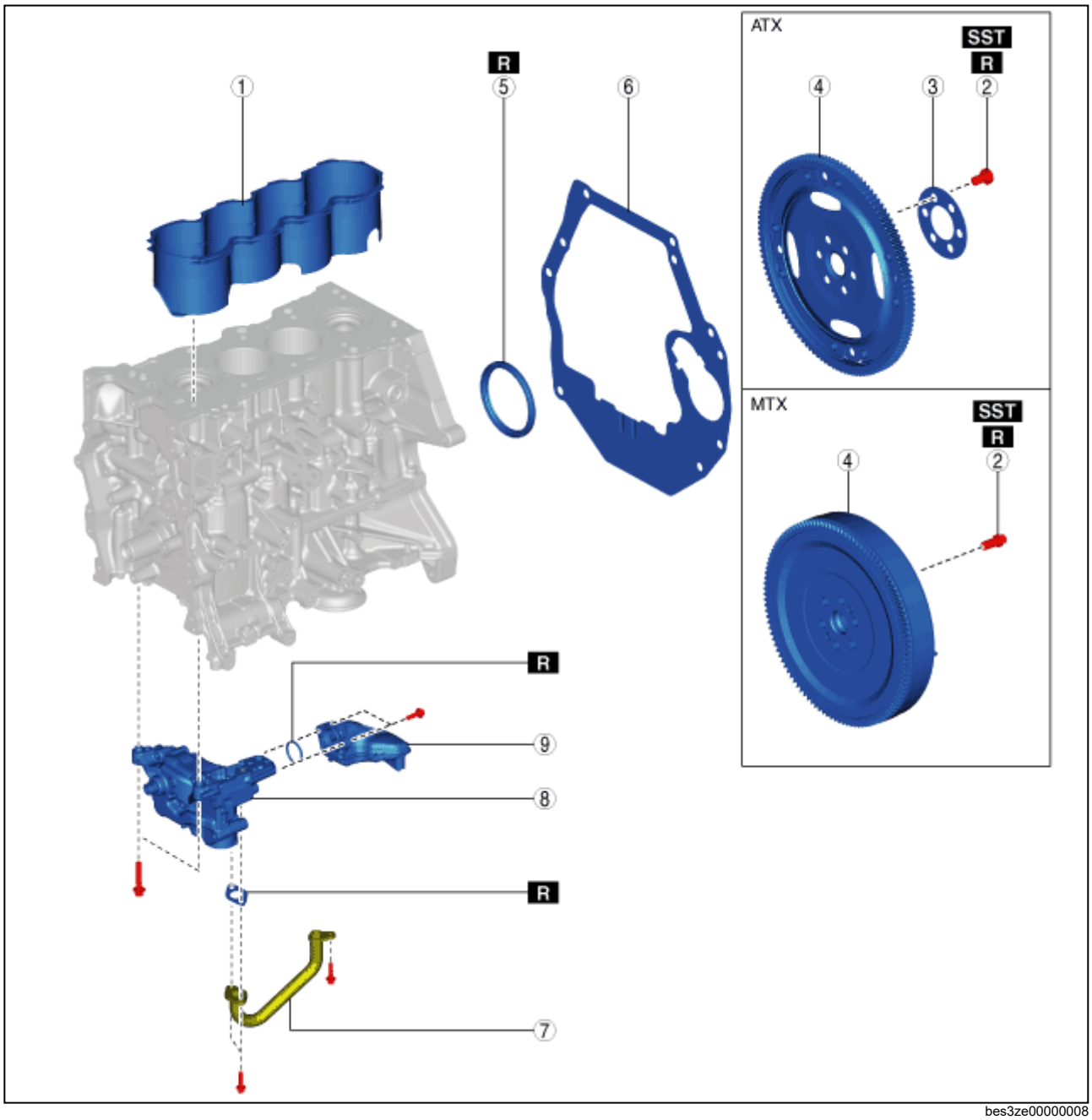
CYLINDER BLOCK DISASSEMBLY (I)

id011000500600

Caution

- Do not disassemble the oil pump because it is a precision unit.

1. Disassemble in the order indicated in the table.



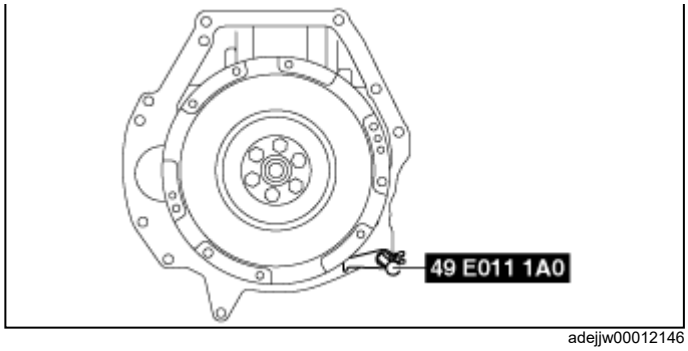
bes3ze00000008

1	Water jacket spacer
2	Dual-mass flywheel (MTX)/drive plate (ATX) installation bolt (See Dual-mass Flywheel (MTX)/Drive Plate (ATX) Installation Bolt Disassembly Note.)
3	Backing plate (ATX)
4	Dual-mass flywheel (MTX)/drive plate (ATX)
5	Rear oil seal (See Rear Oil Seal Disassembly Note.)
6	End plate
7	Oil pipe
8	Oil pump
9	Oil strainer

Dual-mass Flywheel (MTX)/Drive Plate (ATX) Installation Bolt Disassembly Note

MTX

1. Hold the dual-mass flywheel using the SST (49 E011 1A0).

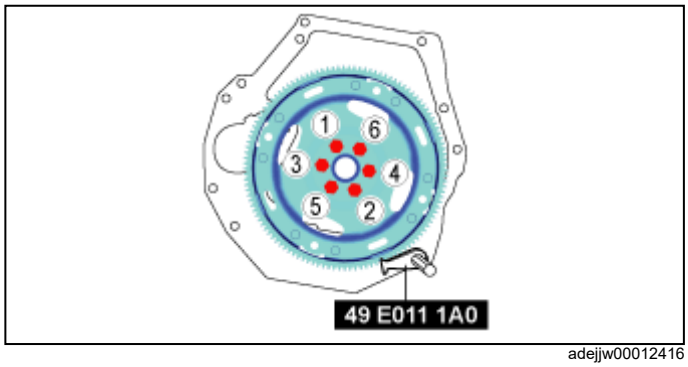


adejjw00012146

2. Remove the lock bolts, and remove the plate and the dual-mass flywheel.

ATX

1. Set the SST against the drive plate.

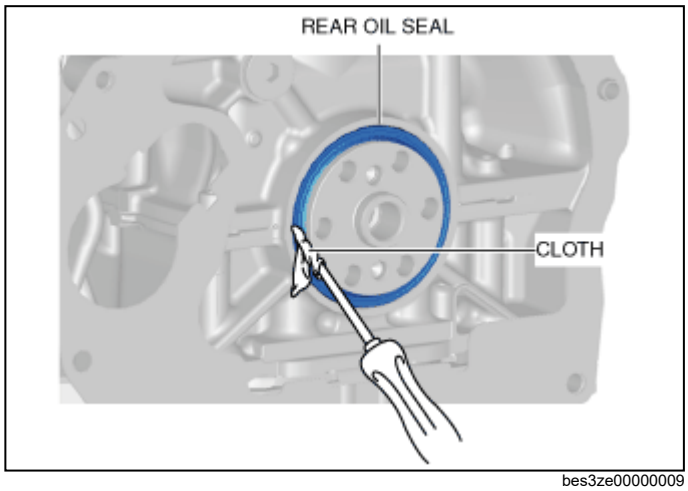


adejjw00012416

2. Loosen the drive plate installation bolts uniformly in two or three steps in the order shown in the figure, then remove the bolts and the drive plate.

Rear Oil Seal Disassembly Note

1. Cut the oil seal lip using a utility knife.
2. Remove the oil seal using a flathead screwdriver with the tip protected by a clean cloth to prevent damage to the oil seal sliding part of the crankshaft.

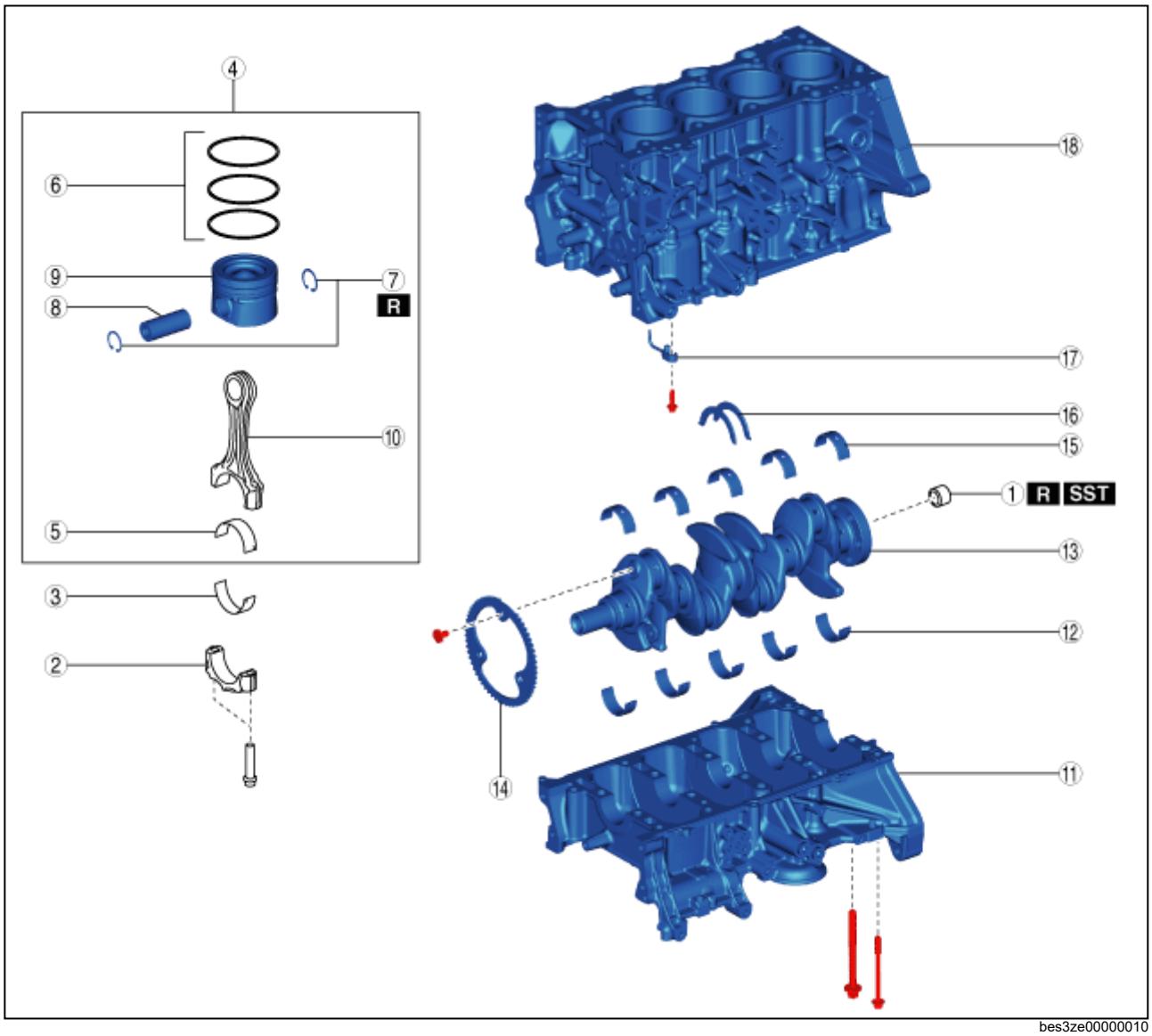


bes3ze00000009

CYLINDER BLOCK DISASSEMBLY (II)

id011000500700

1. Disassemble in the order indicated in the table.



bes3ze00000010

1	Pilot bearing (MTX) (See Pilot Bearing Disassembly Note.)
2	Connecting rod cap (See Connecting Rod Cap Disassembly Note.)
3	Lower connecting rod bearing (See Connecting Rod Bearing Disassembly Note.)
4	Piston, connecting rod (See Piston, Connecting Rod Disassembly Note.)
5	Upper connecting rod bearing (See Connecting Rod Bearing Disassembly Note.)
6	Piston ring
7	Snap ring (See Snap Ring Disassembly Note.)
8	Piston pin
9	Piston
10	Connecting rod
11	Lower cylinder block (See Lower Cylinder Block Disassembly Note.)
12	Lower main bearing (See Thrust Bearing And Main Bearing Disassembly Note.)
13	Crankshaft (See Crankshaft Disassembly Note.)
14	Plate
15	Upper main bearing (See Thrust Bearing And Main Bearing Disassembly Note.)
16	Thrust bearing (See Thrust Bearing And Main Bearing Disassembly Note.)

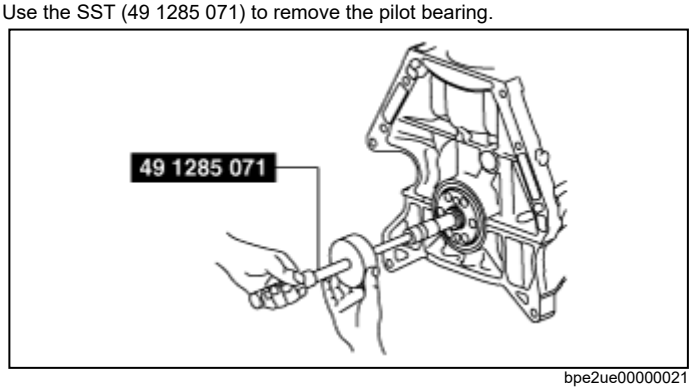
17	Oil jet valve
18	Upper cylinder block

Pilot Bearing Disassembly Note

Note

- The pilot bearing does not need to be removed unless you are replacing it.

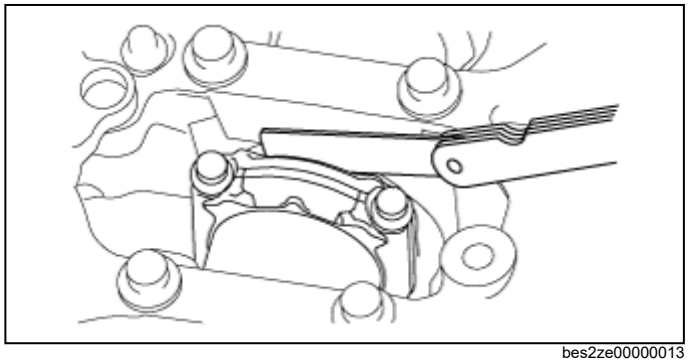
1. Use the SST (49 1285 071) to remove the pilot bearing.



bpe2ue00000021

Connecting Rod Cap Disassembly Note

1. Before removing the connecting rod cap, inspect the connecting rod side clearance. (See CONNECTING ROD CLEARANCE INSPECTION.)



bes2ze00000013

2. The removed connecting rod caps are to be kept so that they can be assembled to the same positions and in the direction as before removal.

Connecting Rod Bearing Disassembly Note

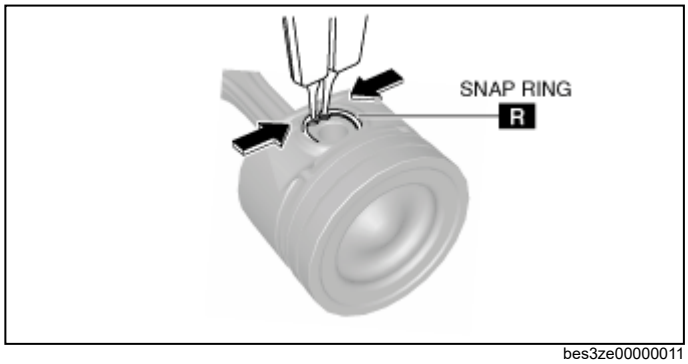
1. The removed connecting rod bearings are to be kept so that they can be assembled to the same positions and in the direction as before removal.

Piston, Connecting Rod Disassembly Note

1. Before removing the piston and connecting rod, remove the carbon in the cylinder.
2. Before removing the piston and connecting rod, inspect the oil clearance at the large end of the connecting rod. (See CONNECTING ROD CLEARANCE INSPECTION.)

Snap Ring Disassembly Note

1. Before removing the snap ring, inspect that the large end of connecting rod drops under its own weight with no resistance. (See PISTON AND CONNECTING ROD INSPECTION.)
2. Remove the snap ring using a pair of thin pliers.



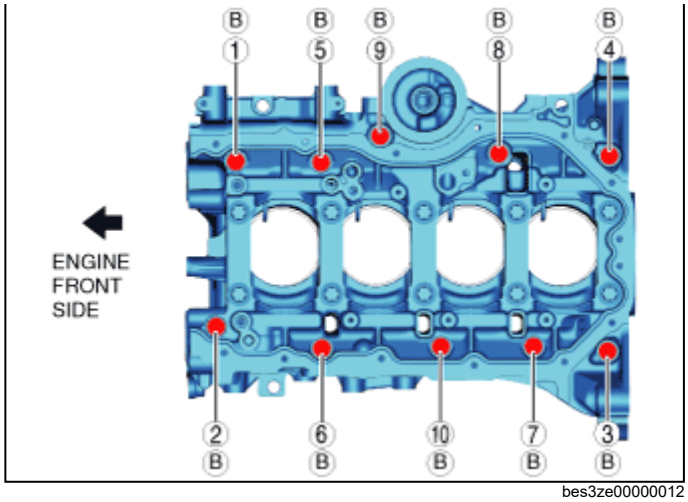
bes3ze00000011

Lower Cylinder Block Disassembly Note

Caution

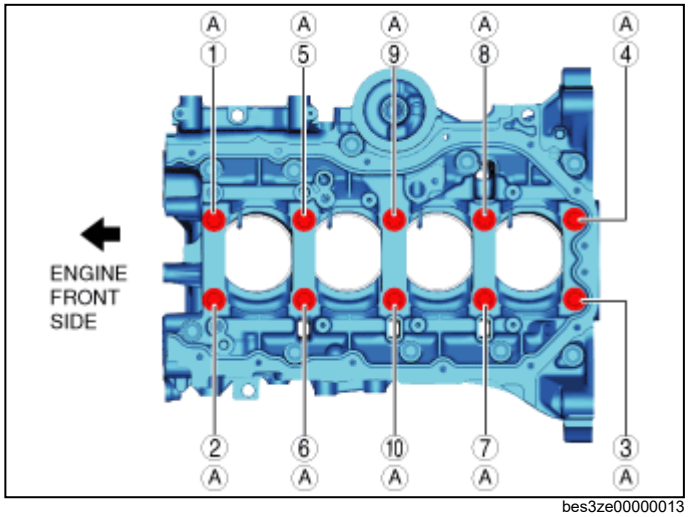
- When the lower cylinder block is removed and re-installed, the fuel injection amount learning must be performed after installing the engine to the vehicle. For the fuel injection amount learning procedure, refer to “FUEL INJECTION AMOUNT LEARNING” in the workshop manual.

1. Before removing the lower cylinder block, inspect the crankshaft end play. (See CRANKSHAFT INSPECTION.)
2. Loosen the lower cylinder block installation bolts B in two or three passes in the order shown in the figure and remove them.



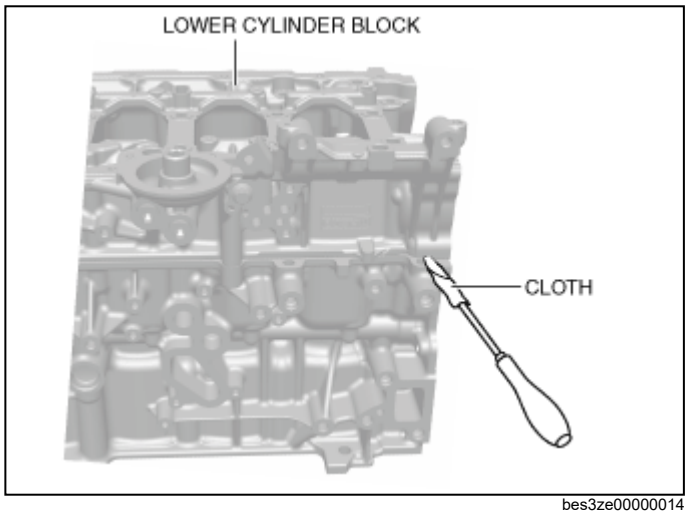
bes3ze00000012

3. Loosen the lower cylinder block installation bolts A in two or three passes in the order shown in the figure and remove them.



bes3ze00000013

4. Using a screwdriver wrapped in a cloth, peel the sealant away a little at a time, and remove the lower cylinder block.



bes3ze00000014

Caution

- Do not apply excessive force to the screwdriver. Otherwise, the lower cylinder block could be damaged.
- Be careful not to scratch or damage the seal surface. Otherwise, it could cause oil leakage.

Thrust Bearing And Main Bearing Disassembly Note

1. The removed thrust bearings and main bearings are to be kept so that they can be assembled to the same positions and in the direction as before removal.

Crankshaft Disassembly Note

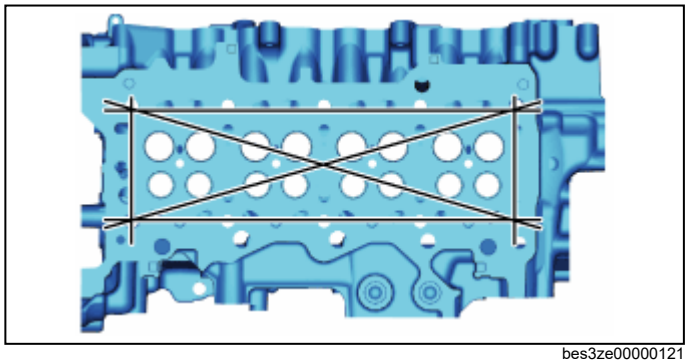
Caution

- If the plate for the crankshaft position sensor signal detection, which is installed to the crankshaft, contacts the disassembly bench, it could be deformed or damaged. Therefore, set wood blocks or similar objects on both sides of the crankshaft so that the plate does not contact the disassembly bench directly when placing the crankshaft on the disassembly bench.

CYLINDER HEAD INSPECTION

id011000507100

1. Inspect the cylinder head surface for cracks and other damage using a red dye penetrant.
    - If there is a malfunction, replace the cylinder head.
2. Measure the combustion chamber side of the cylinder head for distortion in six directions as shown in the figure using a straight edge and feeler gauge.



bes3ze00000121

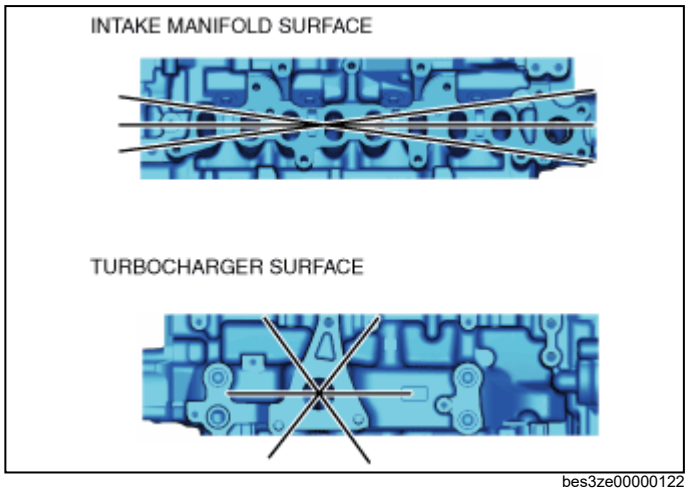
- If the distortion exceeds the maximum specification, replace the cylinder head.

Note
- Polishing and grinding for repair cannot be performed on the combustion chamber side.

Maximum distortion, head gasket side of the cylinder head

0.06 mm {0.002 in}

3. Inspect the distortion of the intake manifold and turbocharger installation surfaces using a straight edge and feeler gauge. Perform the inspection at each of the three locations as shown in the figure.



bes3ze00000122

- If the distortion of the intake manifold and turbocharger surface is more than the maximum specification, replace the cylinder head.

Note
- Polishing and grinding for repair cannot be performed on the intake manifold and turbocharger surface.

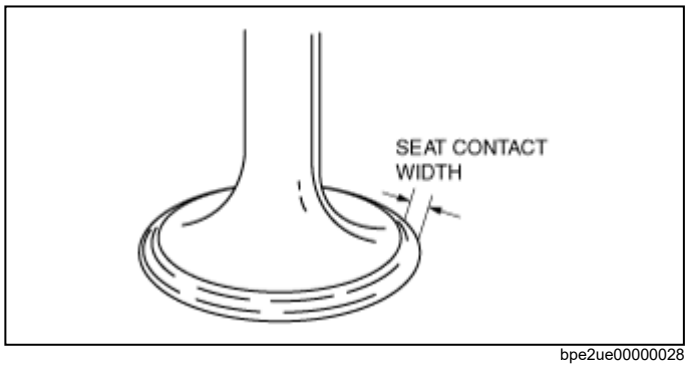
Maximum distortion of intake manifold and turbocharger installation surface

Intake manifold installation surface: 0.05 mm {0.002 in}
Turbocharger installation surface: 0.05 mm {0.002 in}

VALVE SEAT INSPECTION/REPAIR

id011000501300

1. Measure the contact width of the valve face and the valve seat using the valve lapping compound.



bpe2ue00000028

- If it is not within the specification, resurface the valve seat using the 45° valve seat cutter.

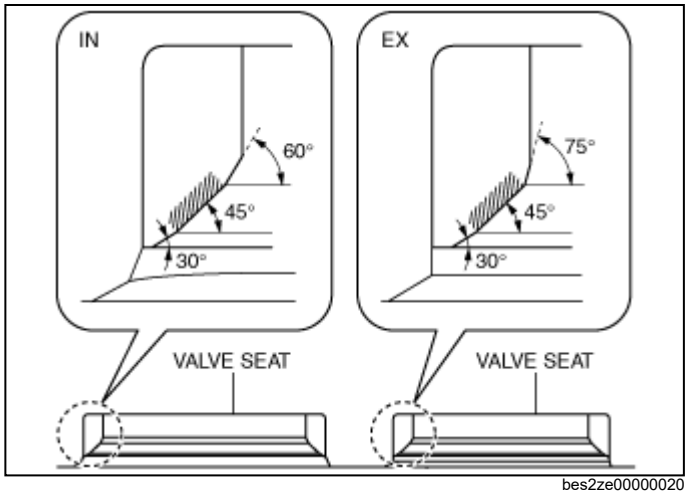
Standard valve seat contact width

1.2—2.0 mm {0.048—0.078 in}

Valve seat angle

45°

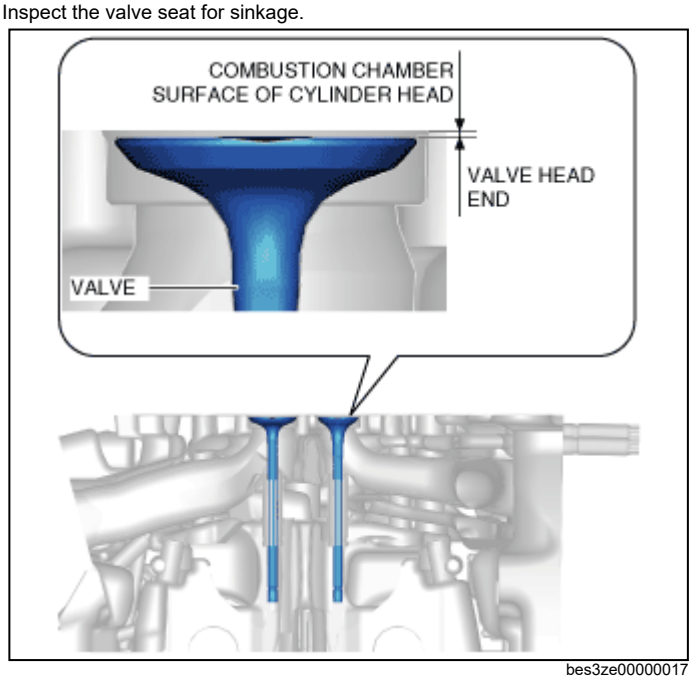
2. Verify that the area where the valve seat contacts the valve face is centered.



bes2ze00000020

- If the seating position is too high, correct the valve seat using a 60° (IN) 75°(EX) valve seat cutter and a 45° valve seat cutter.
- If the seating position is too low, correct the valve seat using a 30° valve seat cutter and a 45° valve seat cutter.

3. Inspect the valve seat for sinkage.



bes3ze00000017

- If it is not within the specification, replace the cylinder head.

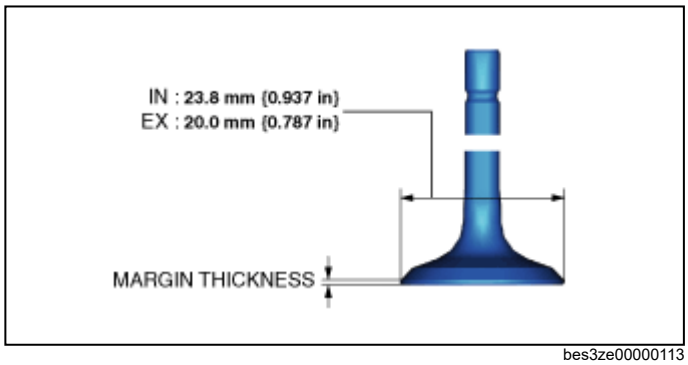
Standard valve seat sinkage amount

IN: 0.39—0.79 mm {0.016—0.031 in}
EX: 0.47—0.87 mm {0.019—0.034 in}

VALVE, VALVE GUIDE INSPECTION

id011000506200

1. Measure the valve head margin thickness using a vernier caliper.



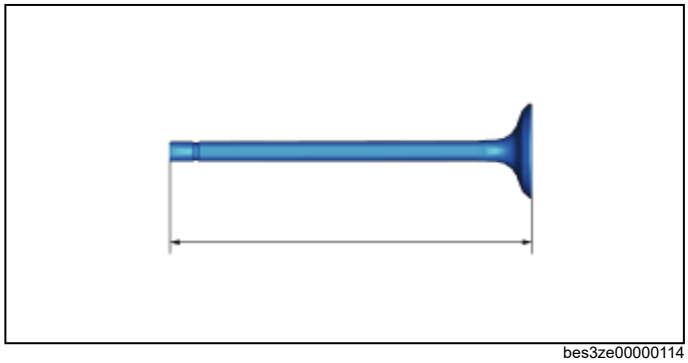
bes3ze00000113

- If it is less than the standard specification, replace the valve.

Standard valve head margin thickness

IN: 1.50—1.70 mm {0.0591—0.0669 in}
EX: 1.68—1.88 mm {0.0662—0.0740 in}

2. Measure the length of the each valve using a vernier caliper.



bes3ze00000114

- If it is less than the minimum specification, replace the valve.

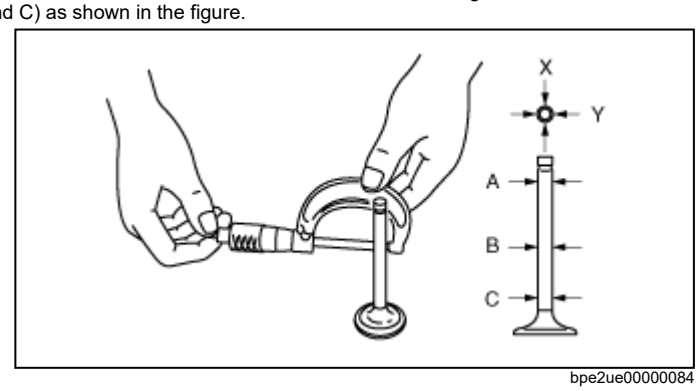
Standard valve length

IN: 97.07—97.37 mm {3.822—3.833 in}
EX: 96.99—97.29 mm {3.819—3.830 in}

Minimum valve length

IN: 96.85 mm {3.813 in}
EX: 96.77 mm {3.810 in}

3. Measure the valve stem diameter of each valve using the micrometer. Measurement positions total six and are in the X and Y directions, at three points (A, B, and C) as shown in the figure.



bpe2ue00000084

- If it is less than the minimum specification, replace the valve.

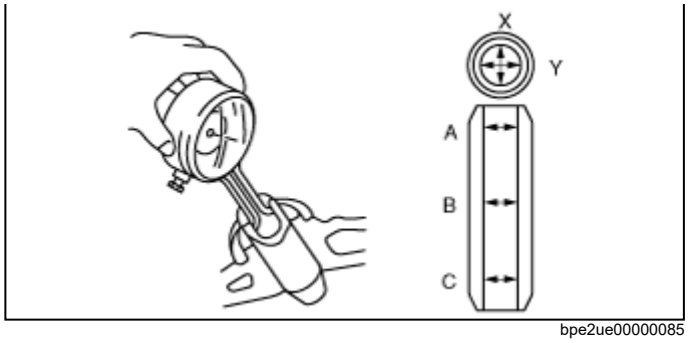
Standard valve stem diameter

IN: 5.470—5.485 mm {0.2154—0.2159 in}
EX: 5.465—5.480 mm {0.2152—0.2157 in}

Minimum valve stem diameter

IN: 5.424 mm {0.2135 in}
EX: 5.419 mm {0.2133 in}

4. Measure the inner diameter of each valve guide using the caliper gauge. Measurement positions total six and are in the X and Y directions, at three points (A, B, and C) as shown in the figure.



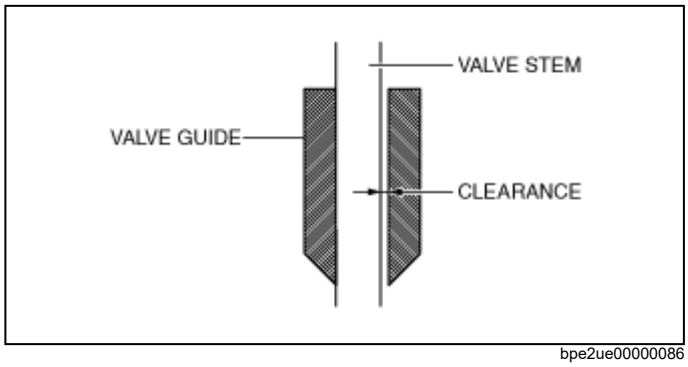
bpe2ue00000085

- If it is not within the specification, replace the valve guide.

Standard valve guide inner diameter

IN: 5.510—5.530 mm {0.2170—0.2177 in}
EX: 5.510—5.530 mm {0.2170—0.2177 in}

5. Calculate the clearance between the valve stem and the valve guide by subtracting the inner diameter of the valve guide from the outer diameter of the corresponding valve stem.



bpe2ue00000086

- If it exceeds the maximum specification, replace the valve or valve guide.

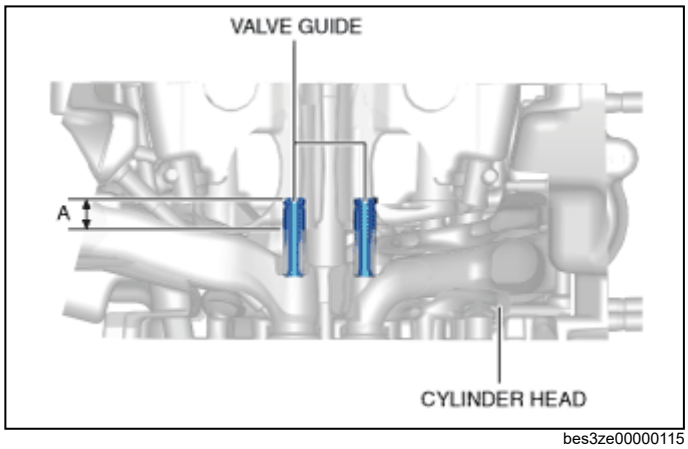
Standard clearance between valve stem and guide

IN: 0.025—0.060 mm {0.0010—0.0023 in}
EX: 0.030—0.065 mm {0.0012—0.0025 in}

Maximum clearance between valve stem and guide

0.10 mm {0.0039 in}

6. Measure the projection height (dimension A) of each valve guide using the vernier caliper.



bes3ze00000115

- If it is not within the specification, replace the valve guide.

Standard valve guide projection height

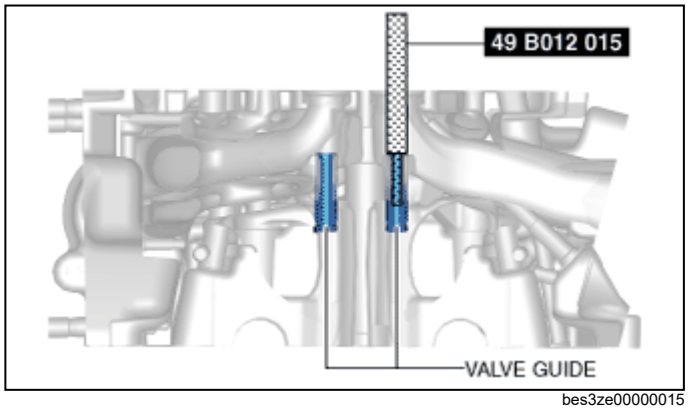
14.8—15.3 mm {0.583—0.602 in}

VALVE GUIDE REPLACEMENT

id011000501200

Removal

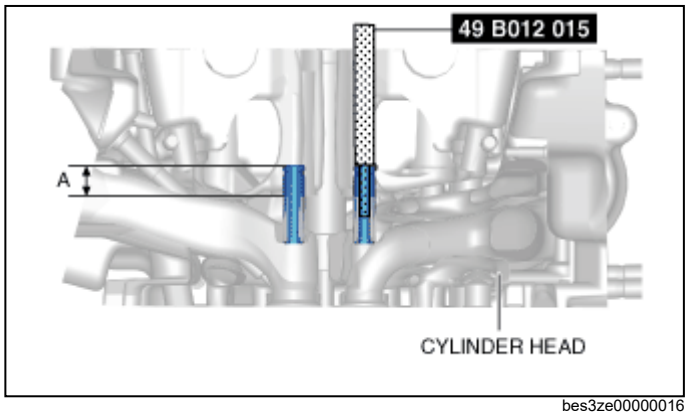
1. Tap the valve guide out from combustion chamber side using the SST.



bes3ze00000015

Installation

1. Apply clean engine oil to the valve guide.
2. Tap the valve guide from the camshaft side using the SST so that the projection height (dimension A) is within the specification.



bes3ze00000016

Standard valve guide projection height

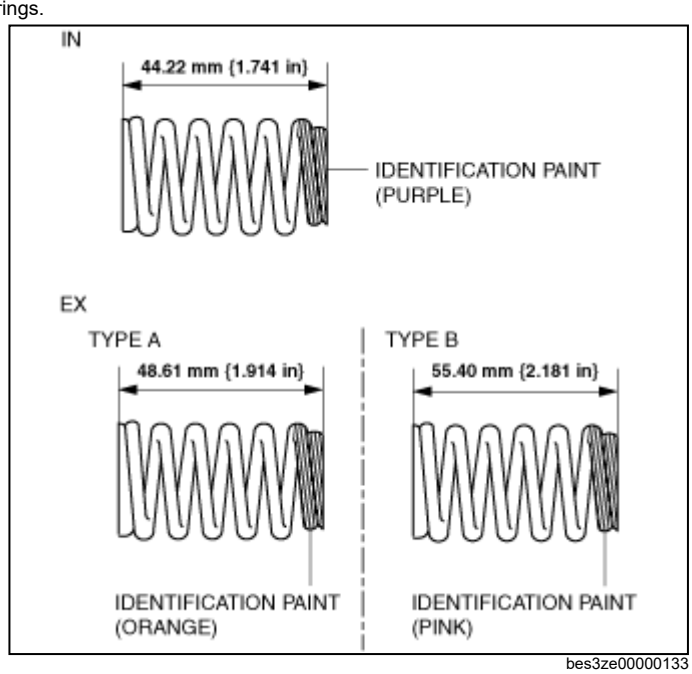
14.8—15.3 mm {0.583—0.602 in}

VALVE SPRING INSPECTION

id011000501400

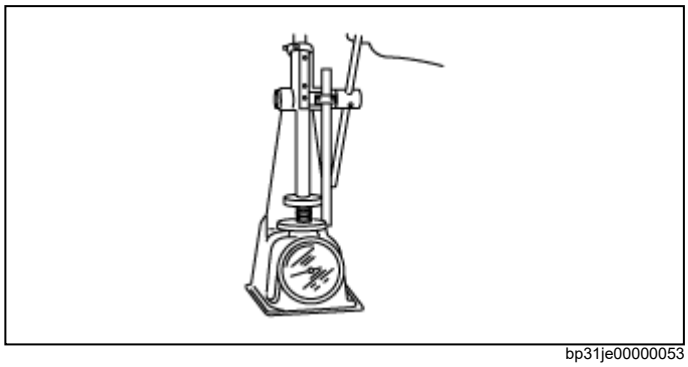
Caution

• The valve springs differ depending on the IN and EX sides. Therefore, verify the free length or identification paint beforehand and inspect the valve springs.



bes3ze00000133

1. Measure the valve spring height using the spring tester.



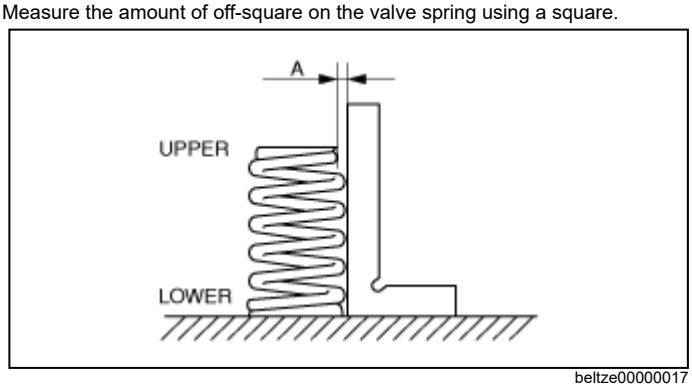
bp31je00000053

• If it is not within the specification, replace the valve spring.

Valve spring installation height

IN: When pressurized with spring force of 128.3—141.7 N {13.09—14.44 kgf, 28.85—31.85 lbf}, spring height is 35.0 mm {1.38 in}
EX (TYPE A): When pressurized with spring force of 137.8—152.2 N {14.06—15.52 kgf, 30.98—34.21 lbf}, spring height is 35.0 mm {1.38 in}
EX (TYPE B): When pressurized with spring force of 209—231 N {21.4—23.5 kgf, 47.0—51.9 lbf}, spring height is 35.0 mm {1.38 in}

2. Measure the amount of off-square on the valve spring using a square.



beltze00000017

(1) Rotate the valve spring one full turn and measure A at the point where the gap is the largest.

• If it exceeds the maximum specification, replace the valve spring.

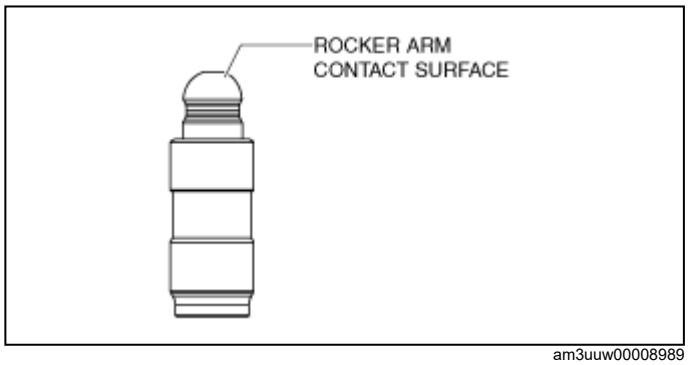
Maximum valve spring off-square

IN: 2.0 ° (1.5 mm {0.059 in})
EX (TYPE A): 2.0 ° (1.7 mm {0.067 in})
EX (TYPE B): 2.0 ° (1.9 mm {0.075 in})

HYDRAULIC LASH ADJUSTER (HLA) INSPECTION

id011000805000

1. Visually inspect the HLA surface where it contacts the rocker arm for wear or damage.



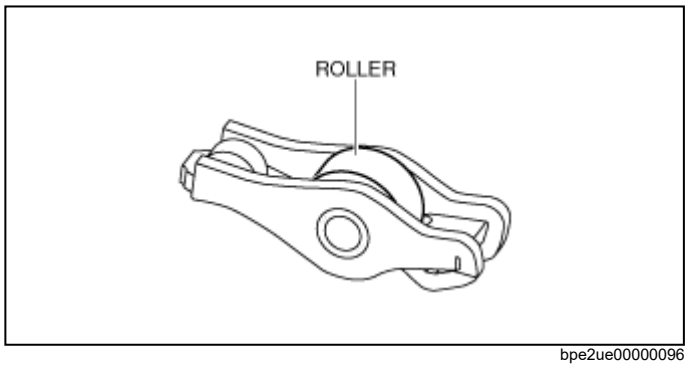
am3uuw00008989

- If there is any malfunction, replace the HLA.

ROCKER ARM INSPECTION

id011000904800

1. Rotate the roller of the rocker arm by hand and verify that rotates smoothly.



bpe2ue00000096

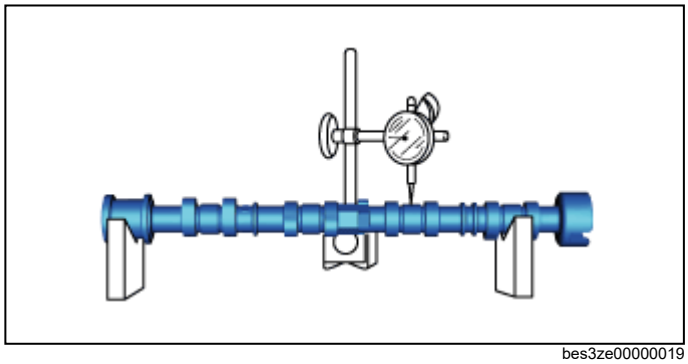
   - If it does not rotate smoothly, replace the rocker arm.

2. Visually inspect the rocker arm surface where it contacts the HLA and valve stem for wear or damage.
   - If there is any malfunction, replace the rocker arm.

CAMSHAFT INSPECTION

id011000501500

1. Set the No.1 and No.5 journals of the camshaft on V-blocks.
2. Measure the camshaft runout using the dial gauge.



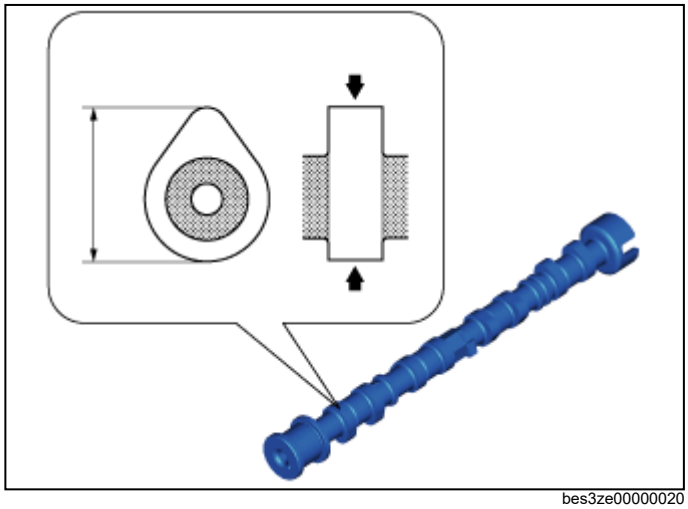
bes3ze00000019

- If it exceeds the maximum specification, replace the camshaft.

Maximum camshaft runout

0.030 mm {0.0012 in}

3. Measure the cam height using the micrometer.



bes3ze00000020

- If it is less than the minimum specification, replace the camshaft.

Standard cam height [intake camshaft]

34.12 mm {1.343 in}

Standard cam height [exhaust camshaft]

34.48 mm {1.357 in}

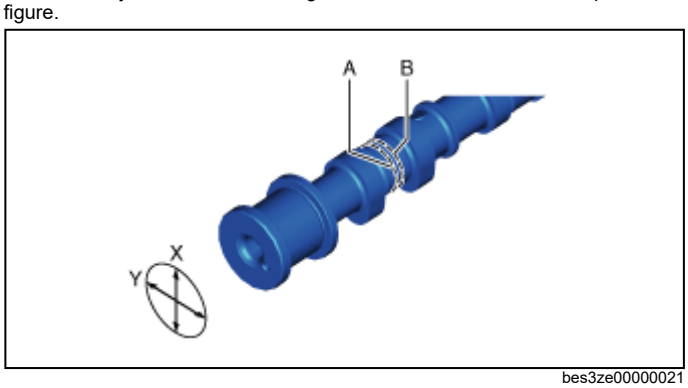
Minimum cam height [intake camshaft]

34.05 mm {1.341 in}

Minimum cam height [exhaust camshaft]

34.41 mm {1.355 in}

4. Measure the journal diameter using the micrometer. Measurement positions total four and are in the X and Y directions, at two points (A and B) as shown in the figure.



bes3ze00000021

- If it is less than the minimum specification, replace the camshaft.

Note

- The journal diameter of only the exhaust camshaft No.1 journal differs.

Standard camshaft journal diameter [intake camshaft]

No.1 journal: 33.950—33.970 mm {1.3367—1.3374 in}
Except No.1 journal: 25.950—25.975 mm {1.0217—1.0226 in}

Standard camshaft journal diameter [exhaust camshaft]

No.1 journal: 33.950—33.975 mm {1.3367—1.3375 in}
Except No.1 journal: 25.950—25.975 mm {1.0217—1.0226 in}

Minimum camshaft journal diameter

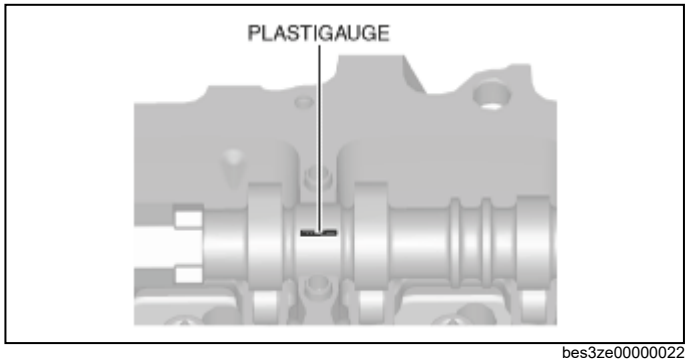
No.1 journal: 33.920 mm {1.3354 in}
Except No.1 journal: 25.920 mm {1.0205 in}

5. Measure the camshaft journal oil clearance using the following procedure:

   (1) Clean the camshaft journal and the journal receptacle part.

   (2) Put the camshaft on the cylinder head with the rocker arm detached.

   (3) Cut the plastigauge to the same length as the journal width and position it parallel to the camshaft.



bes3ze00000022

   (4) Install the camshaft caps. (See CYLINDER HEAD ASSEMBLY (II).)

   (5) Remove the camshaft caps. (See CYLINDER HEAD DISASSEMBLY (I).)

   (6) Measure the camshaft journal oil clearance.

   - If it exceeds the maximum specification, replace the cylinder head.

   Standard camshaft journal oil clearance [intake camshaft]

   No.1 journal: 0.030—0.075 mm {0.0012—0.0029 in}
   Except No.1 journal: 0.025—0.071 mm {0.0010—0.0027 in}

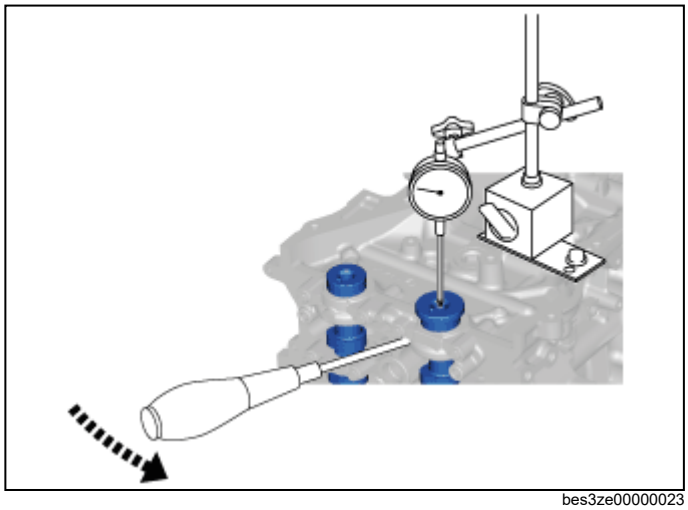
   Standard camshaft journal oil clearance [exhaust camshaft]

   No.1 journal: 0.025—0.075 mm {0.0010—0.0029 in}
   Except No.1 journal: 0.025—0.071 mm {0.0010—0.0027 in}

   Maximum camshaft journal oil clearance

   No.1 journal: 0.105 mm {0.00413 in}
   Except No.1 journal: 0.101 mm {0.00398 in}

6. Measure the camshaft end play using a dial gauge.


bes3ze00000023

- If it exceeds the maximum specification, replace the cylinder head or camshaft.

Standard camshaft end play

0.07—0.22 mm {0.003—0.008 in}

Maximum camshaft end play

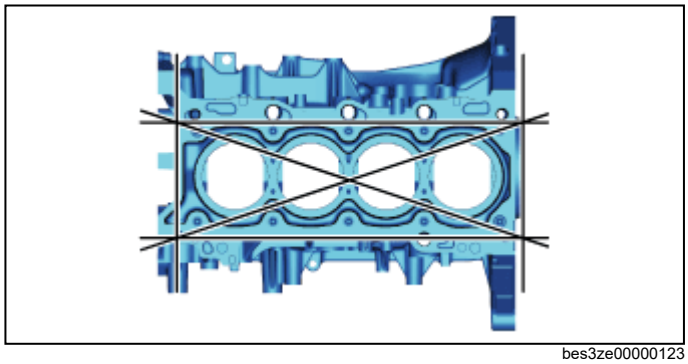
0.98 mm {0.039 in}

CYLINDER BLOCK INSPECTION

id011000507200

1. Measure the distortion of the following using a straight edge and feeler gauge.

   (1) Measure the distortion in six directions as shown in the figure.



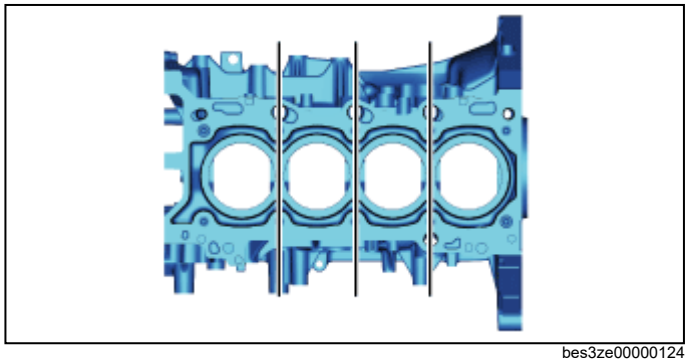
bes3ze00000123

   - If it exceeds the maximum specification, replace the cylinder block.

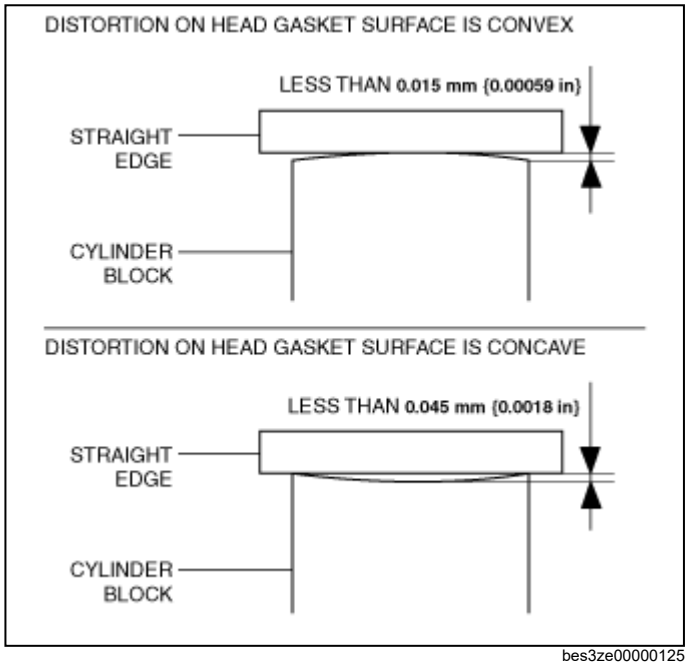
     Maximum distortion, head gasket side of the cylinder block [Other than between cylinder bores]

     0.10 mm {0.0039 in}

   (2) Measure the distortion in three directions (between cylinder bores) as shown in the figure.



bes3ze00000124



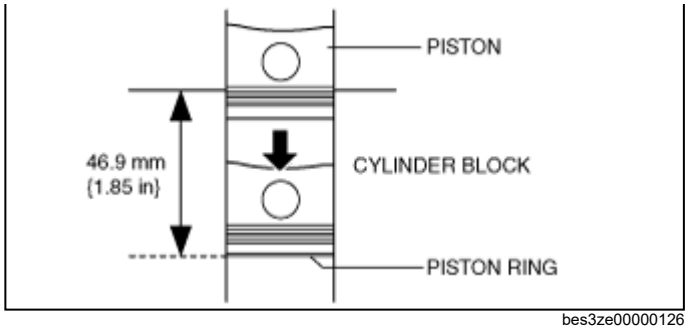
bes3ze00000125

   - If it exceeds the maximum specification, replace the cylinder block.

     Maximum distortion, head gasket side of the cylinder block [between cylinder bores]

     Distortion on head gasket surface is convex: 0.015 mm {0.00059 in}
     Distortion on head gasket surface is concave: 0.045 mm {0.0018 in}

2. Measure the cylinder bore diameter using the cylinder gauge. The measurement position is in the X and Y directions at a point 46.9 mm {1.85 in} below the top surface of the cylinder as shown in the figure.



bes3ze00000126

- If it is not within the specification, replace the cylinder block.

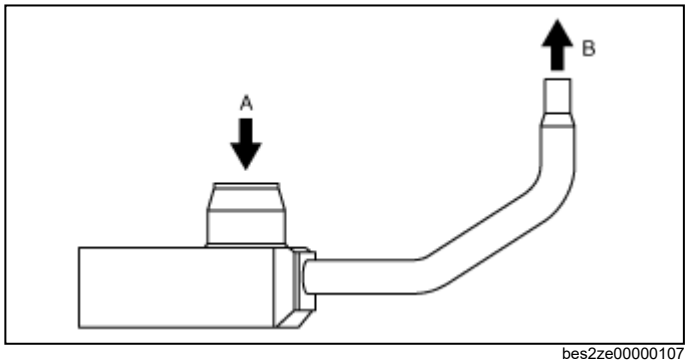
Standard cylinder bore diameter

76.000—76.022 mm {2.9922—2.9929 in}

OIL JET VALVE INSPECTION

id011000505900

1. Apply compressed air to oil jet valve A and verify that air passes through oil jet valve B.



bes2ze00000107

- If air does not flow, replace the oil jet valve.
- If there is air flow with air compressor of less than 70 kPa {0.71 kgf/cm^2, 10 psi}, replace the oil jet valve.

Oil jet valve opening pressure

70—110 kPa {0.72—1.12 kgf/cm^2, 11.0—15.9 psi}

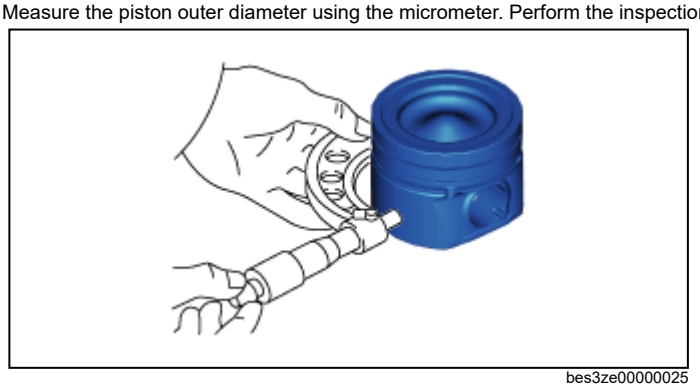
PISTON INSPECTION

id011000502100

Caution

- If the piston is replaced, replace the piston, piston pin, and the snap ring as a single component.

1. Measure the piston outer diameter using the micrometer. Perform the inspection at the following location.



bes3ze00000025

- 10.0 mm {0.394 in} from the lower end of the piston (area with no coating on piston skirt)
  - — If it is not within the specification, replace the piston.

Standard piston outer diameter

75.958—75.984 mm {2.9905—2.9914 in}

2. Measure the cylinder bore diameter. (See CYLINDER BLOCK INSPECTION.)
3. Calculate the cylinder-to-piston clearance from the cylinder bore diameter and the piston outer diameter.

- If the clearance exceeds the maximum specification, replace the piston or cylinder block.

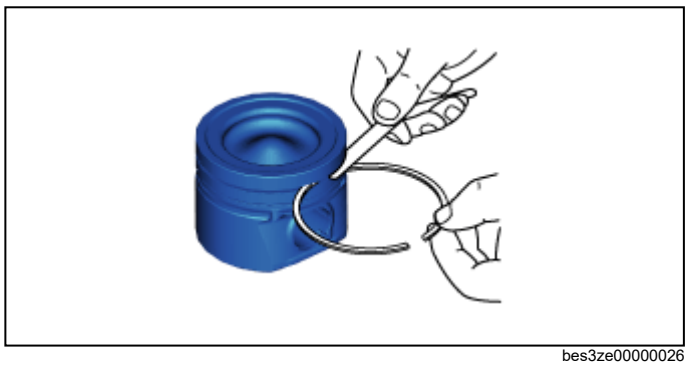
Standard clearance between piston and cylinder

0.031—0.049 mm {0.0013—0.0019 in}

Maximum clearance between piston and cylinder

0.067 mm {0.0026 in}

4. Measure the piston-to-ring groove clearance along the perimeter using a feeler gauge. For the O-ring, measure the clearance with the O-ring assembled to the piston.



bes3ze00000026

- If the clearance exceeds the maximum specification, replace the piston or piston ring.

Standard clearance between piston ring and ring groove

Top: 0.10—0.14 mm {0.0040—0.0055 in}
Second: 0.05—0.09 mm {0.002—0.003 in}
Oil: 0.05—0.09 mm {0.002—0.003 in}

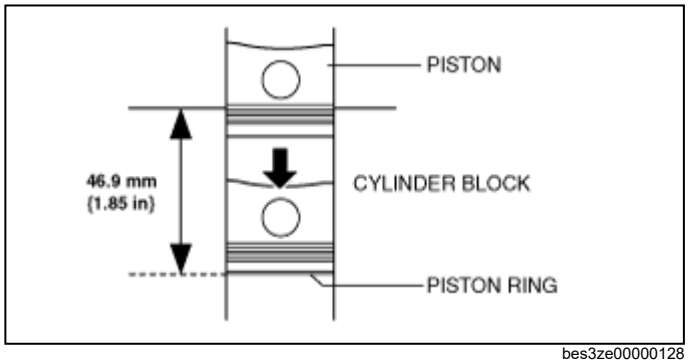
Maximum clearance between piston ring and ring groove

Top: 0.16 mm {0.0063 in}
Second: 0.12 mm {0.0047 in}
Oil: 0.11 mm {0.0043 in}

PISTON RING INSPECTION

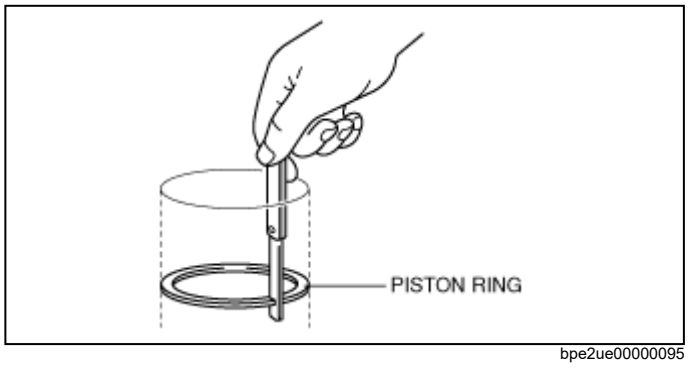
id011000592300

1. Using the piston, press the piston ring parallel into the cylinder to 46.9 mm {1.85 in} from the upper end of the cylinder block.



bes3ze00000128

2. Measure the piston ring end gap using a feeler gauge.



bpe2ue00000095

- If it exceeds the maximum specification, replace the piston ring.

Standard piston ring end gap

Top: 0.17—0.22 mm {0.0067—0.0086 in}
Second: 0.17—0.27 mm {0.007—0.010 in}
Oil (rail): 0.15—0.25 mm {0.0060—0.0098 in}

Maximum piston ring end gap

Top: 0.38 mm {0.015 in}
Second: 0.49 mm {0.019 in}
Oil (rail): 0.46 mm {0.018 in}

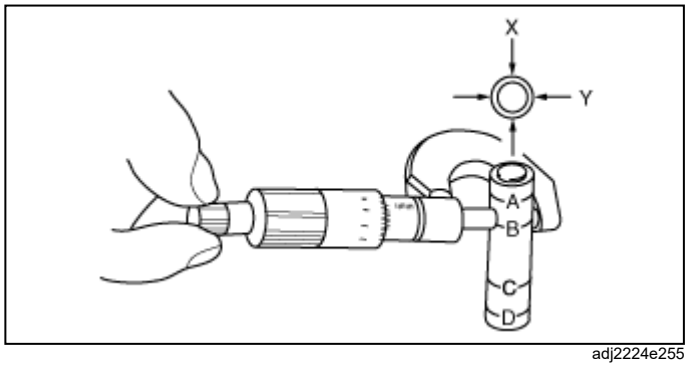
PISTON PIN INSPECTION

id011000507500

Caution

- If the piston or piston pin is replaced, replace the piston, piston pin and the snap ring as a single component.

1. Measure the piston pin outer diameter using the micrometer. Measurement positions total eight and are in the X and Y directions, at four points (A, B, C, and D) as shown in the figure.

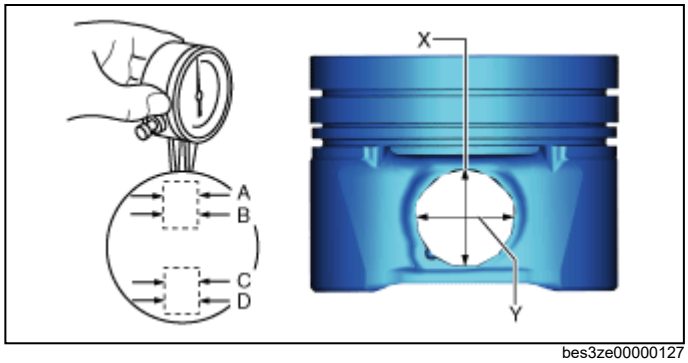


adj2224e255

Standard piston pin outer diameter

23.995—24.000 mm {0.94469—0.94488 in}

2. Measure the piston pin hole diameter using the caliper gauge. Measurement positions total eight and are in the X and Y directions, at four points (A, B, C, and D) as shown in the figure.



bes3ze00000127

Standard piston pin hole diameter

24.005—24.013 mm {0.94508—0.94539 in}

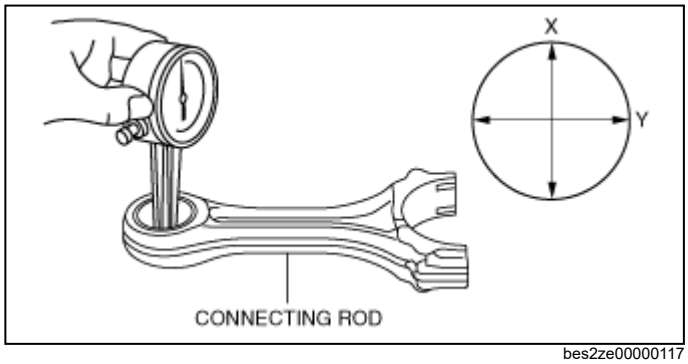
3. Calculate the clearance between the piston pin hole diameter and the piston pin outer diameter.

- If it is not within the specification, replace the piston or the piston pin.

Standard clearance between piston pin hole diameter and piston pin outer diameter

0.005—0.018 mm {0.0002—0.0007 in}

4. Measure the inner diameter on the small end of the connecting rod using the caliper gauge in the X and Y directions as shown in the figure.



bes2ze00000117

Standard connecting rod small end inner diameter

24.010—24.021 mm {0.94528—0.94570 in}

5. Calculate the clearance between the inner diameter on the small end of the connecting rod and the piston pin outer diameter.

- If it is not within the specification, replace the connecting rod or the piston pin.

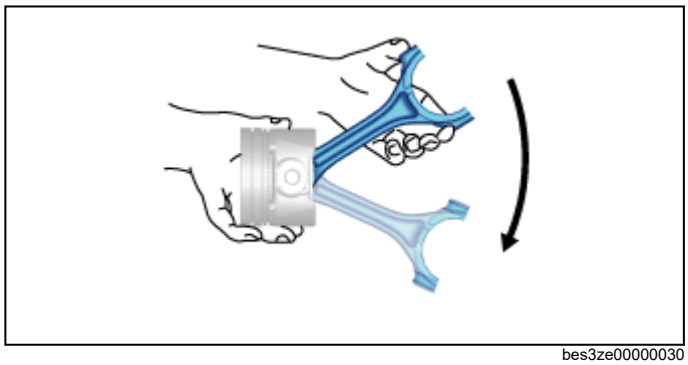
Standard clearance between connecting rod small end inner diameter and piston pin outer diameter

0.010—0.026 mm {0.0004—0.0010 in}

PISTON AND CONNECTING ROD INSPECTION

id011000503100

1. Check the oscillation torque as shown in the figure. Verify that the large end drops under its own weight with no resistance.



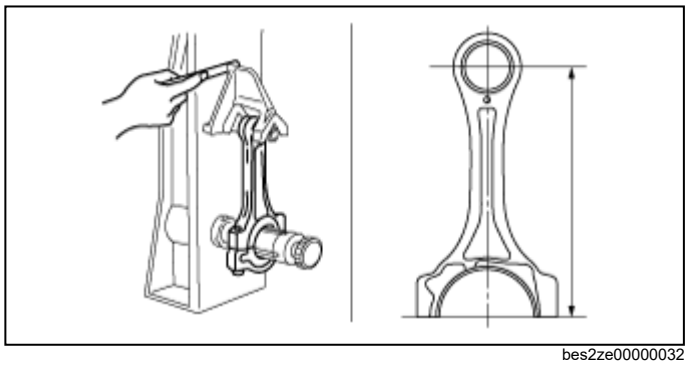
bes3ze00000030

- If the piston shakes heavily or unsmoothly, disassemble the piston and connecting rod, then inspect the following: (See PISTON PIN INSPECTION.)
  - — Clearance between piston pin outer diameter and piston pin hole diameter.
  - — Clearance between piston pin outer diameter and connecting rod small end inner diameter.

CONNECTING ROD INSPECTION

id011000502800

1. Inspect the connecting rod for bending and distortion using the connecting rod aligner.



bes2ze00000032

- If it exceeds the maximum specification, replace the connecting rod.

Maximum connecting rod bending

0.08 mm {0.003 in}

Maximum connecting rod distortion

0.08 mm {0.003 in}

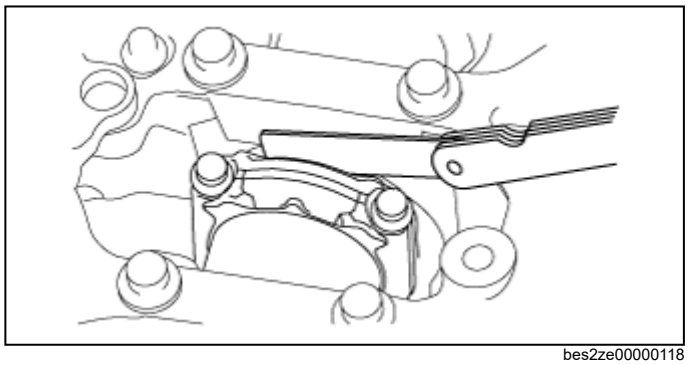
Connecting rod center-to-center distance

138.19 mm {5.4406 in}

CONNECTING ROD CLEARANCE INSPECTION

id011000507600

1. Measure the side clearance at the large end of the connecting rod using a feeler gauge.



bes2ze00000118

- If it exceeds the maximum specification, replace the connecting rod or crankshaft.

Standard side clearance at the large end of connecting rod

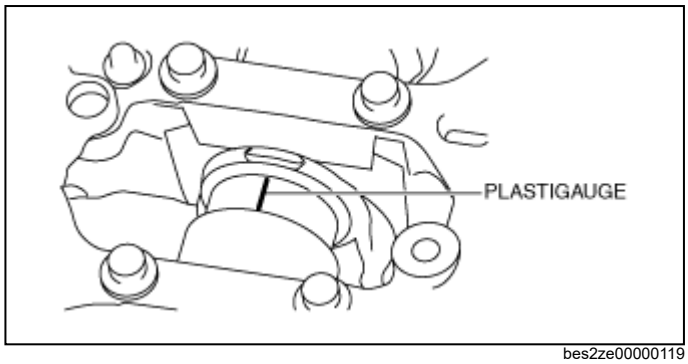
0.14—0.36 mm {0.006—0.014 in}

Maximum side clearance at the large end of connecting rod

0.512 mm {0.0202 in}

2. Measure the oil clearance at the large end of the connecting rod using the following procedure:

(1) Cut the plastigauge as wide as the connecting rod bearing width, place it parallel to the crankshaft, keeping away from the oil hole.



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(2) Install the lower connecting rod bearing and connecting rod cap. (See CYLINDER BLOCK ASSEMBLY (I).)

(3) Remove the connecting rod cap. (See CYLINDER BLOCK DISASSEMBLY (II).)

(4) Measure the oil clearance at the large end of the connecting rod.

- If it exceeds the maximum specification, replace the bearing or grind the crank pin and use oversize bearings so that the specified clearance is obtained.

Standard bearing oil clearance at the large end of the connecting rod

0.026—0.052 mm {0.0011—0.0020 in}

Maximum bearing oil clearance at the large end of the connecting rod

0.10 mm {0.00394 in}

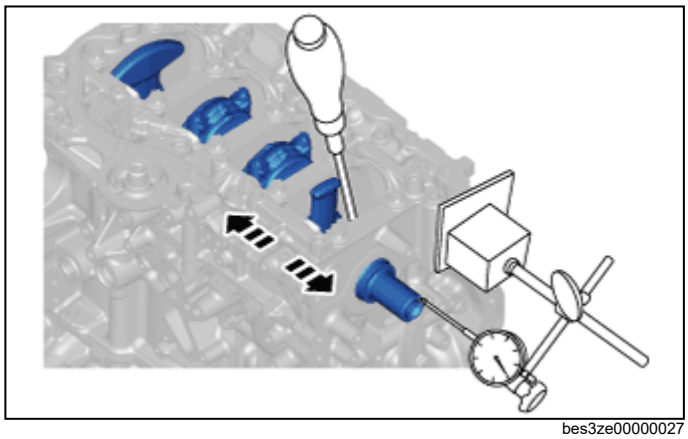
Connecting rod bearing size

STD: 1.501—1.520 mm {0.05910—0.05984 in}
OS 0.25: 1.626—1.633 mm {0.06402—0.06429 in}
OS 0.50: 1.751—1.758 mm {0.06894—0.06921 in}

CRANKSHAFT INSPECTION

id011000502500

1. Measure the crankshaft end play using a dial gauge.



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- If it exceeds the maximum specification, replace the crankshaft or grind the thrust side of crankshaft and use oversize thrust bearing so that the specified end play is obtained.

Standard crankshaft end play

0.08—0.29 mm {0.004—0.011 in}

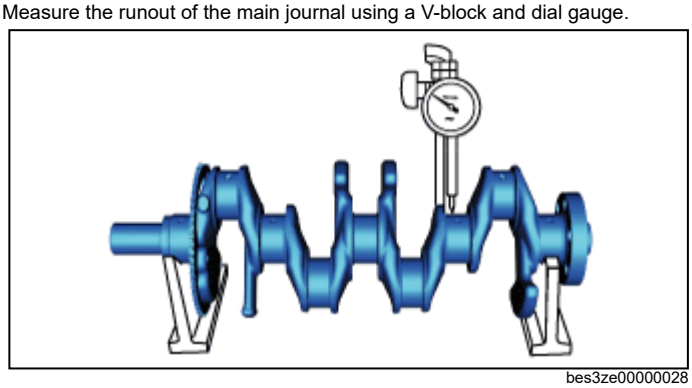
Maximum crankshaft end play

0.30 mm {0.012 in}

Thrust bearing size

STD: 2.500—2.550 mm {0.0985—0.1003 in}
OS 0.25: 2.625—2.675 mm {0.1034—0.1053 in}

2. Measure the runout of the main journal using a V-block and dial gauge.



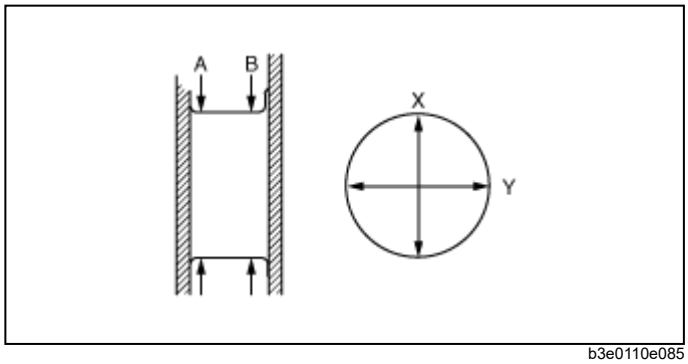
bes3ze00000028

- If it exceeds the maximum specification, replace the crankshaft.

Maximum main journal runout

0.10 mm {0.0039 in}

3. Inspect the main journal diameter and crank pin diameter. Measurement positions total four and are in the X and Y directions, at two points (A and B) as shown in the figure.



b3e0110e085

- If it is not within the specification or if it exceeds the maximum off-round, grind the journal with an oversized bearing.

Standard main journal diameter

46.980—47.000 mm {1.8497—1.8503 in}

Maximum main journal off-round

0.005 mm {0.0002 in}

Standard crank pin diameter

46.980—47.000 mm {1.8497—1.8503 in}

Maximum crank pin off-round

0.005 mm {0.0002 in}

4. Inspect the main journal oil clearance using the following procedure:

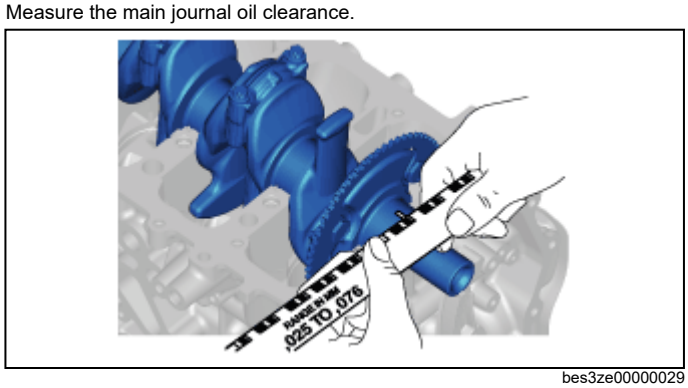
   (1) Install the thrust bearing, upper main bearing and crankshaft.

   (2) Position a plastigauge on the journals.

   (3) Install the lower main bearing and lower cylinder block. (See CYLINDER BLOCK ASSEMBLY (I).)

   (4) Remove the lower cylinder block. (See CYLINDER BLOCK DISASSEMBLY (II).)

   (5) Measure the main journal oil clearance.



bes3ze00000029

   • If it exceeds the maximum specification, replace the main bearing, or grind the main journal and use oversized bearings so that the specified oil clearance is obtained.

   Standard main journal oil clearance

   0.016—0.039 mm {0.0007—0.0015 in}

   Maximum main journal oil clearance

   0.084 mm {0.0033 in}

   Main bearing size

   STD: 2.489—2.510 mm {0.0980—0.0988 in}
   OS 0.25: 2.614—2.620 mm {0.1030—0.1031 in}
   OS 0.50: 2.739—2.745 mm {0.1079—0.1080 in}

DUAL-MASS FLYWHEEL INSPECTION

id011000501900

Caution
- Do not rework the dual-mass flywheel if it is distorted.
- Do not clean the dual-mass flywheel with any kind of fluid. Clean the dual-mass flywheel with a dry cloth only.
- Do not clean the gap between the primary and secondary mass. Only clean the bolt connection surface and the clutch surface.

Inspection Before Removal

1. Inspect the rotation of the dual-mass flywheel for play.

   (1) Lock the dual-mass flywheel using the SST (49 E011 1A0).

   (2) Rotate the secondary flywheel clockwise by hand until resistance can be felt.

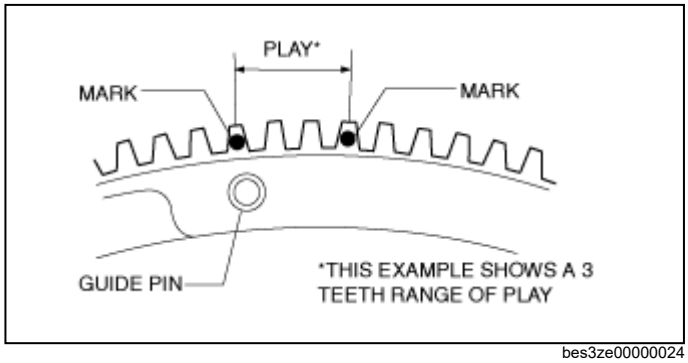
   (3) Mark the guide pin position on the ring gear of the flywheel.

   (4) Rotate the secondary flywheel counterclockwise by hand until resistance can be felt.

   (5) Mark the guide pin position on the ring gear of the primary flywheel again.

   (6) Verify that the play is within the range of 3 teeth.

   - If the play is 4 teeth or more, replace the dual-mass flywheel.

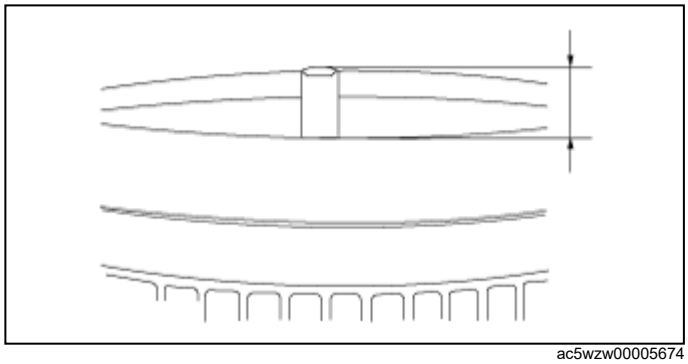


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Caution
- It may not be possible to rotate the secondary flywheel due to the functionality of the dual-mass flywheel.
In this case, the dual-mass flywheel can be considered normal by reason of the play being within the range of 3 teeth.

2. Measure the amount of guide pin projection of the dual-mass flywheel.

   - If it exceeds the maximum amount, replace the dual-mass flywheel.



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Dual-mass flywheel guide pin projection maximum amount

11.0—12.0 mm {0.434—0.472 in}

Inspection After Removal

1. Visually inspect the dual-mass flywheel for crack.

   - If there is any crack, replace the dual-mass flywheel.

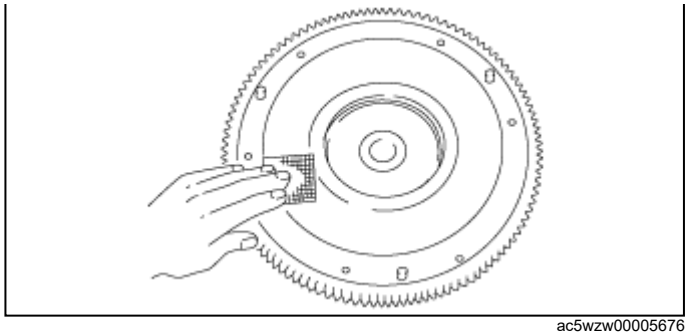
2. Visually inspect the ring gear on the dual-mass flywheel for damage.

   - If there is any damage, replace the dual-mass flywheel.

3. Visually inspect the surface that contacts the clutch disc for scratches, nicks, and discoloration.

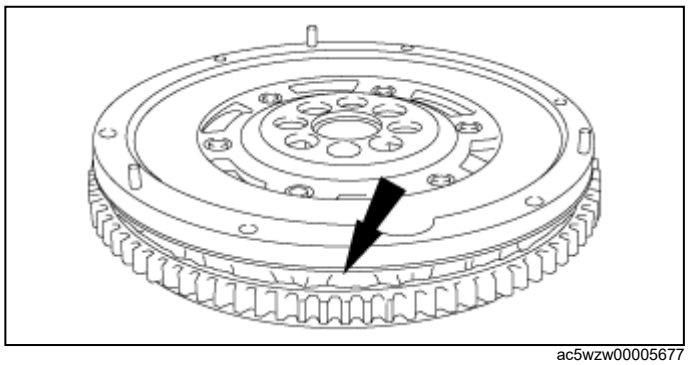
   - If there is any malfunction, replace the dual-mass flywheel.

Note
- Correct slight scratches and discoloration using sandpaper.



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4. Visually inspect for grease leakage between the primary mass and secondary mass.
    - If there is grease leakage, replace the dual-mass flywheel.

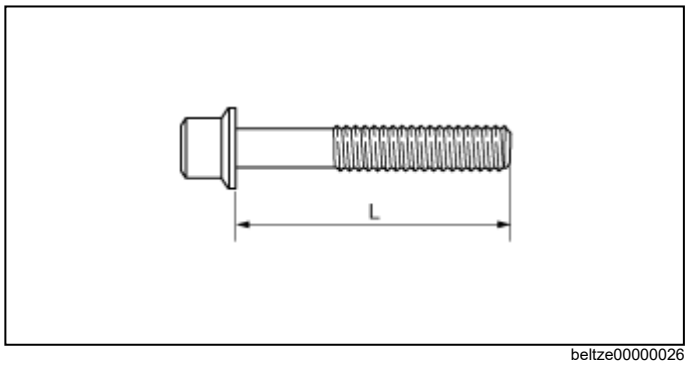


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BOLT INSPECTION

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1. Measure the length of the each bolt.



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- If it exceeds the maximum specification, replace the bolt.

Standard cylinder head bolt length

145.2—145.8 mm {5.717—5.740 in}

Maximum cylinder head bolt length

146.5 mm {5.768 in}

Standard connecting rod bolt length

41.7—42.3 mm {1.65—1.66 in}

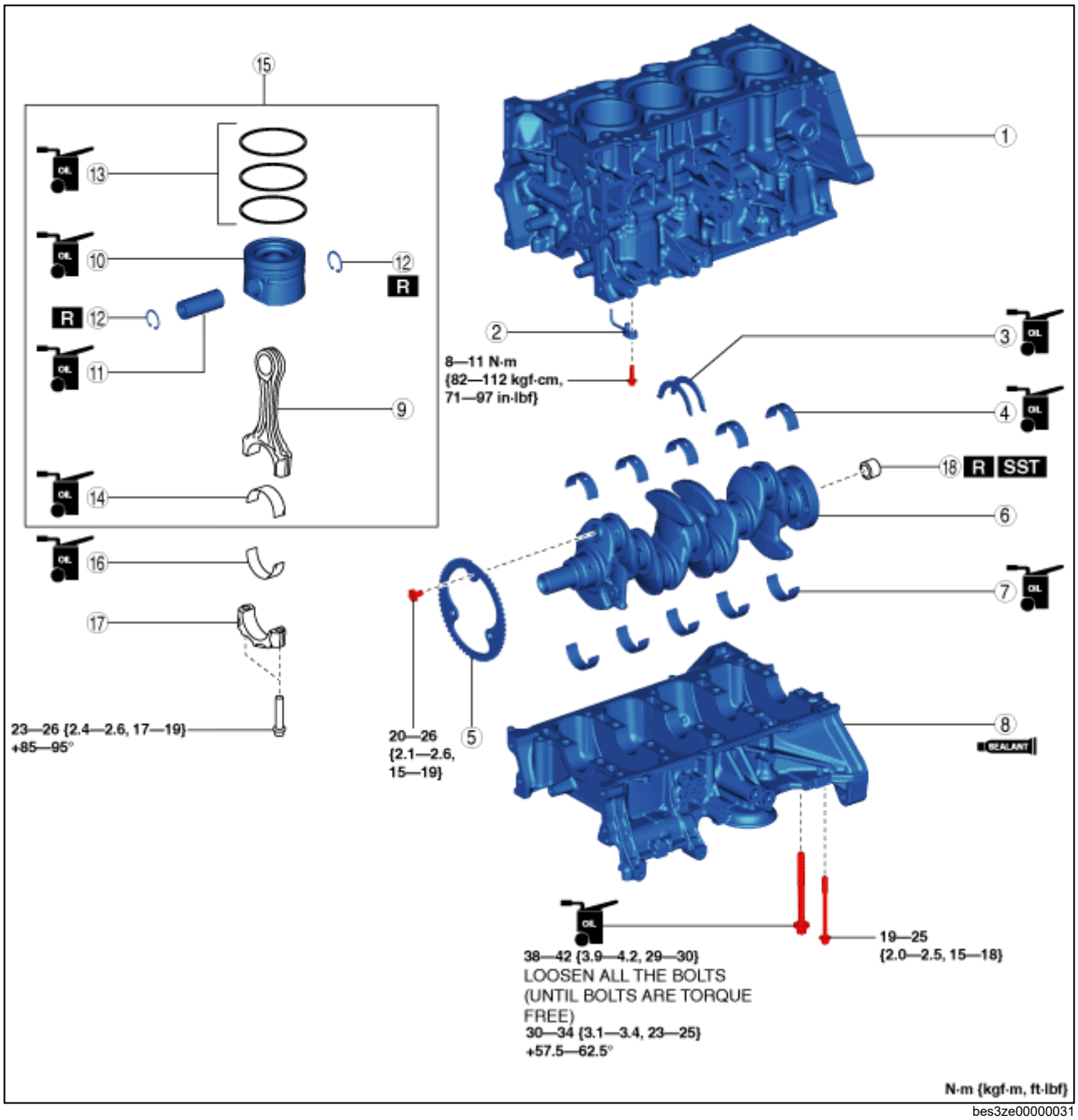
Maximum connecting rod bolt length

42.7 mm {1.68 in}

CYLINDER BLOCK ASSEMBLY (I)

id011000504000

1. Assemble in the order indicated in the table.



bes3ze00000031

1	Upper cylinder block
2	Oil jet valve
3	Thrust bearing (See Thrust Bearing And Main Bearing Assembly Note.)
4	Upper main bearing (See Thrust Bearing And Main Bearing Assembly Note.)
5	Plate (See Plate Assembly Note.)
6	Crankshaft
7	Lower main bearing (See Thrust Bearing And Main Bearing Assembly Note.)
8	Lower cylinder block (See Lower Cylinder Block Assembly Note.)
9	Connecting rod
10	Piston
11	Piston pin (See Piston Pin Assembly Note.)
12	Snap ring (See Snap Ring Assembly Note.)
13	Piston ring (See Piston Ring Assembly Note.)

14	Upper connecting rod bearing (See Connecting Rod Bearing Assembly Note.)
15	Piston, connecting rod (See Piston, Connecting Rod Assembly Note.)
16	Lower connecting rod bearing (See Connecting Rod Bearing Assembly Note.)
17	Connecting rod cap (See Connecting Rod Cap Assembly Note.)
18	Pilot bearing (MTX) (See Pilot Bearing Assembly Note.)

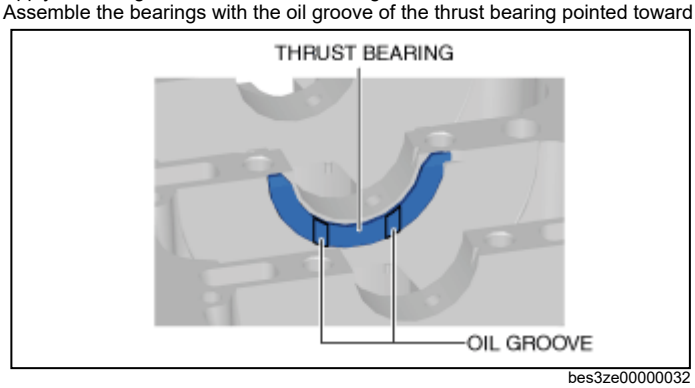
Thrust Bearing And Main Bearing Assembly Note

Caution

- If the thrust bearings and main bearings are reused, assemble the bearings to the same positions and in the direction as before removal to prevent engine damage due to seizure or burning of the bearing.
- To prevent engine damage due to seizure or burning of the bearing, apply engine oil to the sliding part when assembling.

Thrust bearing

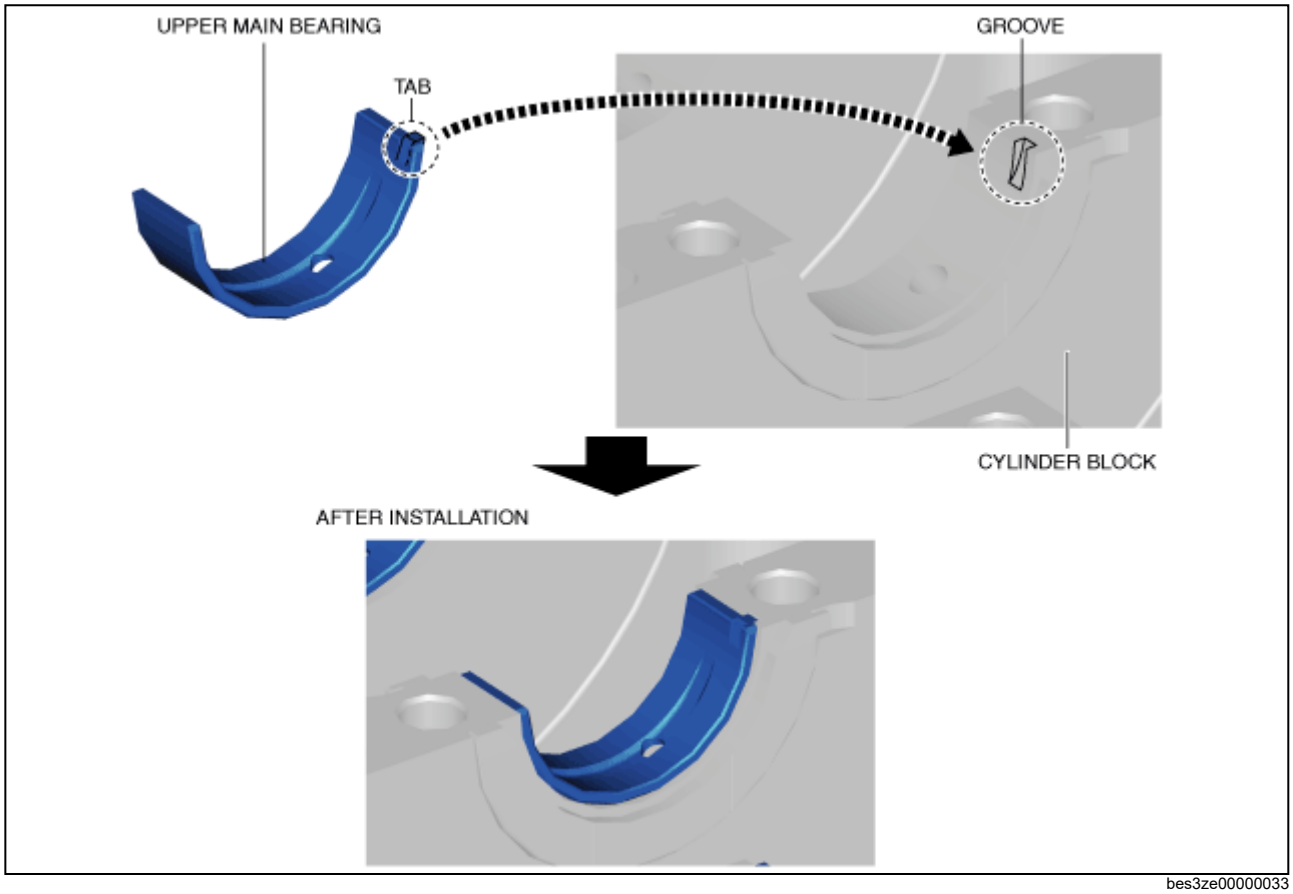
1. Apply clean engine oil to the thrust bearings.
2. Assemble the bearings with the oil groove of the thrust bearing pointed toward the sliding surface.



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Main bearing

1. Apply clean engine oil to the main bearings.
2. Align the positioning tab of the main bearing with the positioning groove of the lower and upper cylinder block, and assemble the bearings.



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Plate Assembly Note

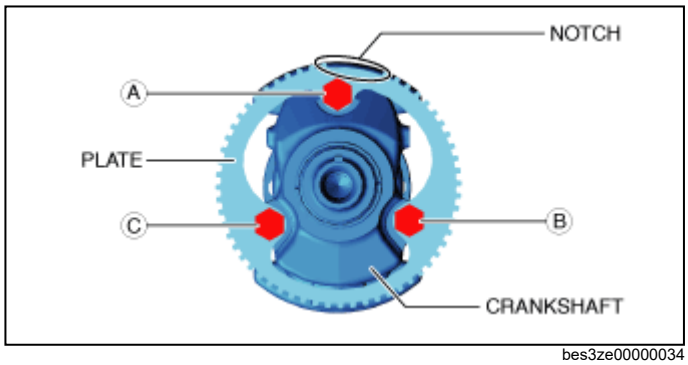
Caution

- If the plate for the crankshaft position sensor signal detection, which is installed to the crankshaft, contacts the disassembly bench, it could be deformed or damaged. Therefore, set wood blocks or similar objects on both sides of the crankshaft so that the plate does not contact the disassembly

bench directly when placing the crankshaft on the disassembly bench.

1. Install the plate using the following procedure:

   (1) Install the groove of the plate tooth to the position shown in the figure and temporarily tighten bolt A.



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   (2) Temporarily tighten the bolt B.

   (3) Install bolt C and tighten the bolts to the specified tightening torque in the order of C, B, and A.

   Tightening torque

   20—26 N·m {2.1—2.6 kgf·m, 15—19 ft·lbf}

Lower Cylinder Block Assembly Note

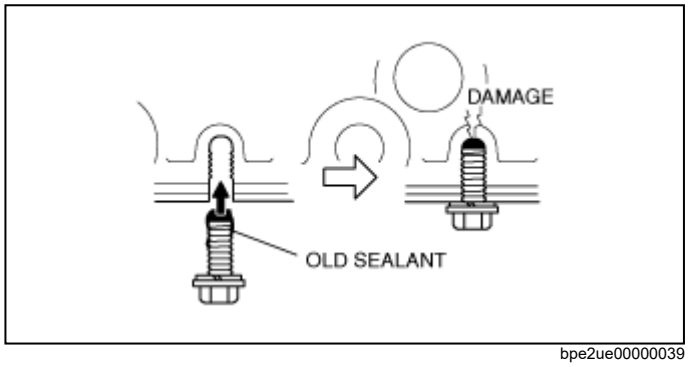
Caution

- When the lower cylinder block is removed and re-installed, the fuel injection amount learning must be performed after installing the engine to the vehicle. For the fuel injection amount learning procedure, refer to “FUEL INJECTION AMOUNT LEARNING” in the workshop manual.

1. Completely clean and remove any dirt, sealant or other foreign material that may be adhering to the upper and lower cylinder block.
2. Degrease the contact surfaces of the lower and upper cylinder blocks.
3. When reusing the lower cylinder block installation bolts, clean any old sealant from the bolts.

Caution

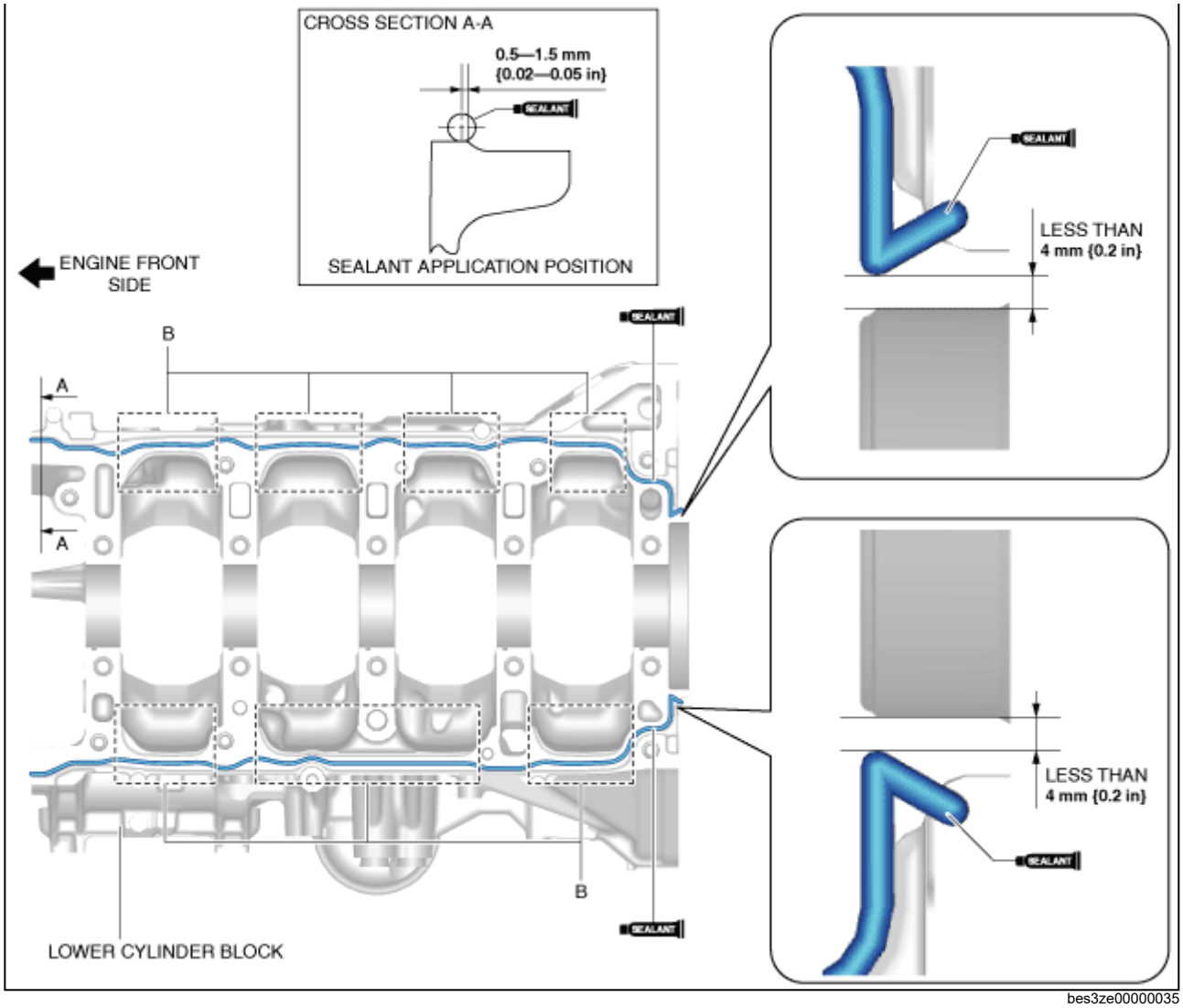
- Apply silicon sealant in a single, unbroken line.
- To prevent silicon sealant from hardening, adhere the lower and upper cylinder block firmly within 10 min. after applying silicon sealant. After adhering them, tighten the installation bolts immediately.
- Using bolts with the old seal adhering could cause cracks in the cylinder block, etc.



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4. Apply silicon sealant (TB1217D or equivalent) to the lower cylinder block shown in the figure.

   - For part B shown in the figure, apply silicone sealant to the center of the contact surface.



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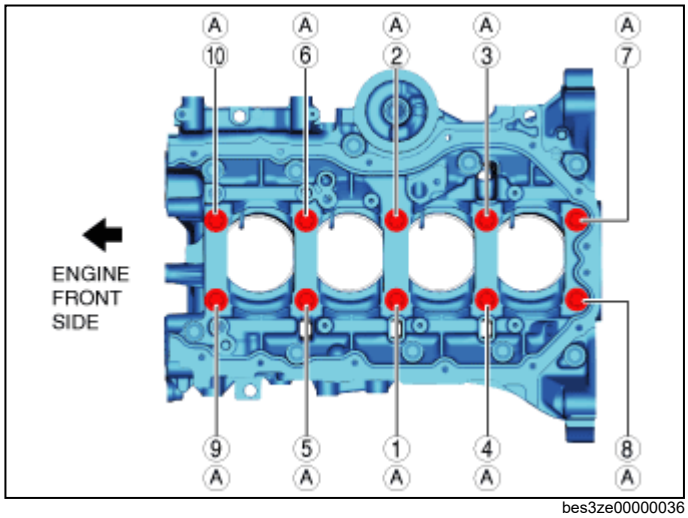
Bead thickness

2—6 mm {0.1—0.2 in}

5. Install the lower cylinder block.
6. Tighten the lower cylinder block installation bolts using the following procedure:

(1) Apply clean engine oil to seating surface and thread of the lower cylinder block installation bolts A.

(2) Tighten the lower cylinder block installation bolts A in the order shown in the figure using the following procedure:

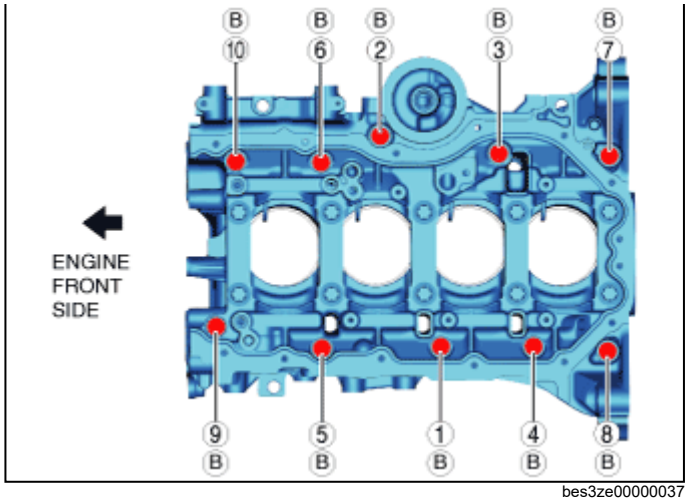


bes3ze00000036

Tightening procedure

Step 1: 38—42 N·m {3.9—4.2 kgf·m, 29—30 ft·lbf}
Step 2: Loosen all the bolts (until bolts are torque-free).
Step 3: 30—34 N·m {3.1—3.4 kgf·m, 23—25 ft·lbf}
Step 4: 57.5—62.5°

(3) Tighten the lower cylinder block installation bolts B in the order shown in the figure.



bes3ze00000037

Tightening torque

19—25 N·m {2.0—2.5 kgf·m, 15—18 ft·lbf}

7. After verifying that silicone sealant protrudes to the rear oil seal press-in part, wipe away the excess silicone sealant.
    - If silicone sealant does not protrude to the rear oil seal press-in part, remove the lower cylinder block and apply silicone sealant again.

Piston Pin Assembly Note

1. Remove dirt adhering to the piston pin.
2. Apply engine oil to the piston pin.

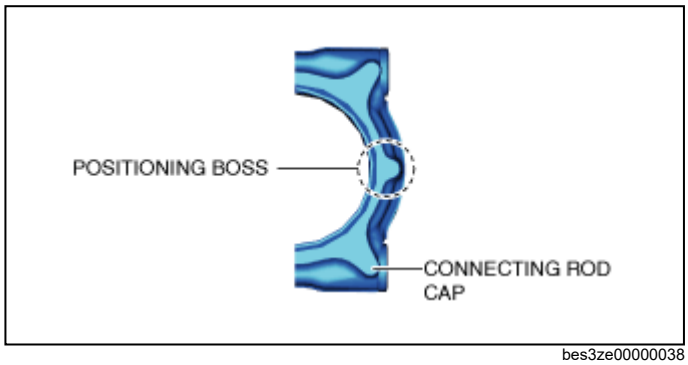
Caution

- When assembling the piston and connecting rod, assemble the connecting rod in the correct direction.

3. Assemble the piston and connecting rod using the following procedure:

Note

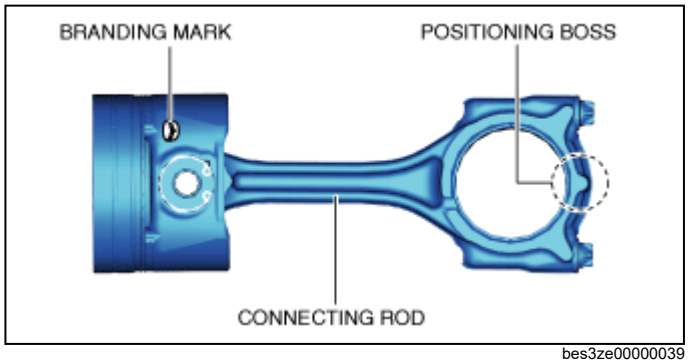
- The connecting rod cap has a positioning boss indicating the assembly direction.



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(1) Position so that the fractured surfaces of the connecting rods and connecting rod caps are aligned exactly, and temporarily assemble the cap to the connecting rod.

(2) Assemble the piston and connecting rod so that the branding mark of the piston and the positioning boss of the connecting rod cap are facing the same direction.



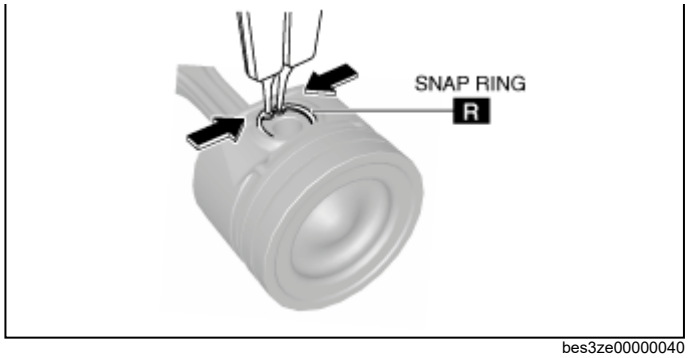
bes3ze00000039

4. Insert the piston pin into the piston and connecting rod.
5. Remove the temporarily assembled connecting rod cap.

Snap Ring Assembly Note

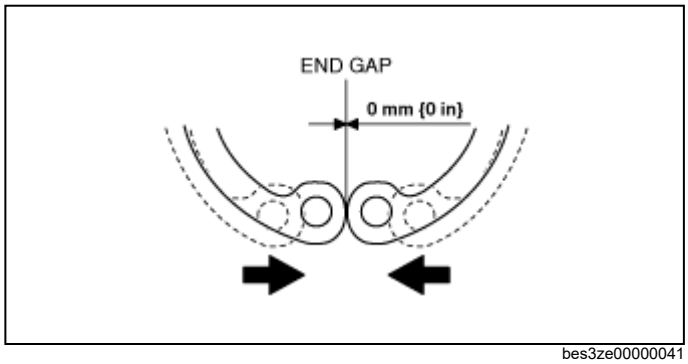
Caution

- When assembling the snap ring, the snap ring end gap can be reduced to 0 mm {0 in}.



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1. Insert a new snap ring using a pair of thin pliers.



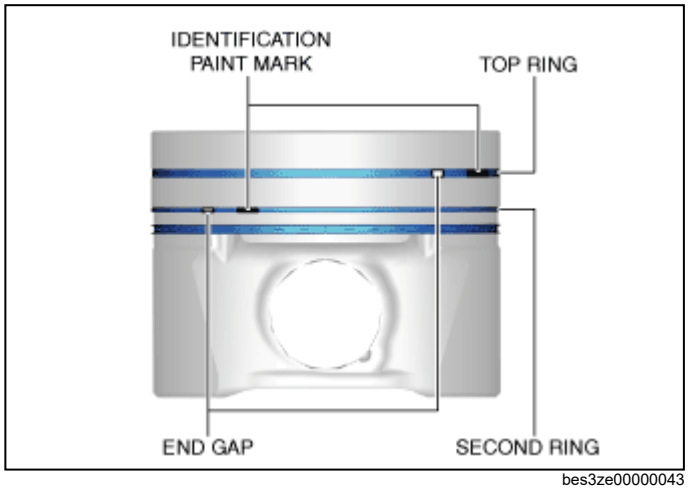
bes3ze00000041

Piston Ring Assembly Note

Note

- It is not necessary to position the piston ring end gap.

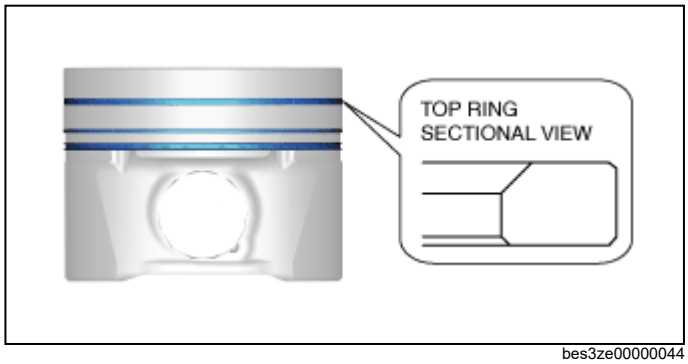
1. Install the oil ring.
2. Assemble so that the identification paint marks on the top and second rings are visible to the right side of each end gap.



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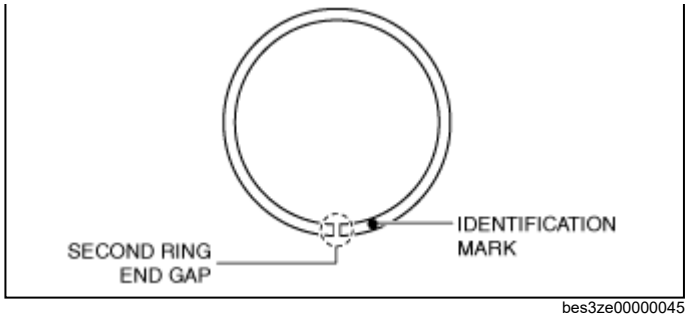
Note

- If there are no identification paint marks on the top rings, place the ring with the bigger inner circumference at the top as shown in the figure.



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- If there is no identification mark on the second ring, assemble the ring with the identification paint mark facing up.



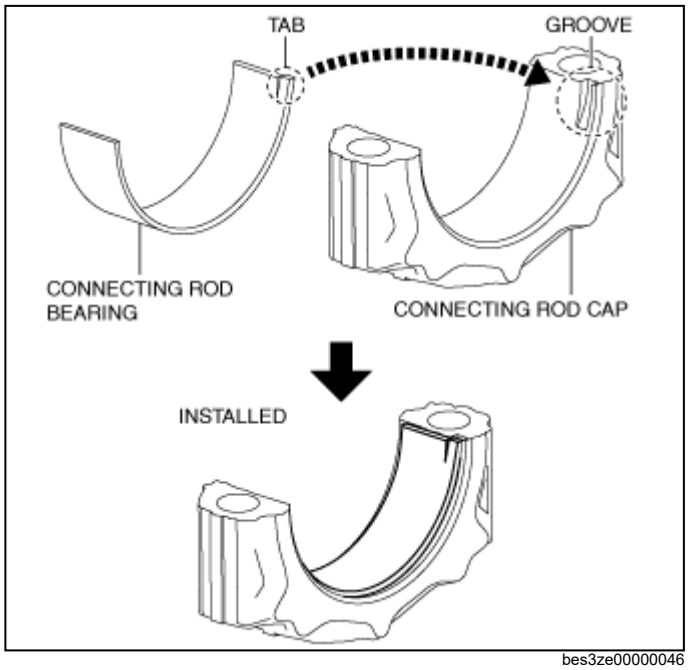
bes3ze00000045

Connecting Rod Bearing Assembly Note

Caution

- If a connecting rod bearing is reused, assemble it to the same position and in the direction as before removal to prevent engine damage due to seizure or burning of the bearing.
- To prevent engine damage due to seizure or burning of the bearing, apply engine oil to the sliding part when assembling.

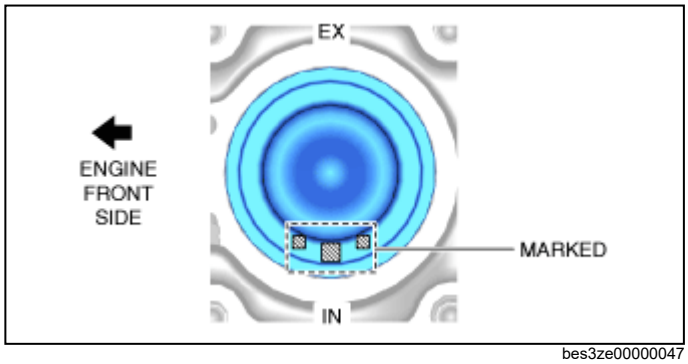
1. Apply engine oil to the connecting rod bearing.
2. Align the positioning tab of the connecting rod bearing with the positioning groove of the connecting rod and connecting rod cap, and assemble the bearings.



bes3ze00000046

Piston, Connecting Rod Assembly Note

1. Remove the dirt adhering to the piston.
2. Apply engine oil to the sliding part.
3. Insert the piston into the cylinder with the mark on top of the piston facing the intake side.



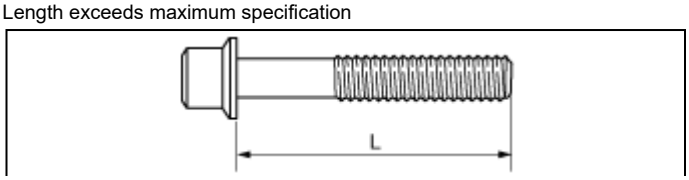
bes3ze00000047

Connecting Rod Cap Assembly Note

Caution

- When assembling the connecting rod caps, align the broken, rough faces of the connecting rods and connecting rod caps.
- If the following condition is met, replace the connecting rod cap bolts.

— Length exceeds maximum specification



Standard connecting rod cap bolt length

41.7—42.3 mm {1.65—1.66 in}

Maximum connecting rod cap bolt length

42.7 mm {1.68 in}

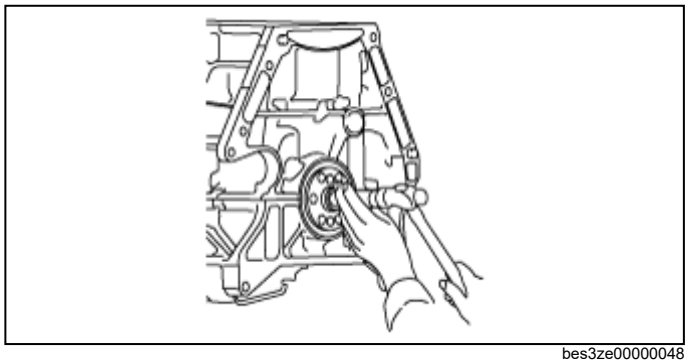
1. Position so that the broken, rough faces of the connecting rods and connecting rod caps are aligned exactly, and assemble the connecting rod caps.
2. Tighten the connecting rod cap bolts in the following two steps.

Tightening procedure

Step 1: 23—26 N·m {2.4—2.6 kgf·m, 17—19 ft·lbf}
Step 2: 85—95°

Pilot Bearing Assembly Note

1. Install new pilot bearing to the specified position using the following tools.



bes3ze00000048

Tool

Snap-on brand millimeter size bushing driver set (A160M) adapter A160M7 (20—22 mm {0.79—0.86 in})

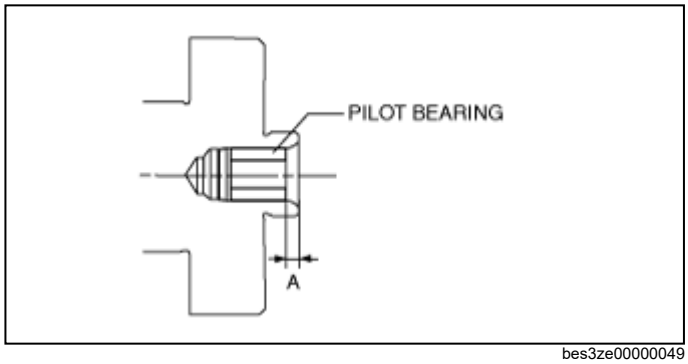
- Use the adapter with the 20 mm {0.79 in} side of the A160M7 (20—22 mm {0.79—0.86 in}) facing the pilot bearing side.

Substitution tool

Outer diameter: 21 mm {0.83 in}
Inner diameter: 19 mm {0.75 in}

Specified position

Distance A of pilot bearing from crankshaft end: 1.5—2.5 mm {0.060—0.098 in}

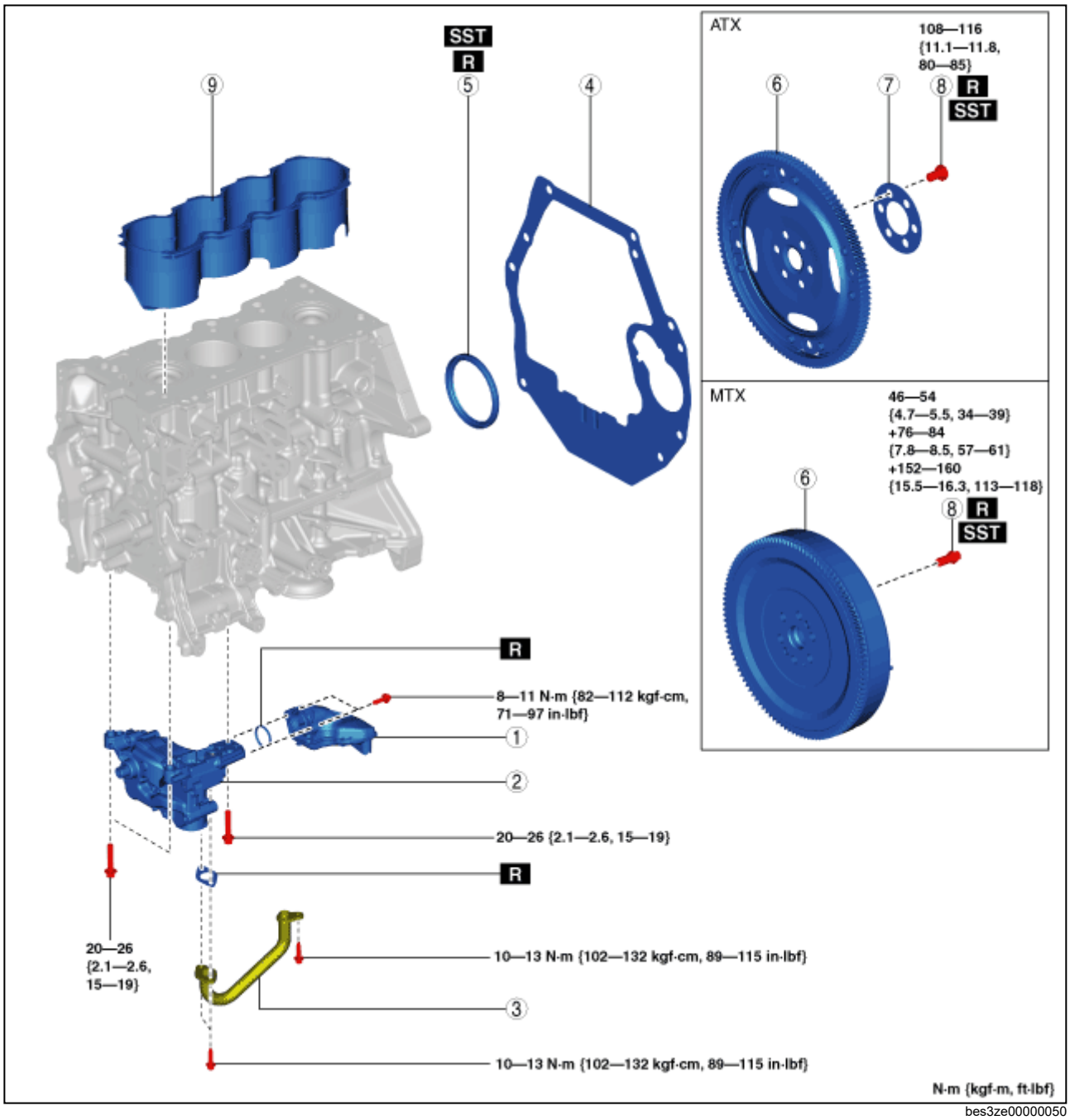


bes3ze00000049

CYLINDER BLOCK ASSEMBLY (II)

id011000504100

1. Assemble in the order indicated in the table.

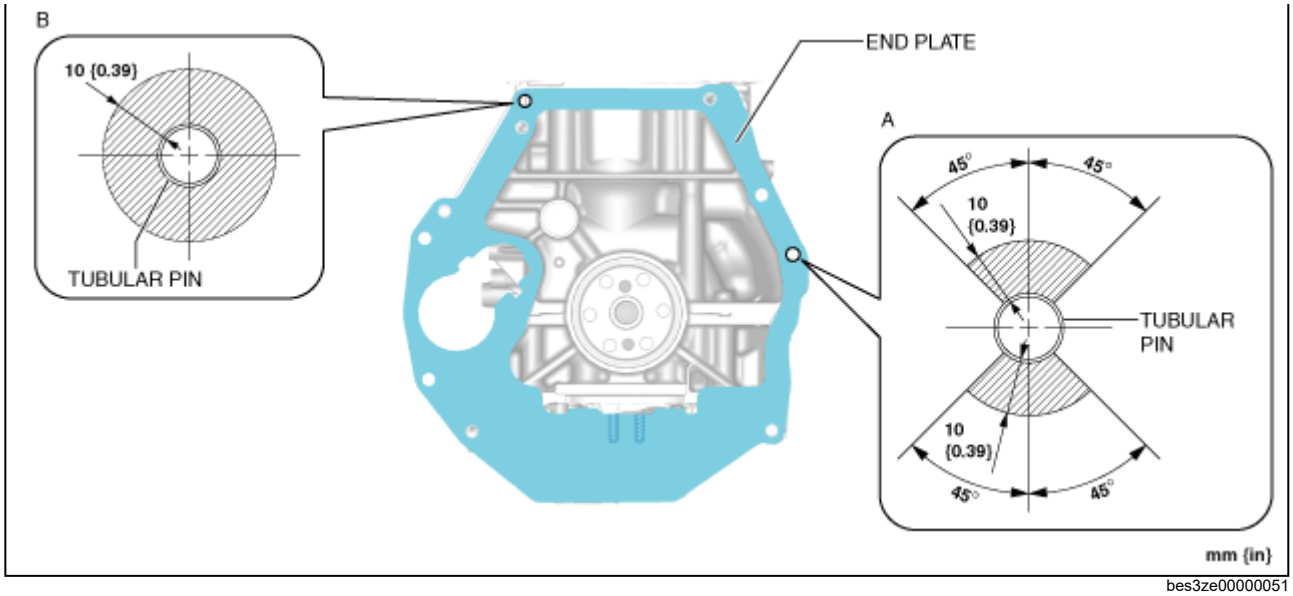


bes3ze00000050

1	Oil strainer
2	Oil pump
3	Oil pipe
4	End plate (See End Plate Assembly Note.)
5	Rear oil seal (See Rear Oil Seal Assembly Note.)
6	Dual-mass flywheel (MTX)/drive plate (ATX)
7	Backing plate (ATX)
8	Dual-mass flywheel (MTX)/drive plate (ATX) installation bolt (See Dual-mass Flywheel (MTX)/Drive Plate (ATX) Installation Bolt Assembly Note.)
9	Water jacket spacer (See Water Jacket Spacer Assembly Note.)

End Plate Assembly Note

1. After end plate assembly, crimp the parts A and B shown in the figure.



bes3ze00000051

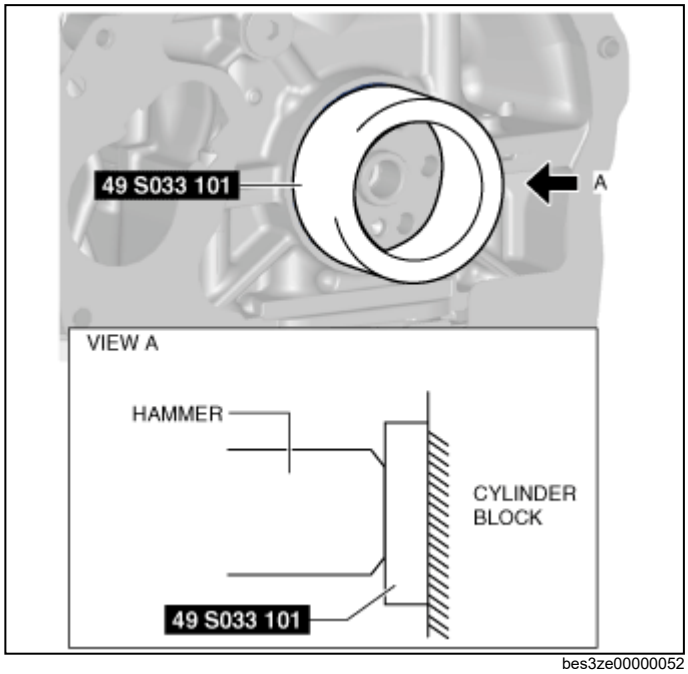
Crimp procedure

Crimp depth: 0.1—1.0 mm {0.004—0.039 in}
Crimp size: 0.5—10.0 mm {0.02—0.39 in}
Crimp locations: Part A is 1 or more on one-side within shaded area and part B is 2 or more within shaded areas

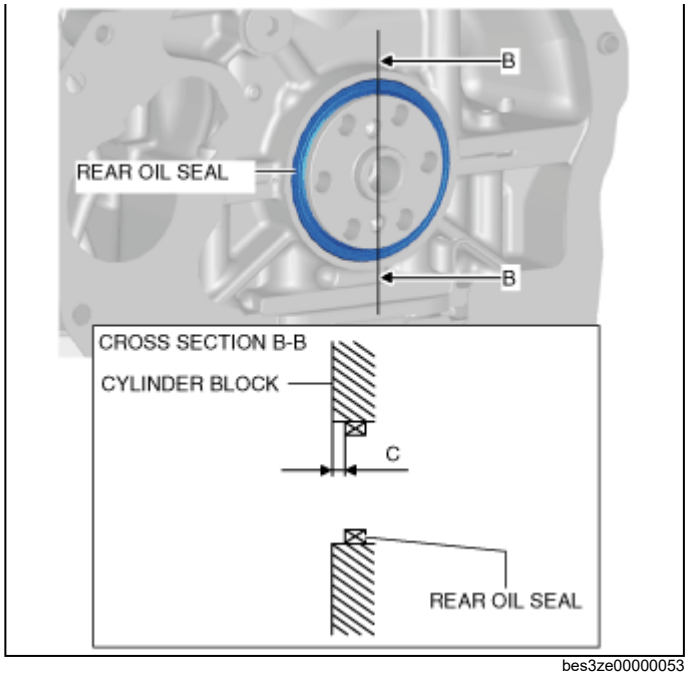
2. After crimping, verify that there is no damage and removal of the end plate.

Rear Oil Seal Assembly Note

1. Apply clean engine oil to the inner surface of a new rear oil seal.
2. Insert the rear oil seal into the cylinder block by hand.
3. Tap the oil seal in evenly using the SST and a hammer.



bes3ze00000052



bes3ze00000053

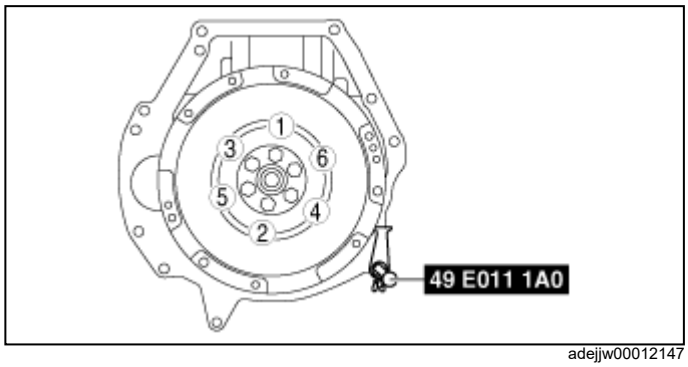
Rear oil seal press-in amount

0—0.5 mm {0—0.019 in} from cylinder block

Dual-mass Flywheel (MTX)/Drive Plate (ATX) Installation Bolt Assembly Note

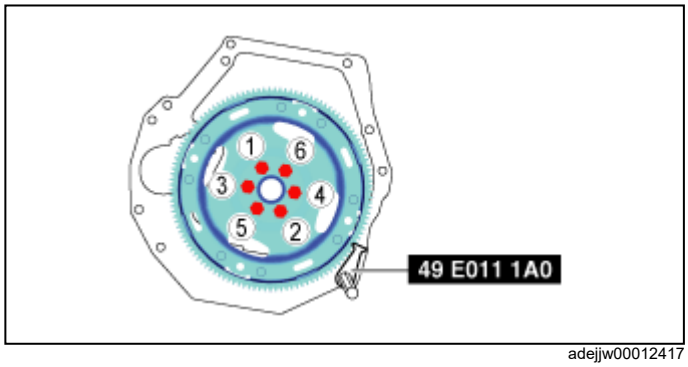
1. Clean the crankshaft thread holes.
2. Hold the crankshaft using the SST (49 E011 1A0).

MTX



adejjw00012147

ATX



adejjw00012417

3. Tighten the installation bolts.

MTX

1. Tighten the lock bolts completely in the order shown in the figure in the following 3 steps.

Tightening torque

Step 1: 46—54 N·m {4.7—5.5 kgf·m, 34—39 ft·lbf}
Step 2: 76—84 N·m {7.8—8.5 kgf·m, 57—61 ft·lbf}
Step 3: 152—160 N·m {15.5—16.3 kgf·m, 113—118 ft·lbf}

ATX

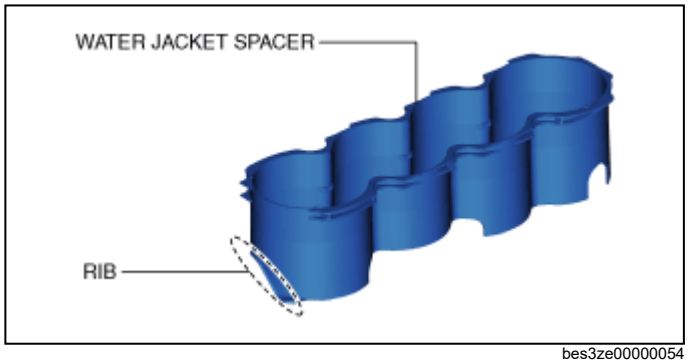
1. Tighten the new bolts in two or three passes in the order shown in the figure.

Tightening torque

108—116 N·m {11.1—11.8 kgf·m, 80—85 ft·lbf}

Water Jacket Spacer Assembly Note

1. Assemble so that the rib of the water jacket spacer is facing the engine front.



bes3ze00000054

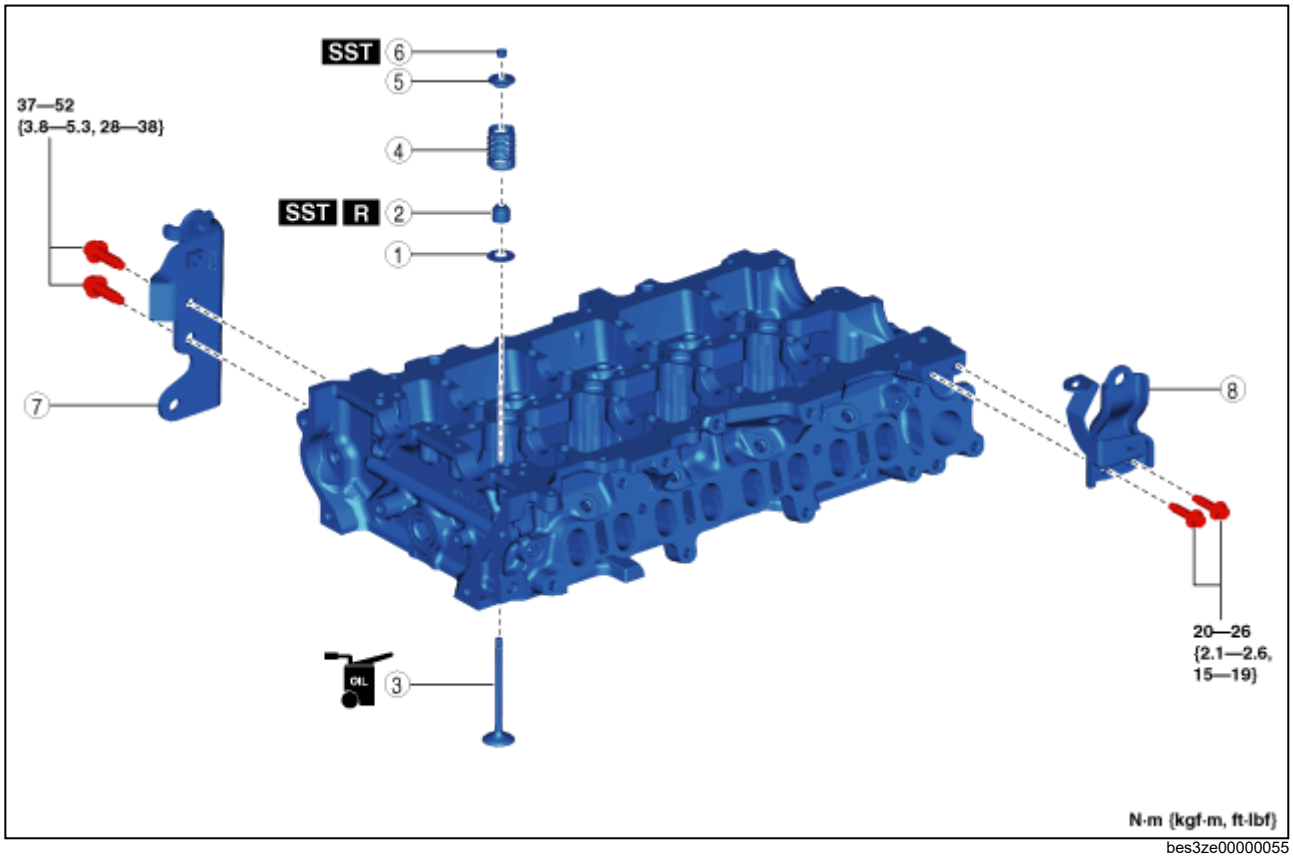
CYLINDER HEAD ASSEMBLY (I)

id011000504300

Caution

• The end of the glow plug protrudes from the cylinder head (tightened surface of cylinder block). When removing/installing the cylinder head and placing the cylinder head on a workbench, be careful not to allow the end of the glow plug to contact the cylinder block or workbench. Otherwise, the end of the glow plug could be damaged.

1. Assemble in the order indicated in the table.



bes3ze00000055

1	Lower valve spring seat
2	Valve seal (See Valve Seal Assembly Note.)
3	Valve
4	Valve spring (See Valve Spring Assembly Note.)
5	Upper valve spring seat
6	Valve keeper (See Valve Keeper Assembly Note.)
7	Hanger bracket
8	Seal cover bracket

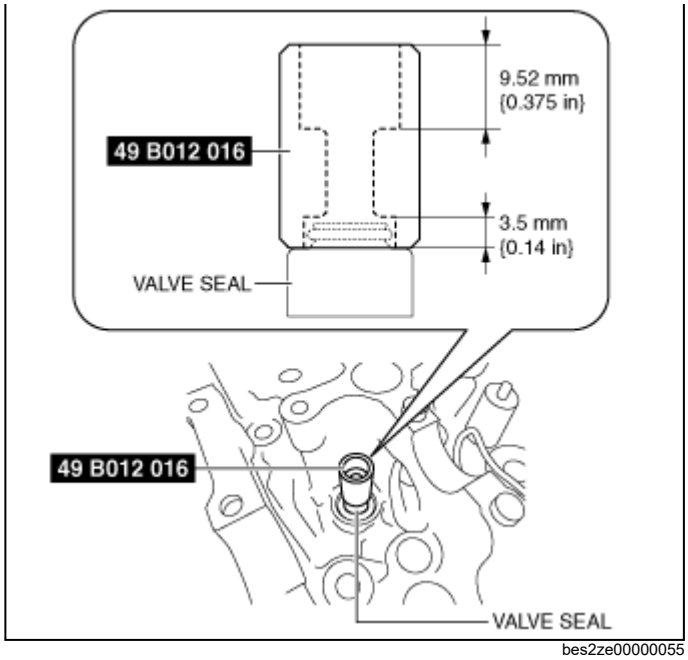
Valve Seal Assembly Note

1. Place the valve seal on the valve guide.

Valve seal identification color

IN: GREEN
EX: GRAY

2. Set the SST to the valve seal in the direction shown in the figure.



bes2ze00000055

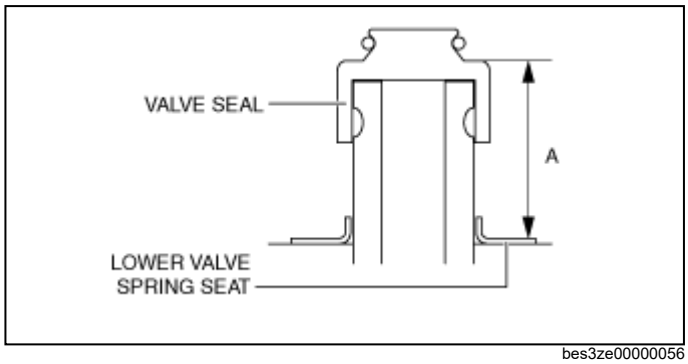
3. Press in the SST by hand and install the valve seal to the valve guide.

Intake side

Insert the valve seal by applying a load of 300—450 N {30.6—45.8 kgf, 67.5—101.0 lbf}.

Exhaust side

Insert the valve seal until height A of the valve seal is 16.63—17.91 mm {0.6548—0.7051 in}.

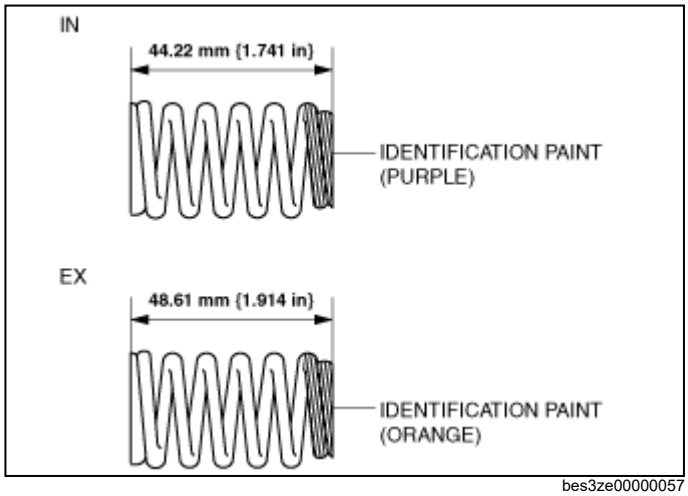


bes3ze00000056

Valve Spring Assembly Note

Caution

• The valve springs differ depending on IN and EX sides. Therefore, verify the free length or identification paint beforehand and assemble the valve springs correctly.



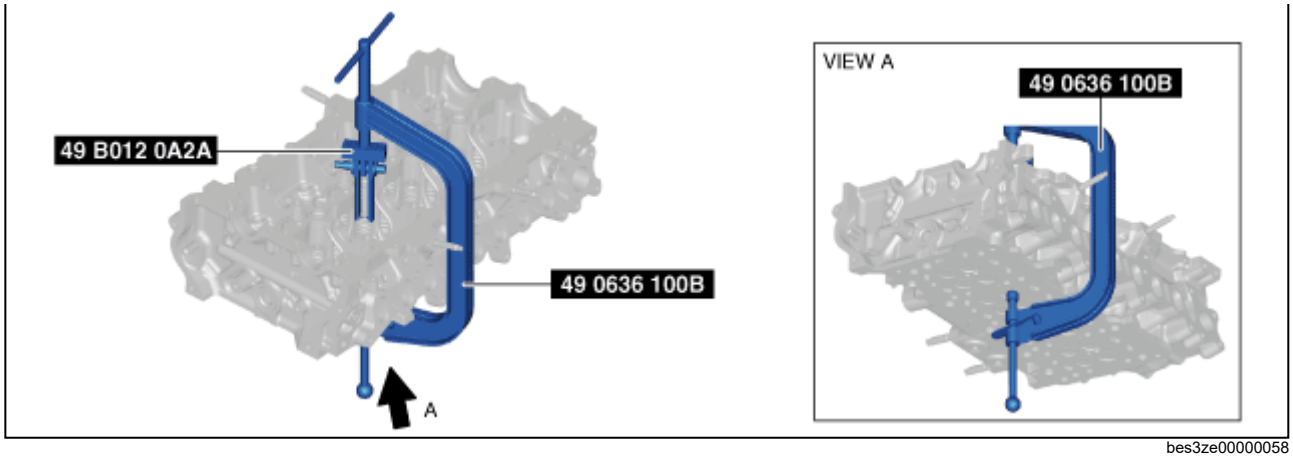
bes3ze00000057

1. Assemble the valve spring with the small diameter side of the valve spring facing upward.

Valve Keeper Assembly Note

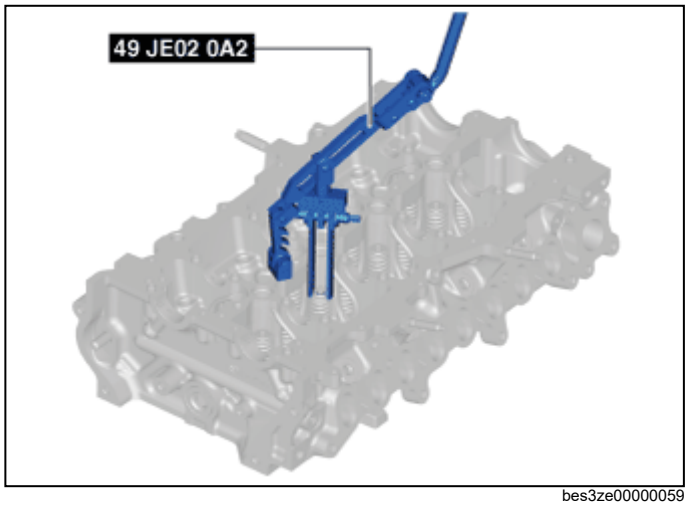
1. Install the valve keeper using the SSTs.

When using the SSTs (49 0636 100B, 49 B012 0A2A)



bes3ze00000058

When using the SST (49 JE02 0A2) (Europe only)



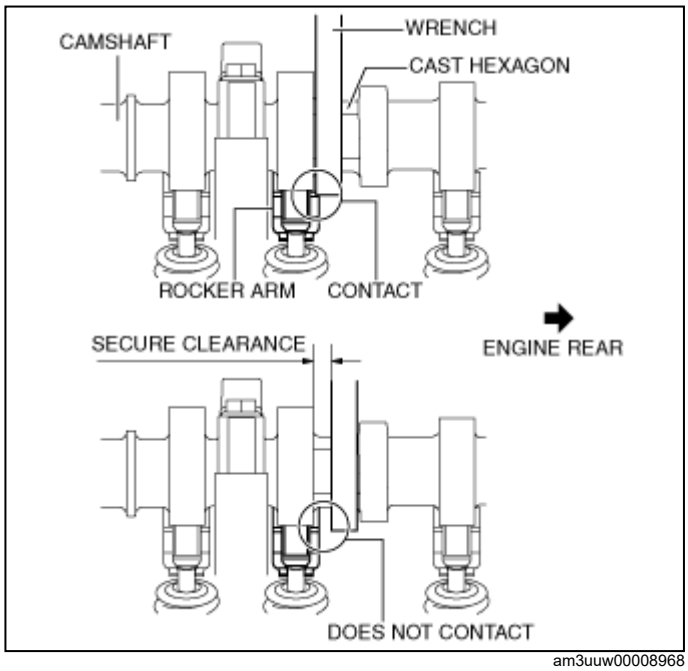
bes3ze00000059

CYLINDER HEAD ASSEMBLY (II)

id011000504400

Caution

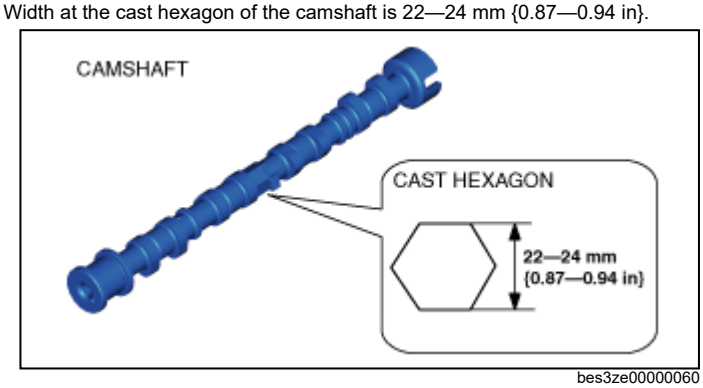
• When rotating the camshaft using a wrench on the cast hexagon, the wrench may contact the rocker arm and damage the rocker arm. To prevent damage to the rocker arm when holding the camshaft on the cast hexagon, use the wrench at engine rear side as shown in the figure to secure a clearance between the cam.



am3uuw00008968

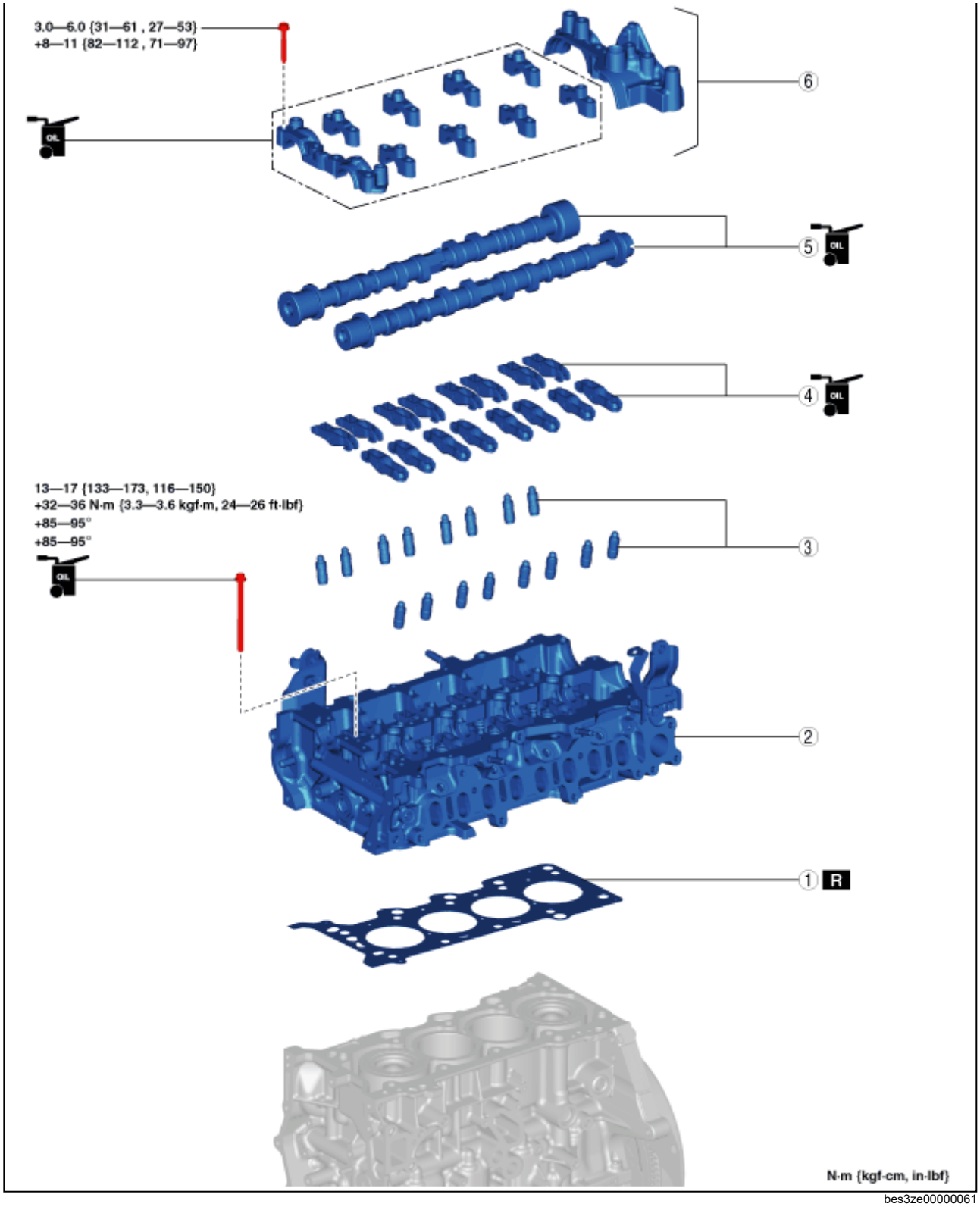
Note

• Width at the cast hexagon of the camshaft is 22—24 mm {0.87—0.94 in}.



bes3ze00000060

1. Assemble in the order indicated in the table.



bes3ze00000061

1	Cylinder head gasket
2	Cylinder head (See Cylinder Head Assembly Note.)
3	HLA (See HLA Assembly Note.)
4	Rocker arm (See Rocker Arm Assembly Note.)
5	Camshaft (See Camshaft Assembly Note.)
6	Camshaft cap (See Camshaft Assembly Note.)

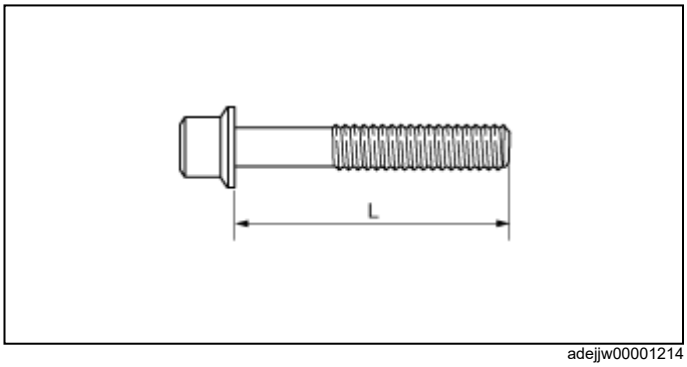
Cylinder Head Assembly Note

Caution

• The end of the glow plug protrudes from the cylinder head (tightened surface of cylinder block). When removing/installing the cylinder head and placing the cylinder head on a workbench, be careful not to allow the end of the glow plug to contact the cylinder block or workbench. Otherwise, the end of the glow plug could be damaged.

- If the following condition is met, replace the cylinder head bolts.

  — Length exceeds maximum specification



adejjw00001214

Standard cylinder head bolt length L

145.2—145.8 mm {5.717—5.740 in}

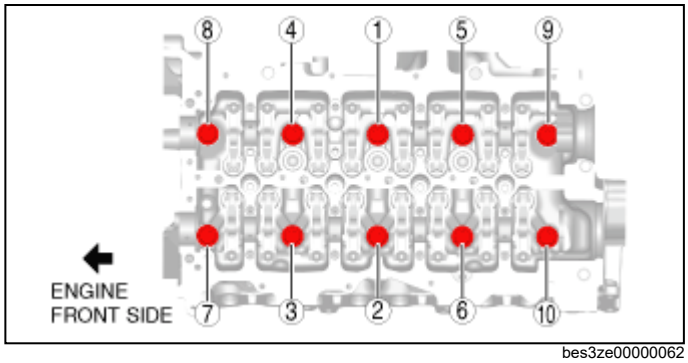
Maximum cylinder head bolt length L

146.5 mm {5.768 in}

1. When a cylinder head bolt is reused, apply engine oil to any part of the following:

   - Bolt seating surface
   - Cylinder head seating surface

2. Tighten the cylinder head bolts in the order shown in the following four steps.



bes3ze00000062

Tightening procedure

Step 1: 13—17 N·m {133—173 kgf·cm, 116—150 in·lbf}
Step 2: 32—36 N·m {3.3—3.6 kgf·m, 24—26 ft·lbf}
Step 3: 85—95°
Step 4: 85—95°

HLA Assembly Note

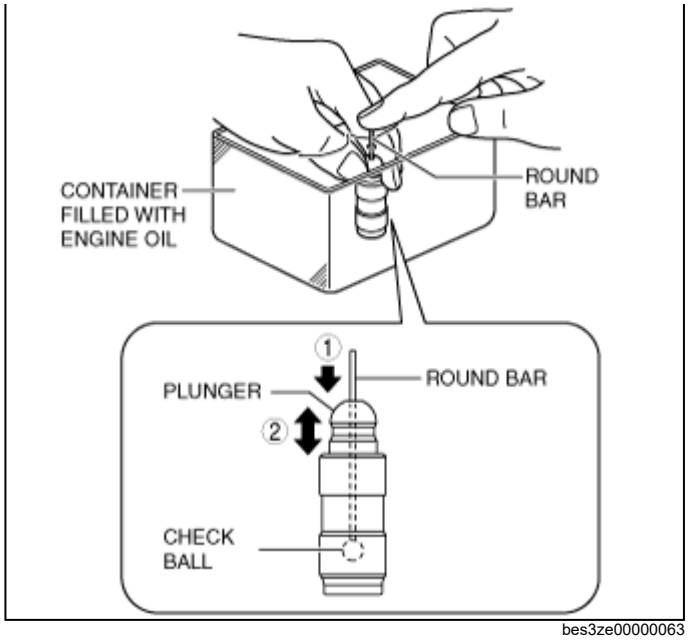
1. Perform HLA air bleeding using the following procedure:

   (1) Put the HLA in a container filled with engine oil.

   Caution

   - Do not insert the round bar firmly because the check ball spring force is extremely weak.

   (2) While lightly pressing the check ball using a round bar (approx. 1.0 mm {0.039 in} diameter), bleed air by moving the plunger up and down.



bes3ze00000063

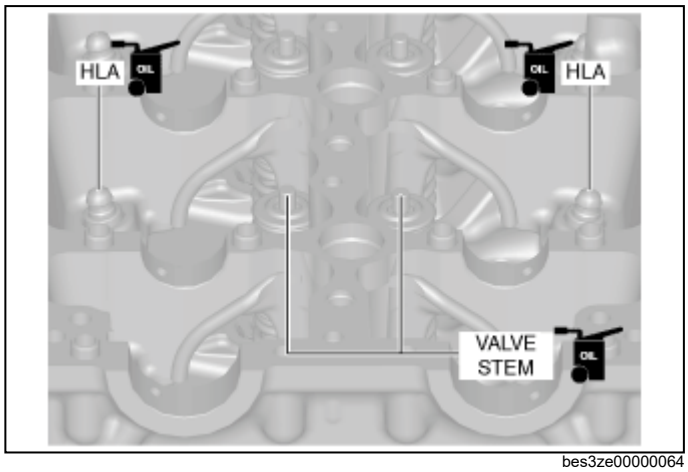
(3) Press the end of the plunger in the oil and verify that there is no rebounding feel.

- If rebounding feel cannot be eliminated, replace the HLA.

2. Install the HLAs to the same positions as before removal.

Rocker Arm Assembly Note

1. Apply engine oil to the HLAs and the end of the valve stems.

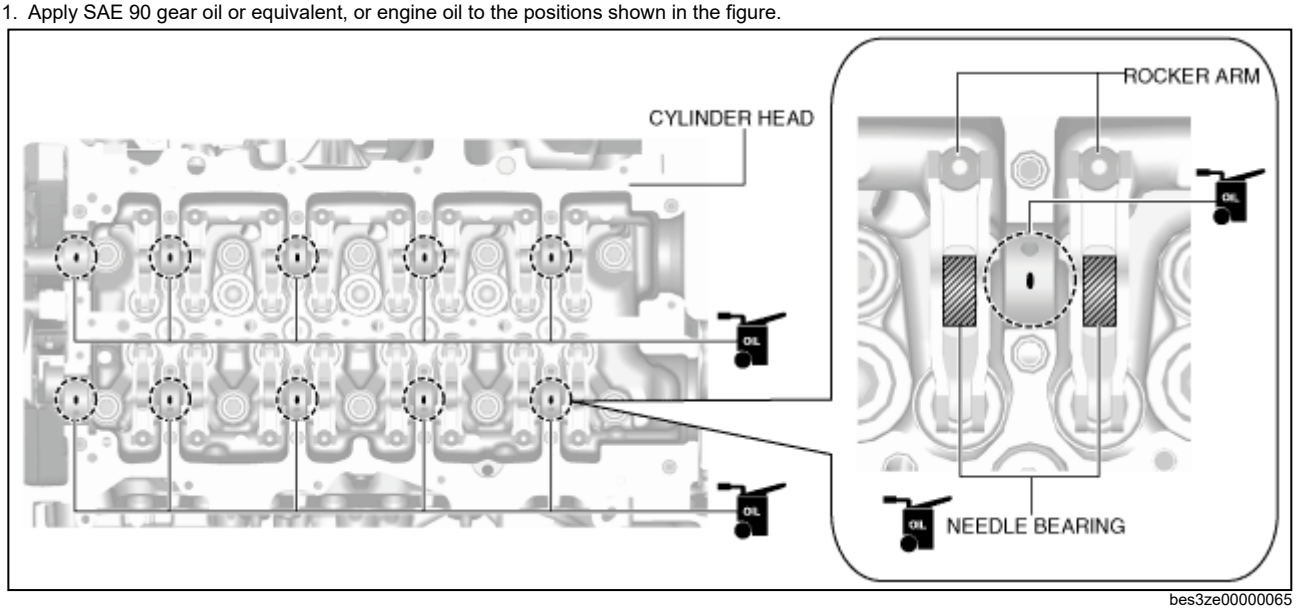


bes3ze00000064

2. Install the rocker arms to the same positions as before removal.

Camshaft Assembly Note

1. Apply SAE 90 gear oil or equivalent, or engine oil to the positions shown in the figure.



bes3ze00000065

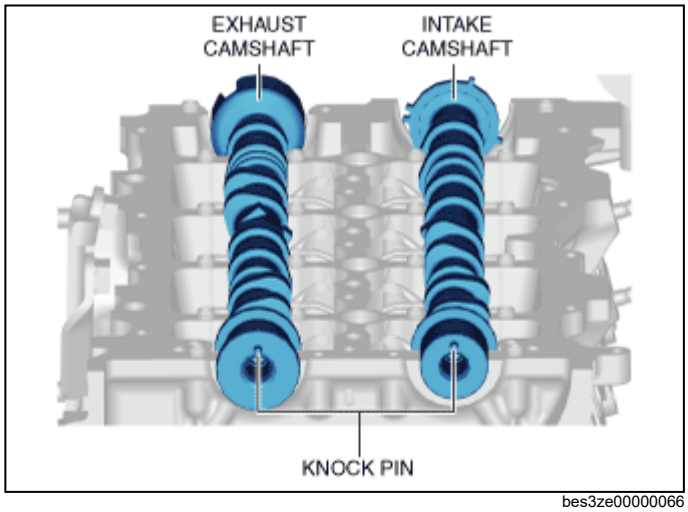
2. Apply SAE 90 gear oil or equivalent, or engine oil to the following locations of each camshaft.

- Gear sliding surfaces

- Thrust surface of front journal (both surfaces front and back)

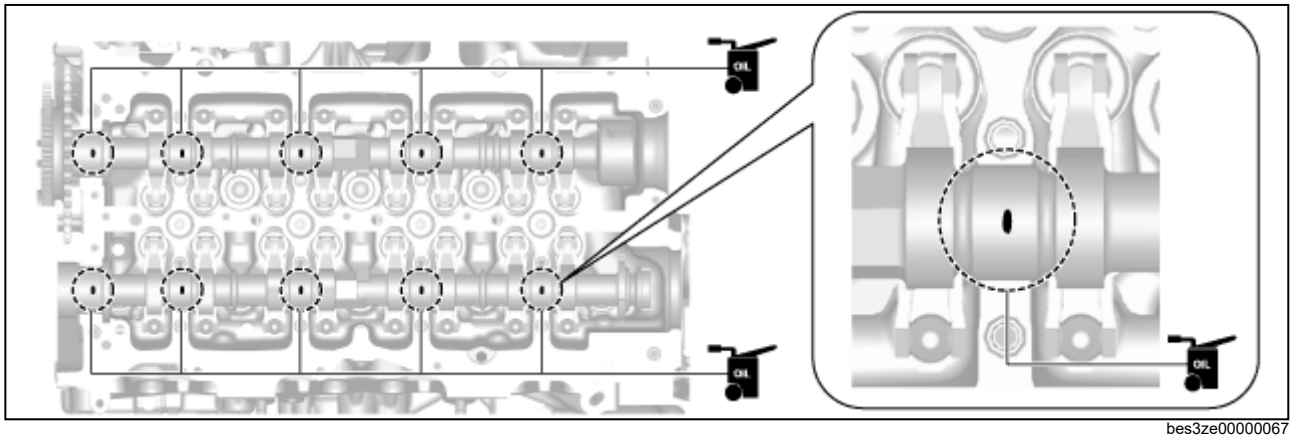
Note
- If oil is applied to the front camshaft cap, oil should not be applied to the thrust surface of the front journal.

3. Set the camshaft on the cylinder heads and align it to the positions shown in the figure.



bes3ze00000066

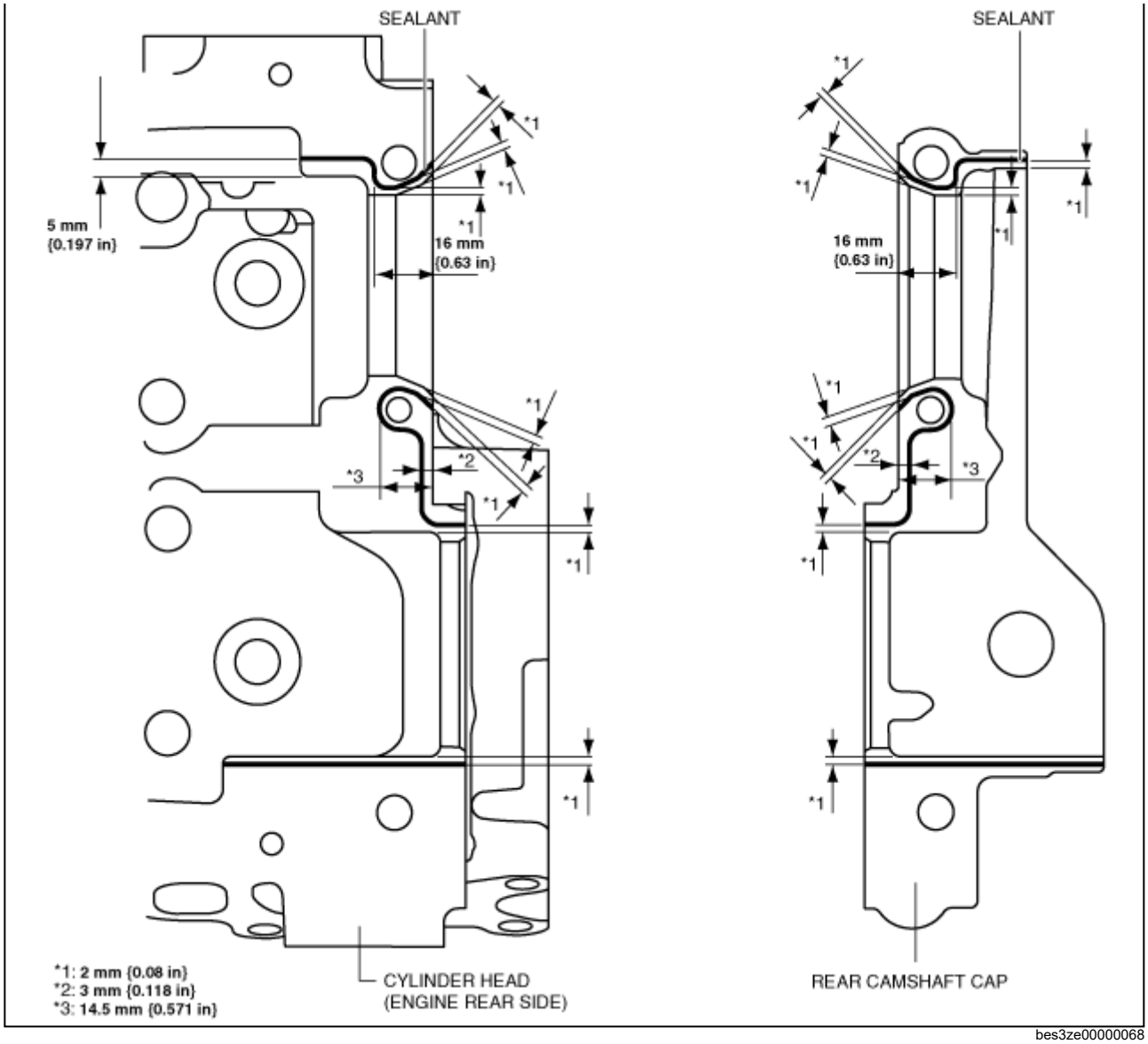
4. Apply SAE 90 gear oil or equivalent, or engine oil to the central area of each journal on the camshaft.



bes3ze00000067

5. Apply sealant (Loctite#962T or equivalent) to the rear side of the cylinder head or rear camshaft cap.

Note
- To prevent engine oil leakage, seal the journal by applying sealant to the rear side of the cylinder head or rear camshaft cap.



bes3ze00000068

Caution

- Do not spill sealant on the journal.
- To prevent sealant from hardening, adhere the rear camshaft cap to the cylinder head within 10 min after sealant is applied. Tighten the installation bolts completely soon after adhering.

Sealant bead width

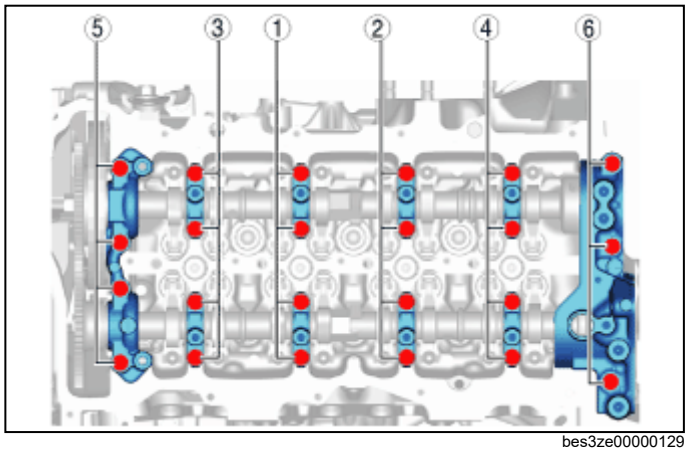
1—3 mm {0.04—0.11 in}

6. Apply gear oil (SAE 90 or equivalent) or engine oil to the thrust surface of the front camshaft cap.

Note

- If oil is applied to the front journal thrust surface of each camshaft, oil should not be applied to the front camshaft cap.

7. Install the camshaft caps in the marked number order, and temporarily tighten the camshaft cap installation bolts in two or three passes evenly.
8. Tighten the camshaft cap installation bolts in two steps in the order shown in the figure.



bes3ze00000129

Tightening torque

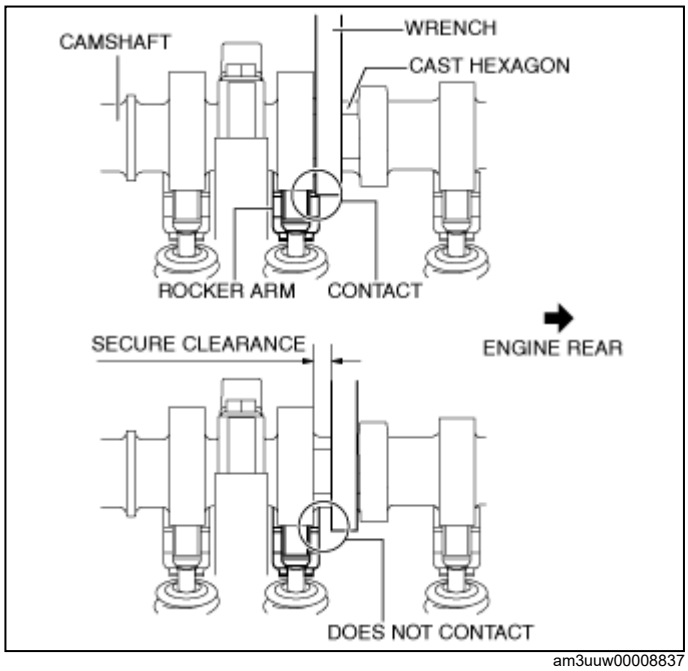
Step 1: 3.0—6.0 N·m {31—61 kgf·cm, 27—53 in·lbf}
Step 2: 8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}

TIMING CHAIN ASSEMBLY

id011000505600

Caution

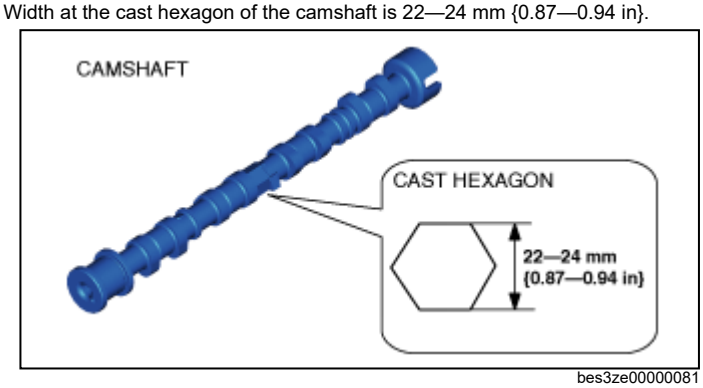
• When rotating the camshaft using a wrench on the cast hexagon, the wrench may contact the rocker arm and damage the rocker arm. To prevent damage to the rocker arm when holding the camshaft on the cast hexagon, use the wrench at engine rear side as shown in the figure to secure a clearance between the cam.



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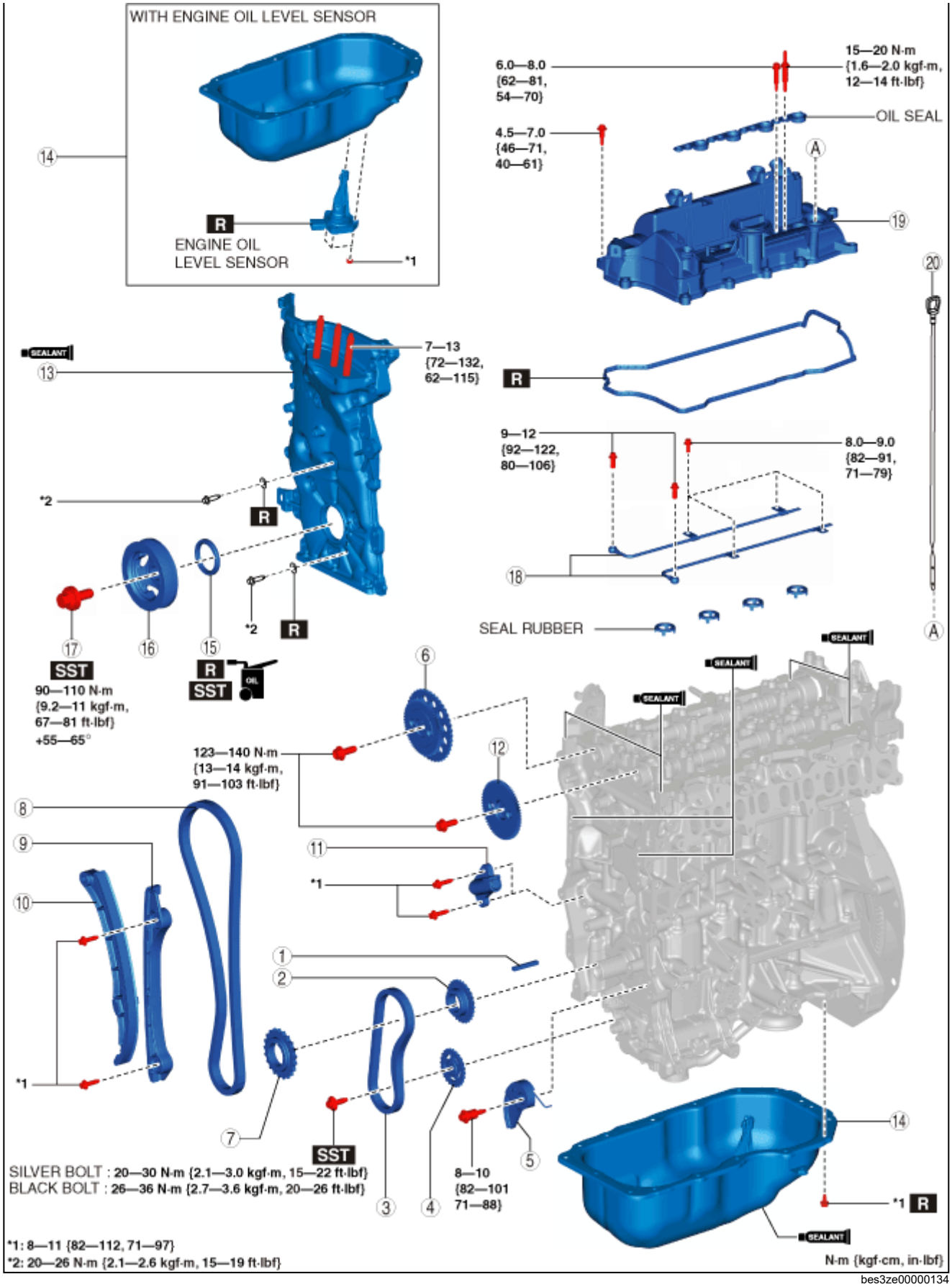
Note

• Width at the cast hexagon of the camshaft is 22—24 mm {0.87—0.94 in}.



bes3ze00000081

1. Assemble in the order indicated in the table.



bes3ze00000134

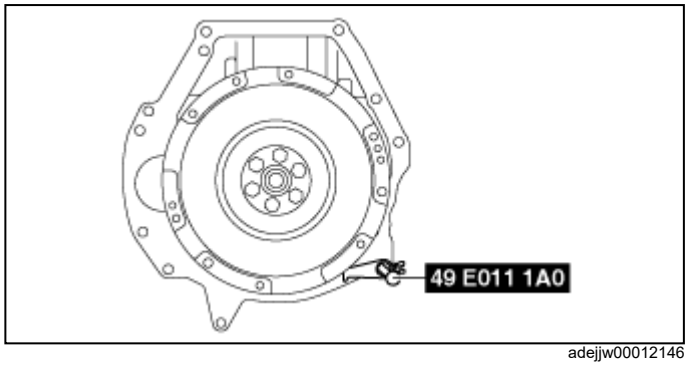
1	Key
2	Oil pump drive sprocket
3	Oil pump chain
4	Oil pump driven sprocket (See Oil Pump Driven Sprocket Assembly Note.)
5	Oil pump chain tensioner (See Oil Pump Chain Tensioner Assembly Note.)
6	Exhaust Camshaft Sprocket (See Exhaust Camshaft Sprocket Assembly Note.)
7	Crankshaft sprocket

	(See Timing Chain Assembly Note.)
8	Timing chain (See Timing Chain Assembly Note.)
9	Timing chain guide (See Timing Chain Assembly Note.)
10	Timing chain tensioner arm (See Timing Chain Assembly Note.)
11	Timing chain tensioner (See Timing Chain Assembly Note.)
12	Intake Camshaft Sprocket (See Intake Camshaft Sprocket Assembly Note.)
13	Engine front cover (See Engine Front Cover Assembly Note.)
14	Oil pan component (See Oil Pan Component Assembly Note.)
15	Front oil seal (See Front Oil Seal Assembly Note.)
16	Crankshaft pulley
17	Crankshaft pulley lock bolt (See Crankshaft Pulley Lock Bolt Assembly Note.)
18	Oil shower pipe (See Oil Shower Pipe Assembly Note.)
19	Cylinder head cover (See Cylinder Head Cover Assembly Note.)
20	Dipstick

Oil Pump Driven Sprocket Assembly Note

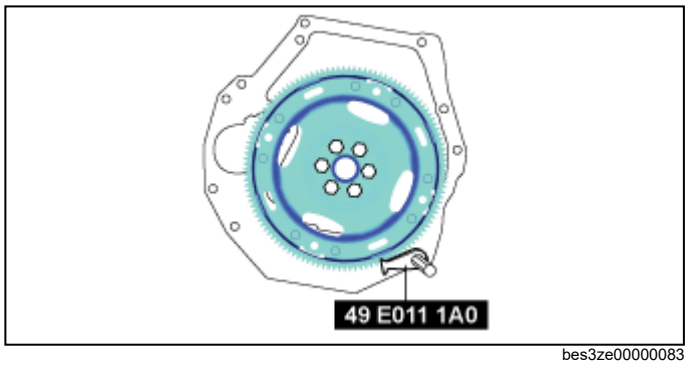
1. Hold the crankshaft using the SST.

MTX



adejjw00012146

ATX



bes3ze00000083

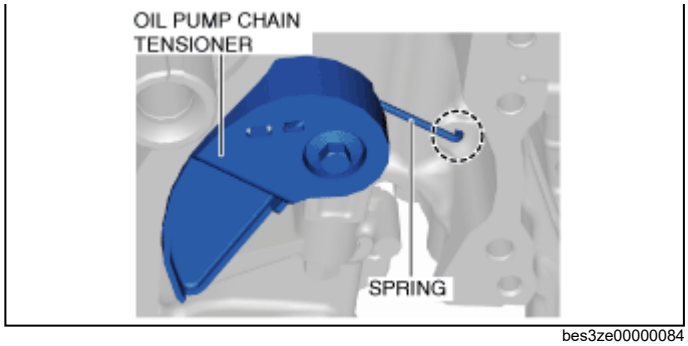
2. Tighten the oil pump driven sprocket installation bolt.

Tightening torque

Silver bolt: 20—30 N·m {2.1—3.0 kgf·m, 15—22 ft·lbf}
Black bolt: 26—36 N·m {2.7—3.6 kgf·m, 20—26 ft·lbf}

Oil Pump Chain Tensioner Assembly Note

1. Temporarily assemble the oil pump chain tensioner.
2. Verify that the spring of the oil pump chain tensioner is in the hole of the lower cylinder block.



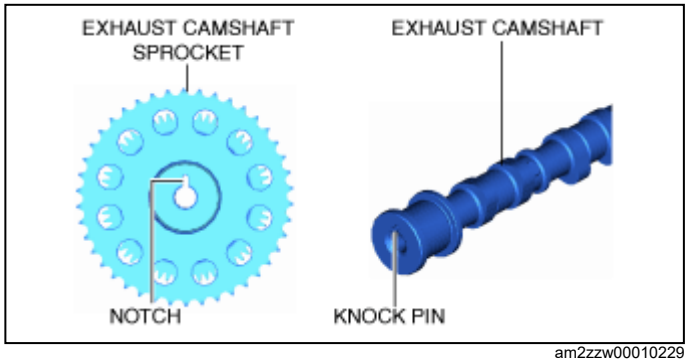
bes3ze00000084

- If the spring is not in the hole, perform the temporary assembly of the oil pump chain tensioner again.

3. Tighten the oil pump chain tensioner installation bolt.

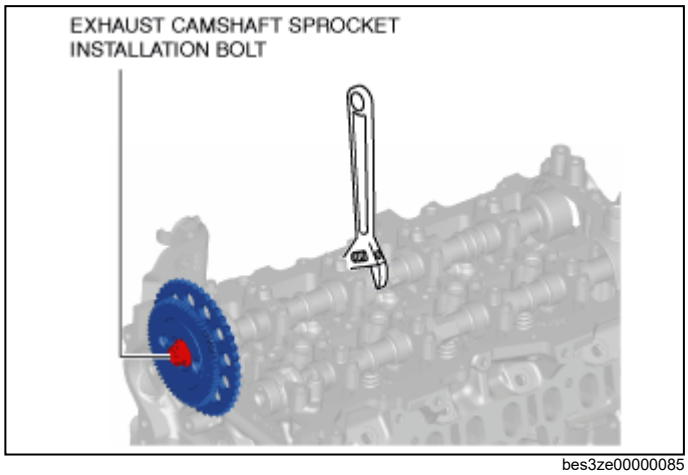
Exhaust Camshaft Sprocket Assembly Note

1. Align the knock pin on the end of the exhaust camshaft with the notch on the exhaust camshaft sprocket side, and temporarily assemble the bolt.



am2zzw00010229

2. Hold the exhaust camshaft using a wrench on the cast hexagon, and tighten the exhaust camshaft sprocket installation bolt.



bes3ze00000085

Tightening torque

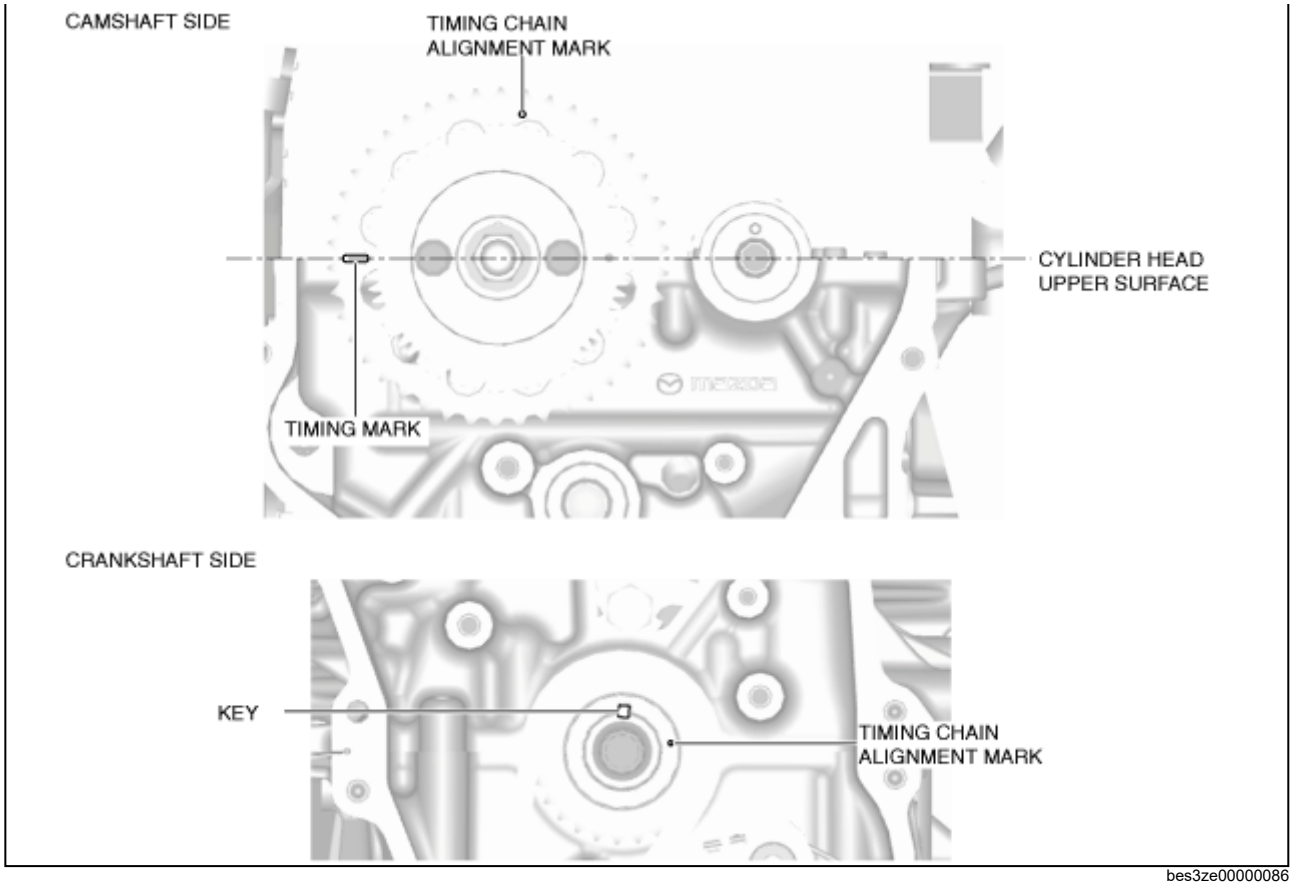
123—140 N·m {13—14 kgf·m, 91—103 ft·lbf}

Timing Chain Assembly Note

Caution

- If the camshaft or crankshaft is rotated before assembling the timing chain, the valve may contact the piston and the engine could be damaged.

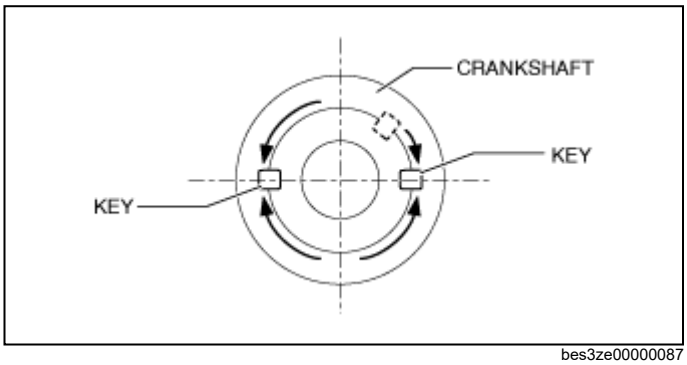
1. Verify that the timing marks and the key are aligned to the position shown in the figure.



bes3ze00000086

• If the key and the timing mark are not in the position shown in the figure, adjust their positions to set cylinder No.1 to top dead center (TDC) using the following procedure:

1. Rotate the crankshaft either left or right so that the key is in the complete horizontal position.

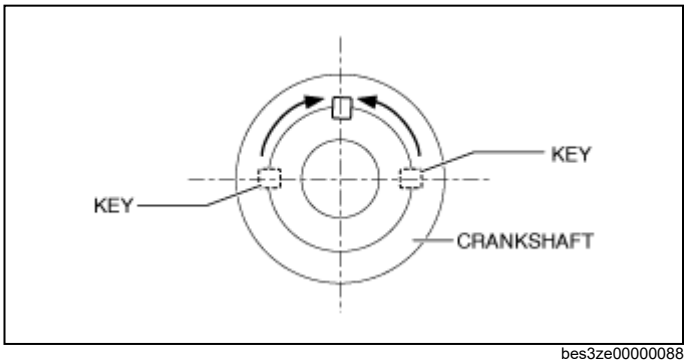


bes3ze00000087

Caution

• Be careful not to rotate the crankshaft to an angle greater than 90 degrees.

2. Rotate the exhaust camshaft to adjust the timing mark position.
3. Rotate the crankshaft as shown in the figure to adjust the key position.

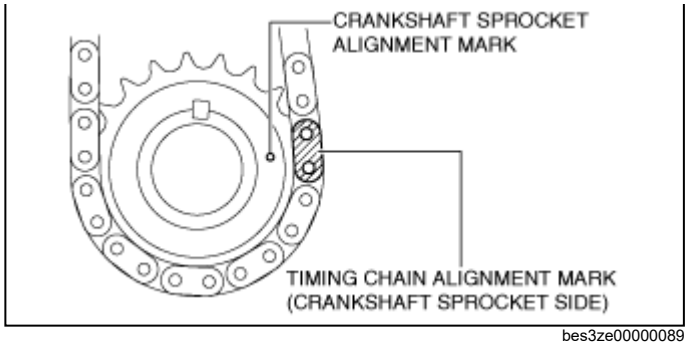


bes3ze00000088

Caution

• Be careful not to rotate the crankshaft to an angle greater than 90 degrees.

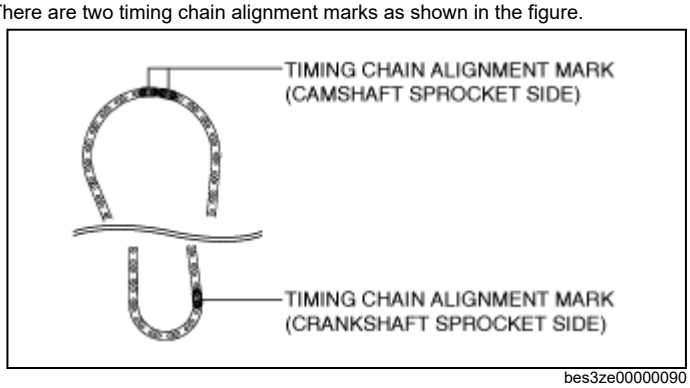
2. Align the timing chain alignment mark (crankshaft sprocket side) with the crankshaft sprocket alignment mark.



bes3ze00000089

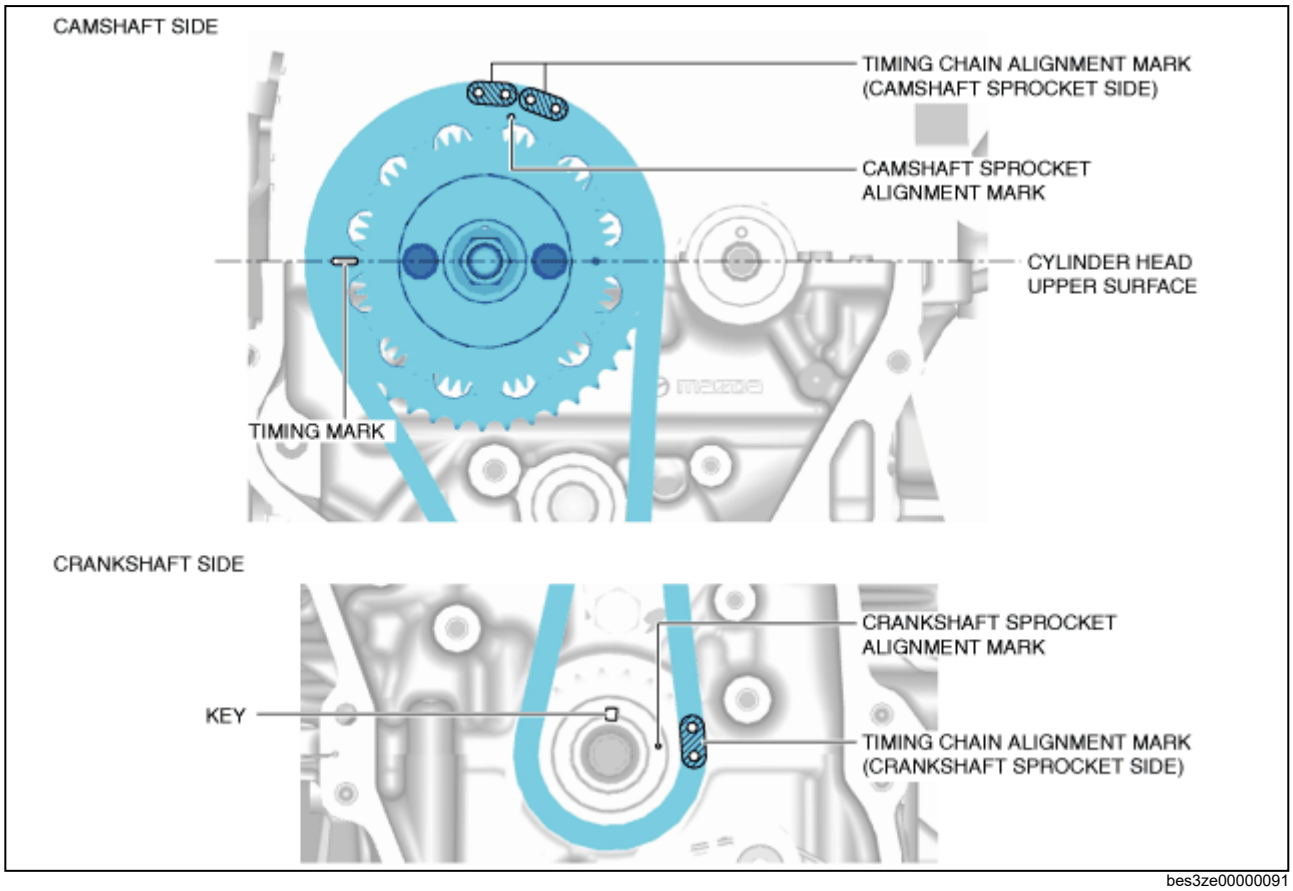
Note

- There are two timing chain alignment marks as shown in the figure.



bes3ze00000090

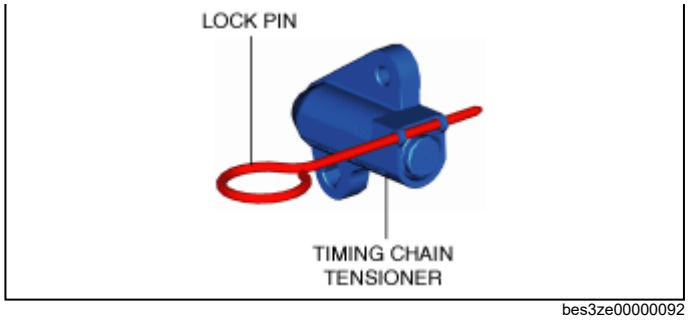
3. Install the timing chain and crankshaft sprocket as a single unit while aligning the marks on each sprocket and the timing chain as shown in the figure.



bes3ze00000091

4. Install the timing chain guide.
5. Install the timing chain tensioner arm.
6. Install the timing chain tensioner.
7. After installing the timing chain tensioner, remove the installed rod, and then apply tension to the timing chain.

    - If a new timing chain tensioner is used, remove the installed lock pin.



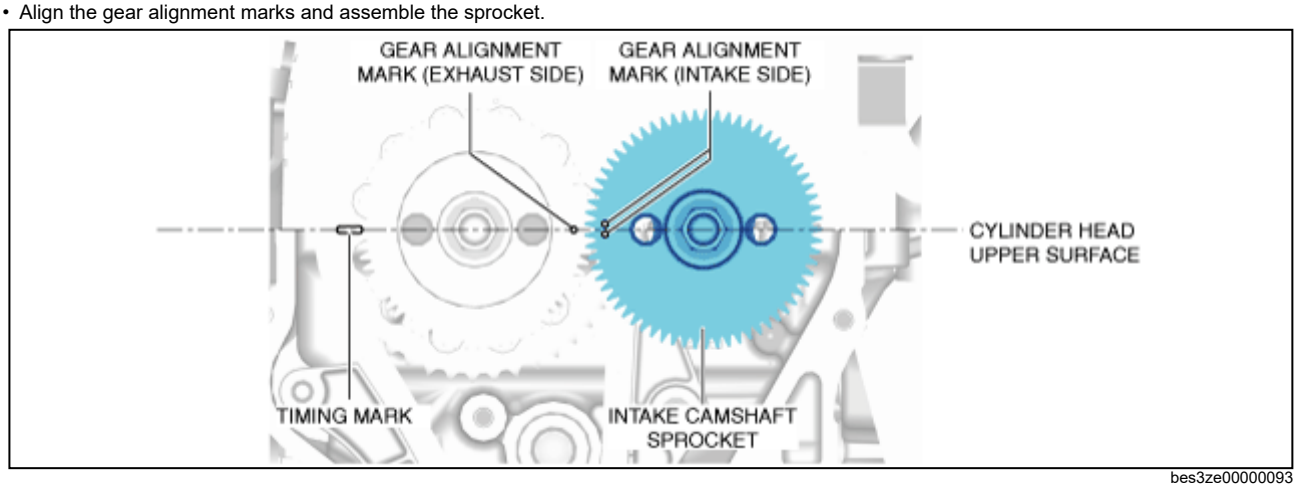
bes3ze00000092

8. Verify that there is no looseness in the timing chain, and re-verify that each sprocket is in the specified location.

Intake Camshaft Sprocket Assembly Note

Caution

- Align the gear alignment marks and assemble the sprocket.



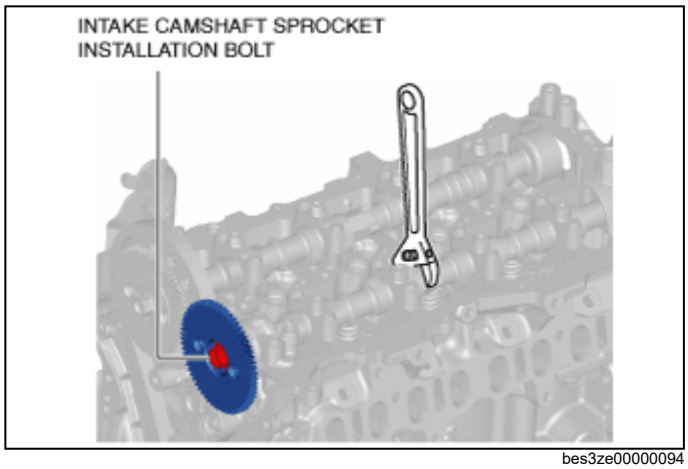
bes3ze00000093

1. Align the knock pin on the end of the intake camshaft with the knock pin installation hole on the intake camshaft sprocket side, and temporarily assemble the bolt.

Note

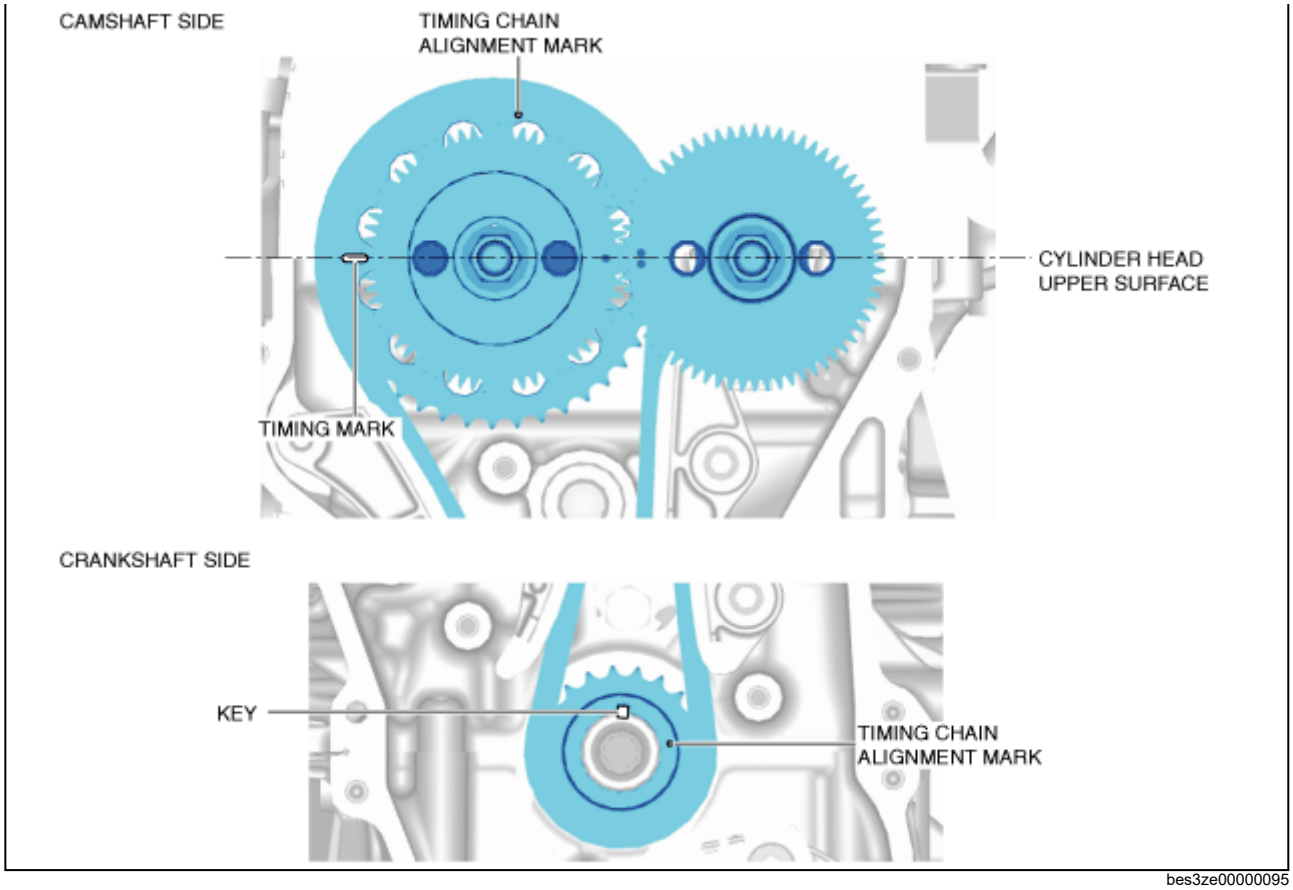
- Finely adjust the knock pin position by rotating the camshaft and align the knock pin with the installation hole.

2. Hold the intake camshaft using a wrench on the cast hexagon, and tighten the intake camshaft sprocket installation bolt.



bes3ze00000094

3. Rotate the crankshaft clockwise two turns and inspect the valve timing.

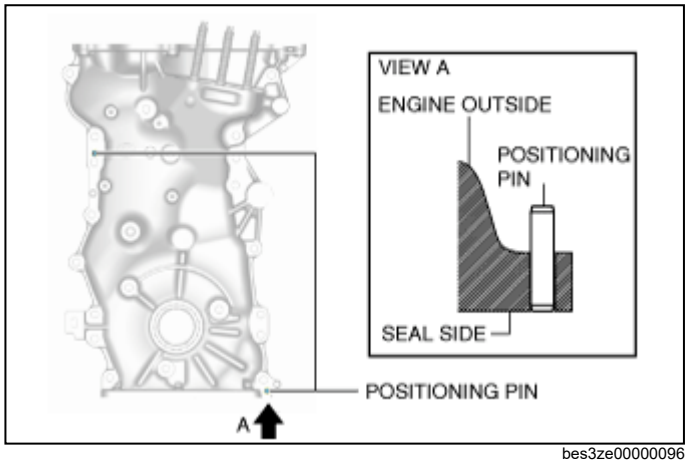


bes3ze00000095

Engine Front Cover Assembly Note

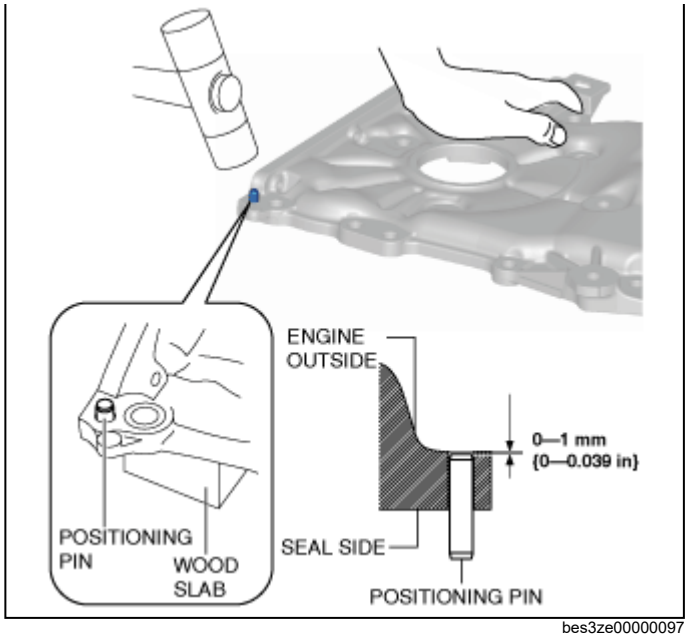
Note

- For a new engine front cover, the positioning pins in the two locations shown in the figure project to the outside of the engine.



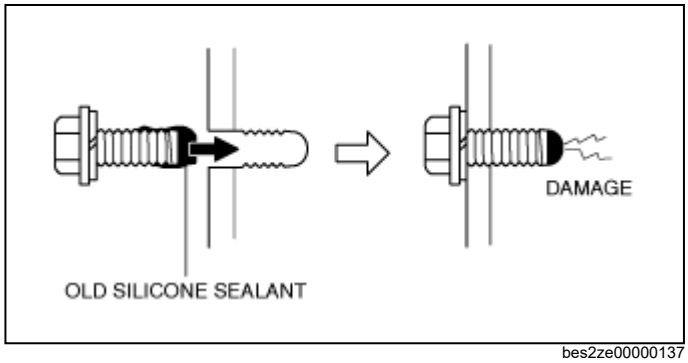
bes3ze00000096

1. If the engine front cover is newly replaced, tap the positioning pins in the two locations to the seal surface side.



bes3ze00000097

Caution
- If a bolt with sealant adhering to it is used, it could result in cracks in the cylinder head and cylinder block.



bes2ze00000137

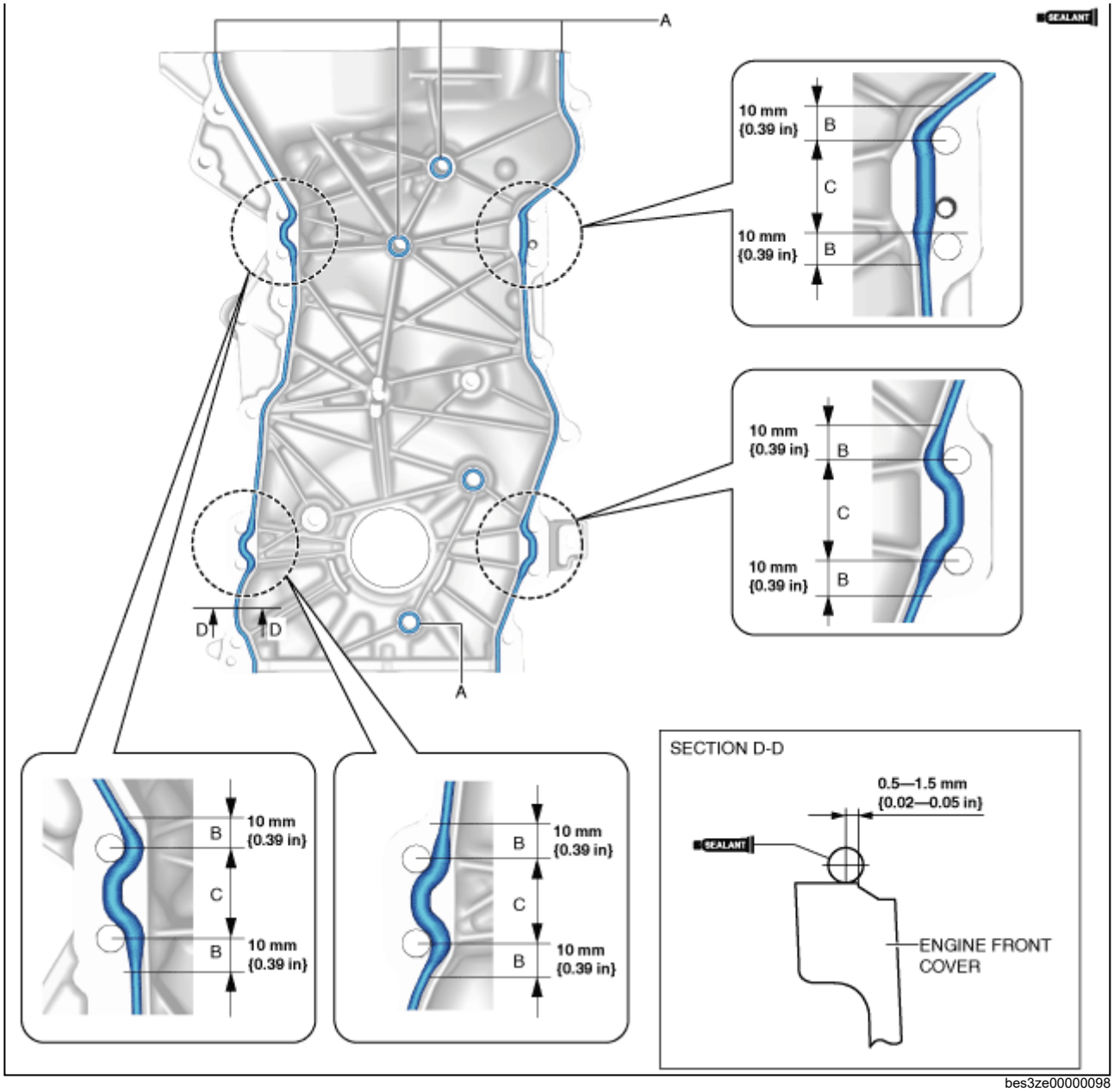
2. When reusing an engine front cover installation bolts, remove sealant adhering to the bolts.

Caution
- If oil, dirt and sealant remains on the sealant application area, the silicone sealant will not seal which will cause oil leakage.

3. Completely clean and remove any oil, dirt, sealant or other foreign matter that may be adhering to the engine front cover, cylinder head, and cylinder block.

Caution
- Apply the silicon sealant in a single, unbroken line.
- To prevent silicone sealant from hardening, adhere the engine front cover and the cylinder block firmly within 10 min. after applying silicone sealant. After adhering them, tighten the installation bolts immediately.

4. Apply silicone sealant (TB1217D or equivalent) to the engine front cover as shown in the figure.

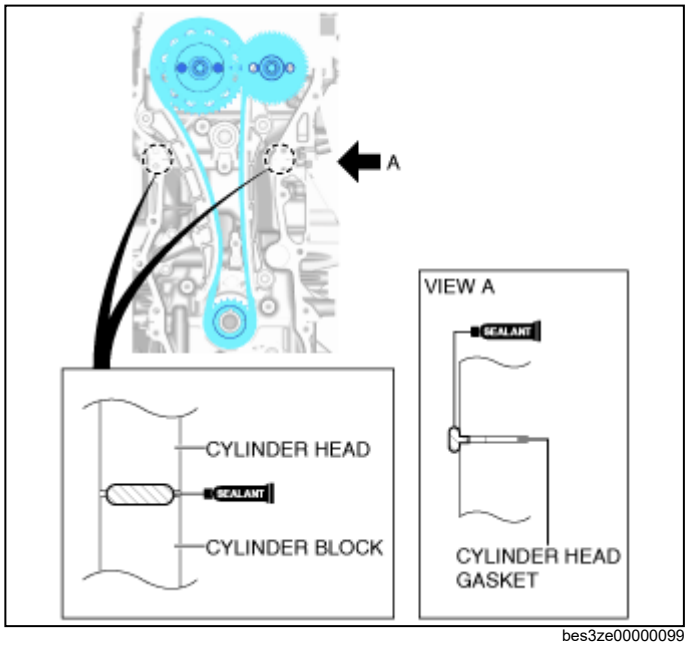


bes3ze00000098

Bead thickness

A: 2—6 mm {0.1—0.2 in}
B: 4—6 mm {0.16—0.23 in}
C: 4—8 mm {0.2—0.3 in}

5. Apply silicone sealant to the areas shown in the figure.



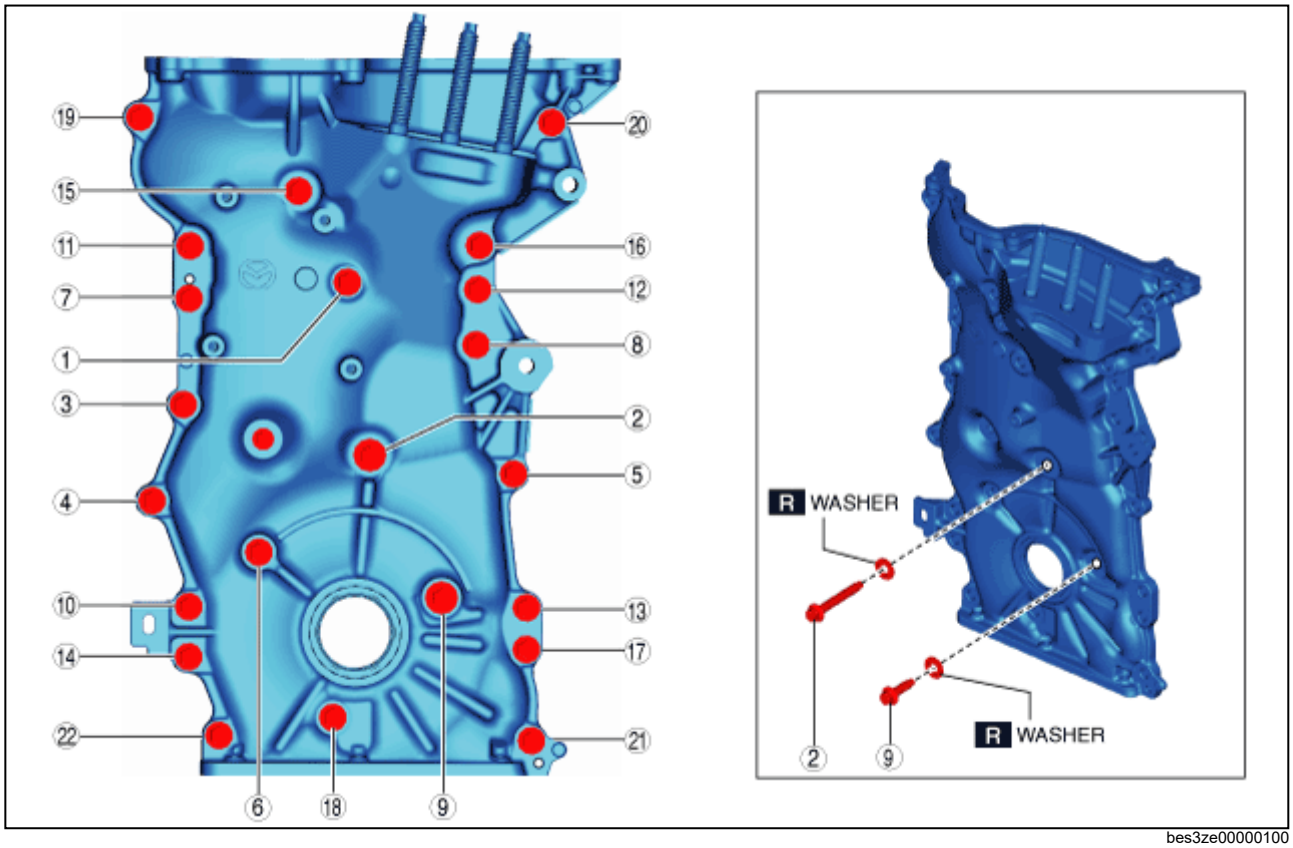
bes3ze00000099

Caution
- Apply the silicone sealant so that it goes into the cylinder head gasket.

6. Install the engine front cover to the engine.

Caution
- For the number 2 and 9 bolts of the tightening order, install the bolts with new washers.

7. Tighten the engine front cover installation bolts in the order shown in the figure.



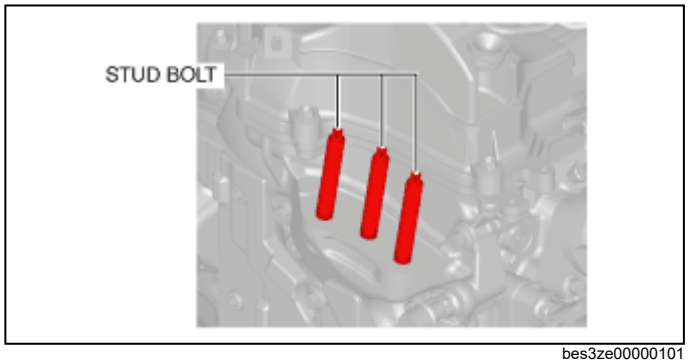
bes3ze00000100

Tightening torque

20—26 N·m {2.1—2.6 kgf·m, 15—19 ft·lbf}

Note
- Install stud bolts because there are no stud bolts installed on a new engine front cover.
- If the engine front cover is reused, tighten the stud bolts because they may have loosened.

8. Tighten the engine front cover stud bolts.



bes3ze00000101

Tightening torque

7—13 N·m {72—132 kgf·cm, 62—115 in·lbf}

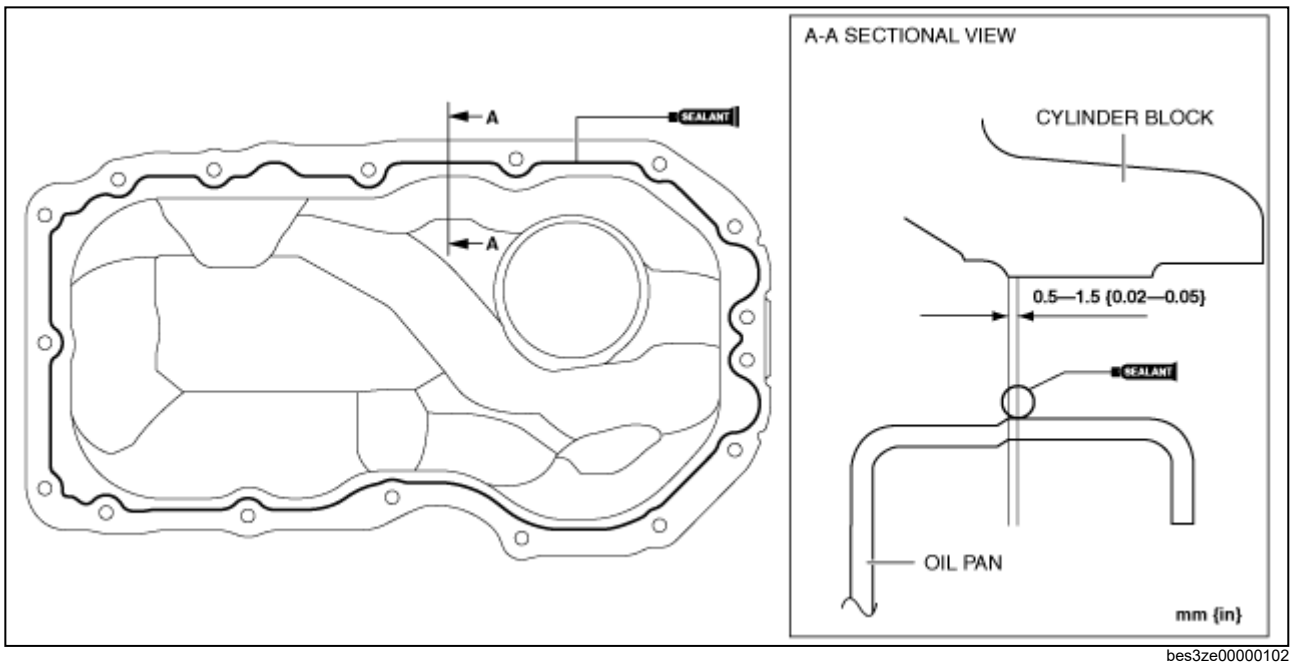
Oil Pan Component Assembly Note

Caution
- If the engine oil level sensor is removed, it is necessary to replace it with a new one. (with engine oil level sensor)

1. Completely clean and remove any oil, dirt, sealant or other foreign material that may be adhering to the cylinder block and oil pan.

Caution
- Apply the silicon sealant in a single, unbroken line around the whole perimeter.
- To prevent silicone sealant from hardening, adhere the oil pan to the cylinder block within 10 min. after silicone sealant is applied. Tighten the installation bolts completely soon after adhering.

2. Apply silicone sealant (TB1217D or equivalent) to the oil pan and the cylinder block.



bes3ze00000102

Thickness

3.0—7.0 mm {0.12—0.27 in}

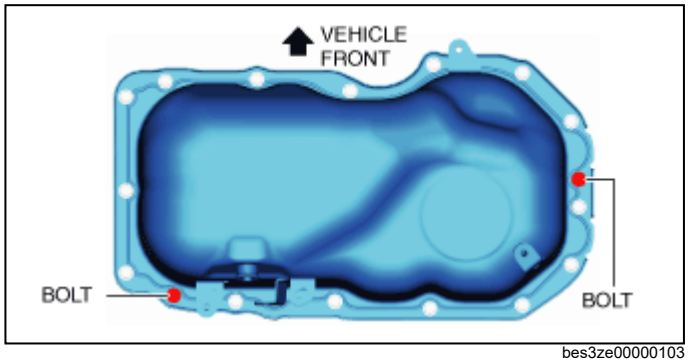
3. Install the oil pan to the cylinder block.

Caution

- Use new bolts because the oil pan installation bolts cannot be reused.

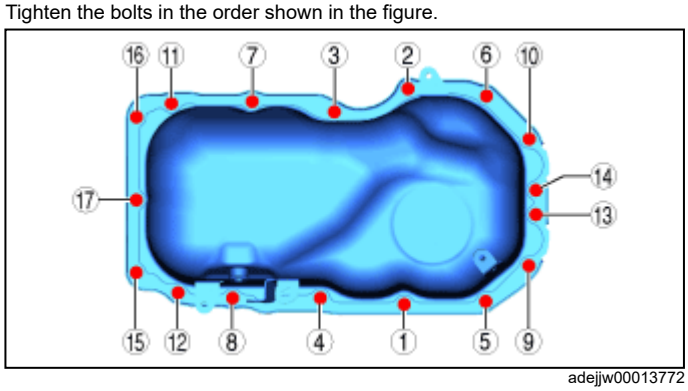
4. Install the oil pan using the following procedure:

(1) Temporarily tighten the two bolts shown in the figure.



bes3ze00000103

(2) Tighten the bolts in the order shown in the figure.



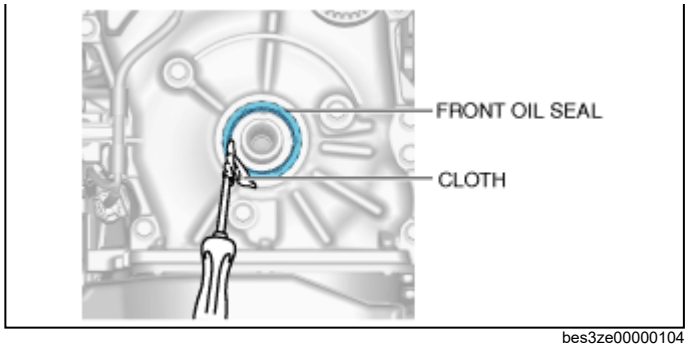
adejjw00013772

Tightening torque

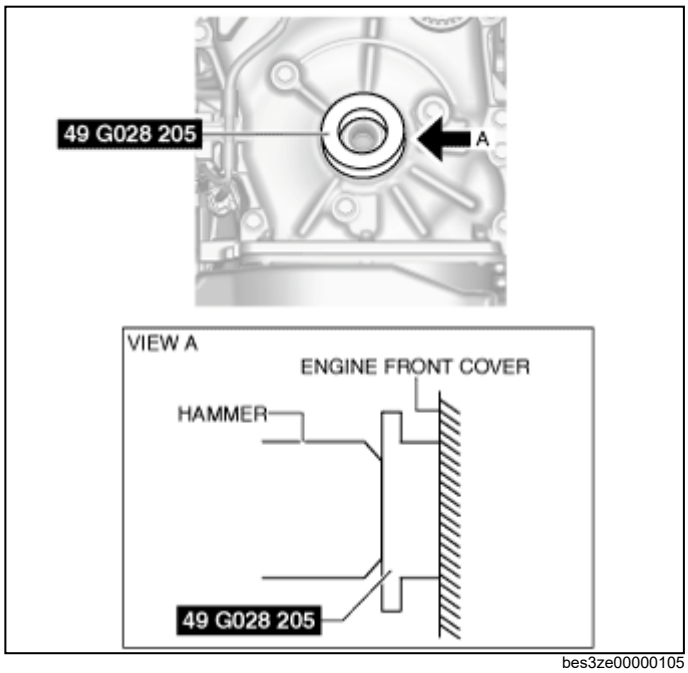
8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}

Front Oil Seal Assembly Note

1. Apply clean engine oil to the inner surface of a new front oil seal.
2. Insert the front oil seal into the engine front cover by hand.
3. Tap the oil seal in evenly using the SST and a hammer.



bes3ze00000104



bes3ze00000105

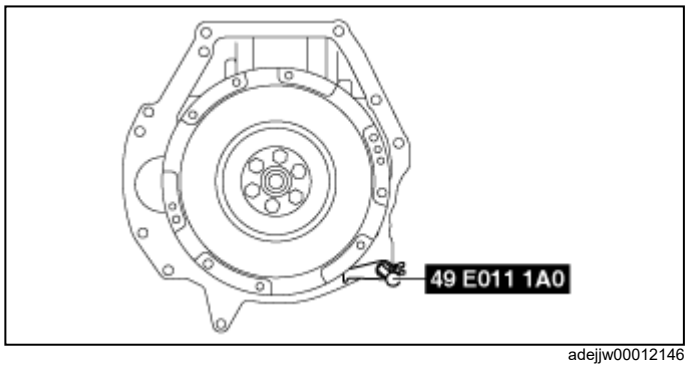
Front oil seal press-in amount

0—0.5 mm {0—0.019 in} from engine front cover

Crankshaft Pulley Lock Bolt Assembly Note

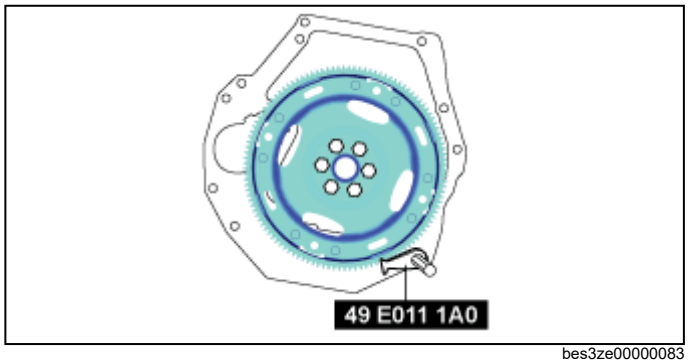
1. Hold the crankshaft using the SST.

MTX



adejjw00012146

ATX



bes3ze00000083

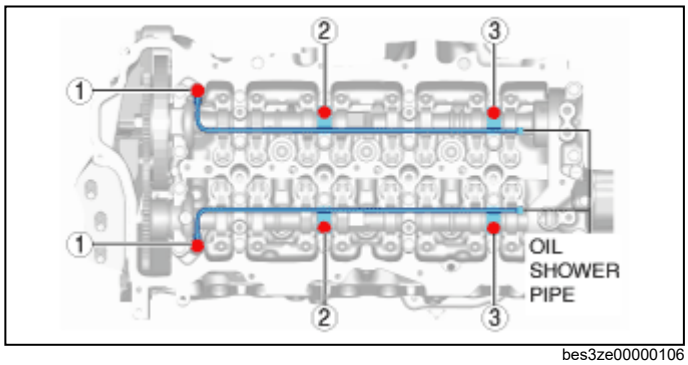
2. Tighten the crankshaft pulley lock bolt in the order shown in the following two steps.

Tightening torque

Step 1: 90—110 N·m {9.2—11 kgf·m, 67—81 ft·lbf}
Step 2: 55—65°

Oil Shower Pipe Assembly Note

1. Install the oil shower pipe in the order shown in the figure.



bes3ze00000106

Tightening torque

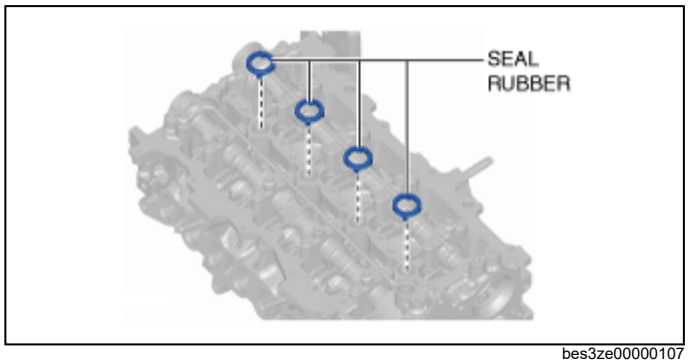
Installation position	Tightening torque
1	9—12 N·m {92—122 kgf·cm, 80—106 in·lbf}
2, 3	8.0—9.0 N·m {82—91 kgf·cm, 71—79 in·lbf}

Cylinder Head Cover Assembly Note

Caution

- To assure the sealing performance of the cylinder head cover, be careful of the following:
  - — Verify that the cylinder head cover gasket is inserted into the cylinder head cover groove and install the cylinder head cover.
  - — Completely clean and remove any oil, dirt, sealant or other foreign material from the seal surface.

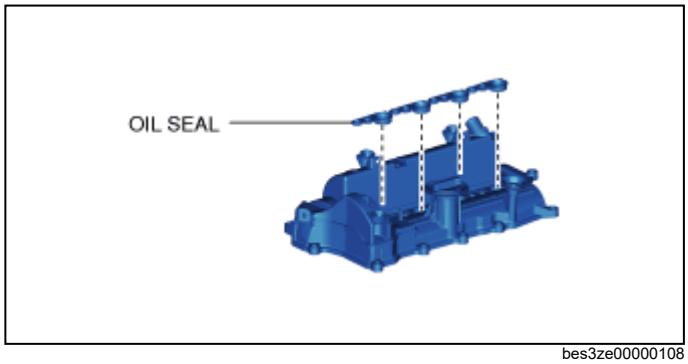
1. Install the seal rubber on the cylinder head.



bes3ze00000107

- Verify if the seal rubber has been damaged and if so replace it.
  - — The seal rubber can be reused even though the cylinder head cover causes contact marks on the seal rubber. However, if there is damage such as cracks and chips which may cause the seal rubber to tear, replace it.

2. Install the oil seal on the cylinder head cover.



bes3ze00000108

- Verify if the oil seal shown in the figure has been damaged and if so replace it.

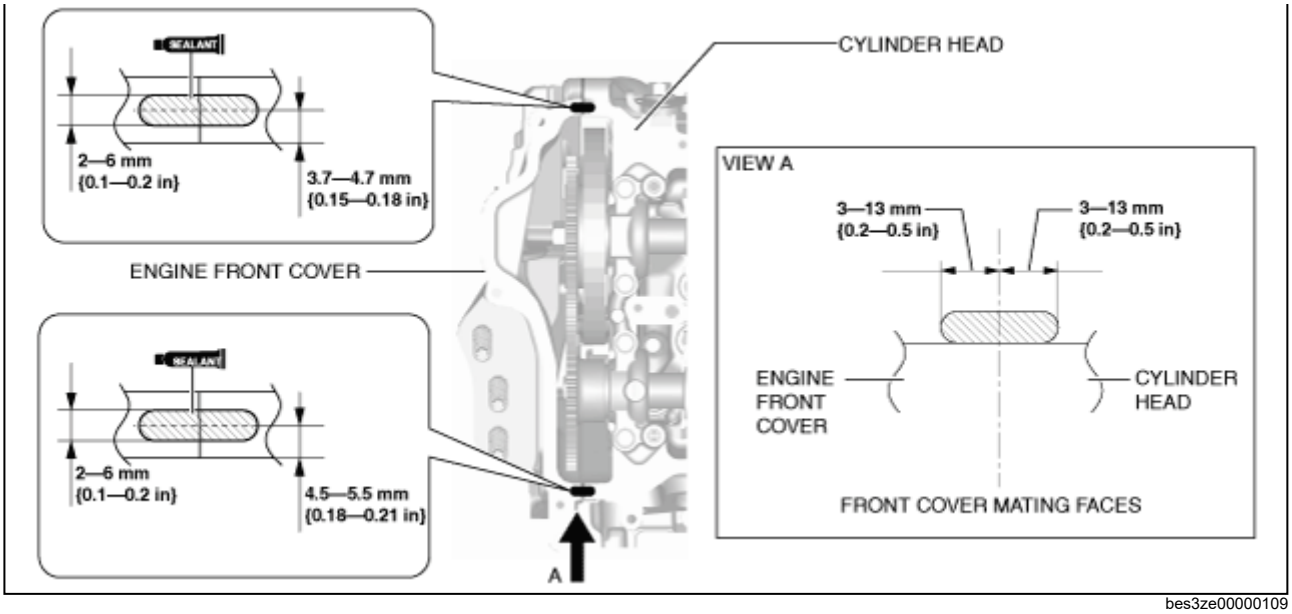
3. Insert a new cylinder head cover gasket into the cylinder head cover groove.

Caution

- To prevent silicone sealant from hardening, adhere the cylinder head cover and the cylinder head firmly within 10 min. after applying silicone sealant. After adhering them, tighten the installation bolts immediately.

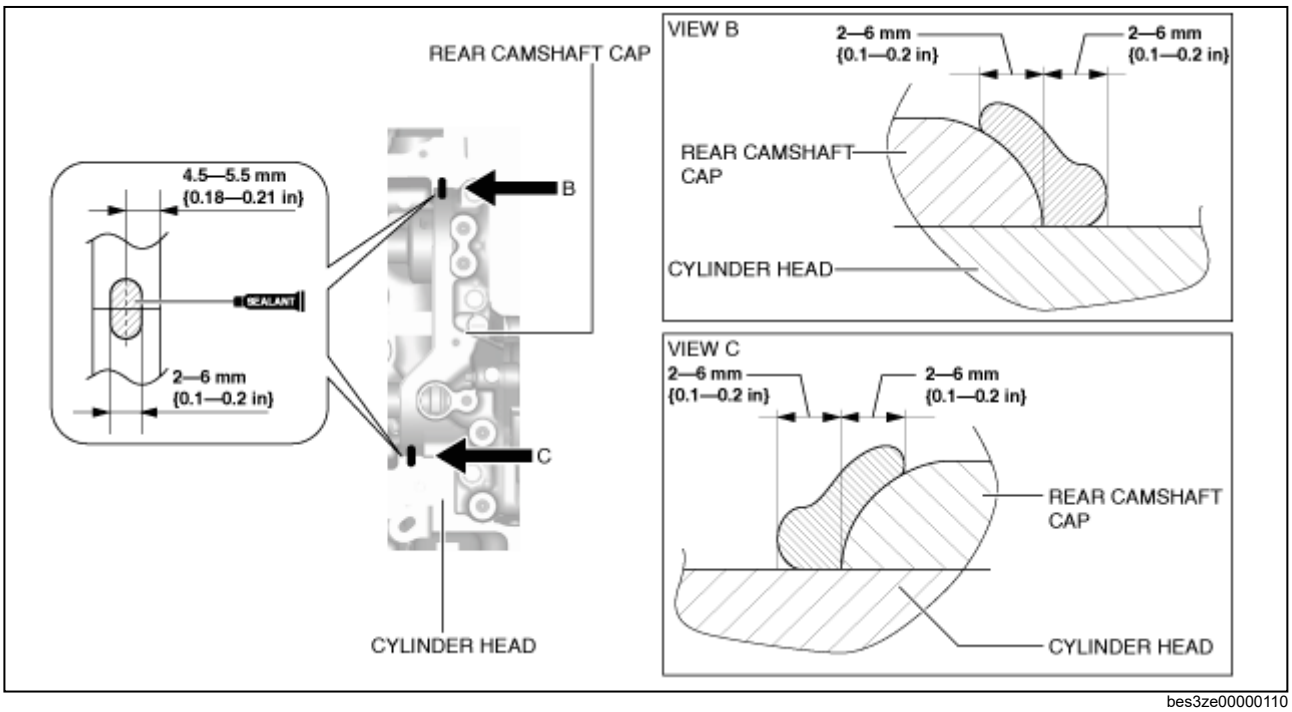
4. Apply silicone sealant (TB1217D or equivalent) to the areas shown in the figure.

Engine front side



bes3ze00000109

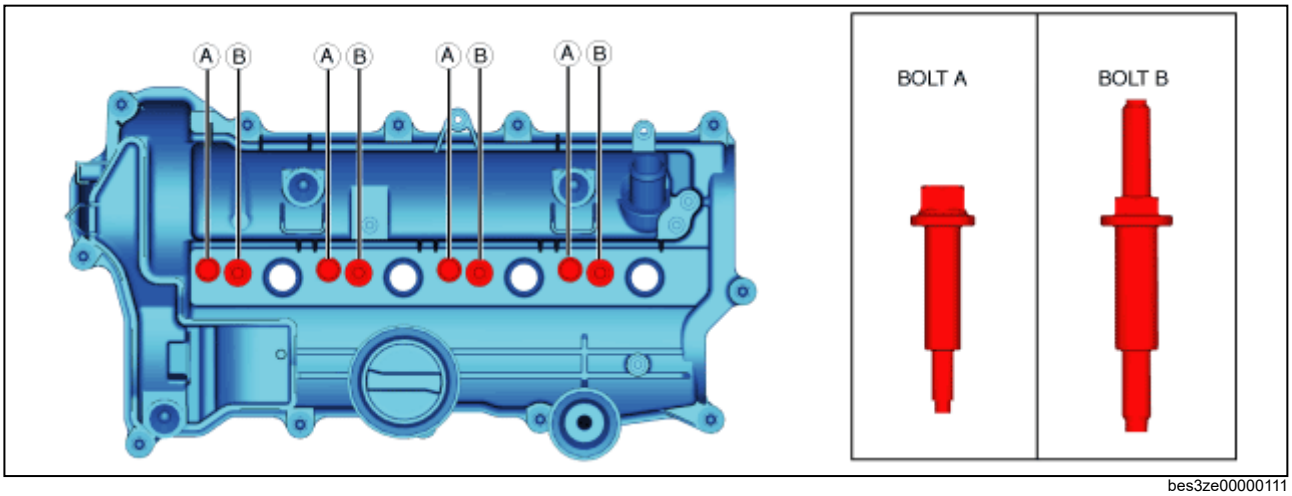
Engine rear side



bes3ze00000110

5. Install the cylinder head cover.
6. Tighten the cylinder head cover installation bolts using the following procedure:

(1) Tighten the cylinder head cover installation bolts A and B.



bes3ze00000111

Tightening torque

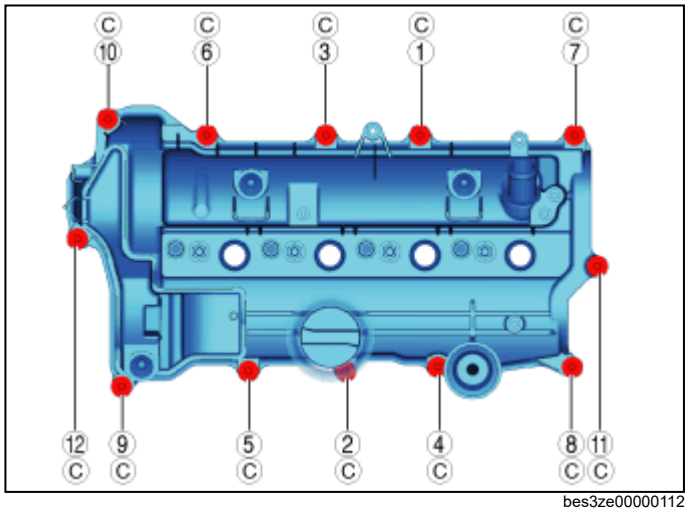
Bolt A: 6.0—8.0 N·m {62—81 kgf·cm, 54—70 in·lbf}

Bolt B: 15—20 N·m {1.6—2.0 kgf·m, 12—14 ft·lbf}

Note

- The tightening order for installation bolts A and B is optional.

(2) Tighten cylinder head cover installation bolts C in the order shown in the figure.



bes3ze00000112

Tightening torque

4.5—7.0 N·m {46—71 kgf·cm, 40—61 in·lbf}

ENGINE TECHNICAL DATA

id015000800100

Item	Specification
Maximum distortion, head gasket side of the cylinder head	0.06 mm {0.002 in}
Maximum distortion of intake manifold and turbocharger installation surface	Intake manifold installation surface: 0.05 mm {0.002 in} Turbocharger installation surface: 0.05 mm {0.002 in}
Standard valve seat contact width	1.2—2.0 mm {0.048—0.078 in}
Valve seat angle	45°
Standard valve seat sinkage amount	IN: 0.39—0.79 mm {0.016—0.031 in} EX: 0.47—0.87 mm {0.019—0.034 in}
Standard valve head margin thickness	IN: 1.50—1.70 mm {0.0591—0.0669 in} EX: 1.68—1.88 mm {0.0662—0.0740 in}
Standard valve length	IN: 97.07—97.37 mm {3.822—3.833 in} EX: 96.99—97.29 mm {3.819—3.830 in}
Minimum valve length	IN: 96.85 mm {3.813 in} EX: 96.77 mm {3.810 in}
Standard valve stem diameter	IN: 5.470—5.485 mm {0.2154—0.2159 in} EX: 5.465—5.480 mm {0.2152—0.2157 in}
Minimum valve stem diameter	IN: 5.424 mm {0.2135 in} EX: 5.419 mm {0.2133 in}
Standard valve guide inner diameter	IN: 5.510—5.530 mm {0.2170—0.2177 in} EX: 5.510—5.530 mm {0.2170—0.2177 in}
Standard clearance between valve stem and guide	IN: 0.025—0.060 mm {0.0010—0.0023 in} EX: 0.030—0.065 mm {0.0012—0.0025 in}
Maximum clearance between valve stem and guide	0.10 mm {0.0039 in}
Standard valve guide projection height	14.8—15.3 mm {0.583—0.602 in}
Valve spring installation height	IN: When pressurized with spring force of 128.3—141.7 N {13.09—14.44 kgf, 28.85—31.85 lbf}, spring height is 35.0 mm {1.38 in} EX (TYPE A): When pressurized with spring force of 137.8—152.2 N {14.06—15.52 kgf, 30.98—34.21 lbf}, spring height is 35.0 mm {1.38 in} EX (TYPE B): When pressurized with spring force of 209—231 N {21.4—23.5 kgf, 47.0—51.9 lbf}, spring height is 35.0 mm {1.38 in}
Maximum valve spring off-square	IN: 2.0 ° (1.5 mm {0.059 in}) EX (TYPE A): 2.0 ° (1.7 mm {0.067 in}) EX (TYPE B): 2.0 ° (1.9 mm {0.075 in})
Maximum camshaft runout	0.030 mm {0.0012 in}
Standard cam height [intake camshaft]	34.12 mm {1.343 in}
Standard cam height [exhaust camshaft]	34.48 mm {1.357 in}
Minimum cam height [intake camshaft]	34.05 mm {1.341 in}
Minimum cam height [exhaust camshaft]	34.41 mm {1.355 in}
Standard camshaft journal diameter [intake camshaft]	No.1 journal: 33.950—33.970 mm {1.3367—1.3374 in} Except No.1 journal: 25.950—25.975 mm {1.0217—1.0226 in}
Standard camshaft journal diameter [exhaust camshaft]	No.1 journal: 33.950—33.975 mm {1.3367—1.3375 in} Except No.1 journal: 25.950—25.975 mm {1.0217—1.0226 in}
Minimum camshaft journal diameter	No.1 journal: 33.920 mm {1.3354 in} Except No.1 journal: 25.920 mm {1.0205 in}
Standard camshaft journal oil clearance [intake camshaft]	No.1 journal: 0.030—0.075 mm {0.0012—0.0029 in} Except No.1 journal: 0.025—0.071 mm {0.0010—0.0027 in}
Standard camshaft journal oil clearance [exhaust camshaft]	No.1 journal: 0.025—0.075 mm {0.0010—0.0029 in} Except No.1 journal: 0.025—0.071 mm {0.0010—0.0027 in}
Maximum camshaft journal oil clearance	No.1 journal: 0.105 mm {0.00413 in} Except No.1 journal: 0.101 mm {0.00398 in}
Standard camshaft end play	0.07—0.22 mm {0.003—0.008 in}
Maximum camshaft end play	0.98 mm {0.039 in}
Maximum distortion, head gasket side of the cylinder block [Other than between cylinder bores]	0.10 mm {0.0039 in}
Maximum distortion, head gasket side of the cylinder block [between cylinder bores]	Distortion on head gasket surface is convex: 0.015 mm {0.00059 in} Distortion on head gasket surface is concave: 0.045 mm {0.0018 in}
Standard cylinder bore diameter	76.000—76.022 mm {2.9922—2.9929 in}
Oil jet valve opening pressure	70—110 kPa {0.72—1.12 kgf/cm^2, 11.0—15.9 psi}
Standard piston outer diameter	75.958—75.984 mm {2.9905—2.9914 in}
Standard clearance between piston and cylinder	0.031—0.049 mm {0.0013—0.0019 in}
Maximum clearance between piston and cylinder	0.067 mm {0.0026 in}
Standard clearance between piston ring and ring groove	Top: 0.10—0.14 mm {0.0040—0.0055 in} Second: 0.05—0.09 mm {0.002—0.003 in} Oil: 0.05—0.09 mm {0.002—0.003 in}
Maximum clearance between piston ring and ring groove	Top: 0.16 mm {0.0063 in} Second: 0.12 mm {0.0047 in} Oil: 0.11 mm {0.0043 in}

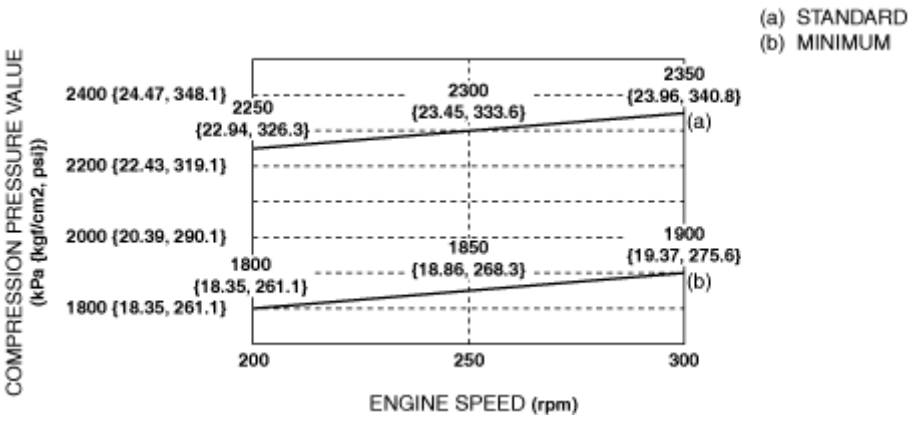
Item	Specification
Standard piston ring end gap	Top: 0.17—0.22 mm {0.0067—0.0086 in} Second: 0.17—0.27 mm {0.007—0.010 in} Oil (rail): 0.15—0.25 mm {0.0060—0.0098 in}
Maximum piston ring end gap	Top: 0.38 mm {0.015 in} Second: 0.49 mm {0.019 in} Oil (rail): 0.46 mm {0.018 in}
Standard piston pin outer diameter	23.995—24.000 mm {0.94469—0.94488 in}
Standard piston pin hole diameter	24.005—24.013 mm {0.94508—0.94539 in}
Standard clearance between piston pin hole diameter and piston pin outer diameter	0.005—0.018 mm {0.0002—0.0007 in}
Standard connecting rod small end inner diameter	24.010—24.021 mm {0.94528—0.94570 in}
Standard clearance between connecting rod small end inner diameter and piston pin outer diameter	0.010—0.026 mm {0.0004—0.0010 in}
Maximum connecting rod bending	0.08 mm {0.003 in}
Maximum connecting rod distortion	0.08 mm {0.003 in}
Connecting rod center-to-center distance	138.19 mm {5.4406 in}
Standard side clearance at the large end of connecting rod	0.14—0.36 mm {0.006—0.014 in}
Maximum side clearance at the large end of connecting rod	0.512 mm {0.0202 in}
Standard bearing oil clearance at the large end of the connecting rod	0.026—0.052 mm {0.0011—0.0020 in}
Maximum bearing oil clearance at the large end of the connecting rod	0.10 mm {0.00394 in}
Connecting rod bearing size	STD: 1.501—1.520 mm {0.05910—0.05984 in} OS 0.25: 1.626—1.633 mm {0.06402—0.06429 in} OS 0.50: 1.751—1.758 mm {0.06894—0.06921 in}
Standard crankshaft end play	0.08—0.29 mm {0.004—0.011 in}
Maximum crankshaft end play	0.30 mm {0.012 in}
Thrust bearing size	STD: 2.500—2.550 mm {0.0985—0.1003 in} OS 0.25: 2.625—2.675 mm {0.1034—0.1053 in}
Maximum main journal runout	0.10 mm {0.0039 in}
Standard main journal diameter	46.980—47.000 mm {1.8497—1.8503 in}
Maximum main journal off-round	0.005 mm {0.0002 in}
Standard crank pin diameter	46.980—47.000 mm {1.8497—1.8503 in}
Maximum crank pin off-round	0.005 mm {0.0002 in}
Standard main journal oil clearance	0.016—0.039 mm {0.0007—0.0015 in}
Maximum main journal oil clearance	0.084 mm {0.0033 in}
Main bearing size	STD: 2.489—2.510 mm {0.0980—0.0988 in} OS 0.25: 2.614—2.620 mm {0.1030—0.1031 in} OS 0.50: 2.739—2.745 mm {0.1079—0.1080 in}
Dual-mass flywheel guide pin projection maximum amount	11.0—12.0 mm {0.434—0.472 in}
Standard cylinder head bolt length	145.2—145.8 mm {5.717—5.740 in}
Maximum cylinder head bolt length	146.5 mm {5.768 in}
Standard connecting rod bolt length	41.7—42.3 mm {1.65—1.66 in}
Maximum connecting rod bolt length	42.7 mm {1.68 in}
Rear oil seal press-in amount	0—0.5 mm {0—0.019 in} from cylinder block
Front oil seal press-in amount	0—0.5 mm {0—0.019 in} from engine front cover

ENGINE TECHNICAL DATA [SKYACTIV-D 1.5]

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Item	Specification
Difference between target idle speed and the engine speed	Within 50 rpm
Target idle speed (reference)	[MTX] Normal: 800 rpm DPF auto regeneration: 800 rpm Injector learning: 800 rpm [ATX] Normal: 800 rpm DPF auto regeneration: 800 rpm Injector learning: 800 rpm
Standard cylinder head bolt length	145.2—145.8 mm {5.717—5.740 in}
Maximum cylinder head bolt length	146.5 mm {5.768 in}
Front oil seal press on amount	0—1.0 mm {0—0.039 in} from edge surface of engine front cover
Rear oil seal press on amount	0—0.5 mm {0—0.019 in} from edge surface of cylinder block
Engine oil capacity (approx. quantity)	Oil replacement: 4.7 L {5.0 US qt, 4.1 Imp qt} Oil and oil filter replacement: 5.1 L {5.4 US qt, 4.5 Imp qt} Total (dry engine): 5.8 L {6.1 US qt, 5.1 Imp qt}
Oil pressure (reference value) [oil temperature: 80—90 °C {176—194 °F}, coolant temperature: 80—90 °C {176—194 °F}]	Lo: 110—175 kPa {1.13—1.78 kgf/cm^2, 16.0—25.3 psi} [1,500 rpm] Hi: 300—430 kPa {3.06—4.38 kgf/cm^2, 43.6—62.3 psi} [3,500 rpm]
Engine coolant capacity (approx. quantity)	MTX: 6.6 L {7.0 US qt, 5.8 Imp qt} ATX: 6.8 L {7.2 US qt, 6.0 Imp qt}
Water-cooled charge air cooler coolant capacity (approx. quantity)	2.1 L {2.2 US qt, 1.8 Imp qt}
Cooling system cap valve opening pressure	93.2—122.6 kPa {0.951—1.250 kgf/cm^2, 13.6—17.7 psi}
Water-cooled charge air cooler reserve tank cap valve opening pressure	24—36 kPa {0.25—0.36 kgf/cm^2, 3.5—5.2 psi}
Cooling fan motor standard current	16.0—20.0 A [12 V]
Glow plug resistance (specification)	Approx. 1.0 ohms [20 °C {68 °F}]
Turbocharger solenoid valve resistance (specification)	9.8—11.2 ohms [20 °C {68 °F}]
Fuel injector resistance (specification)	Approx. 0.59 ohms [20 °C {68 °F}]
Fuel pressure relief valve resistance (specification)	Approx. 0.47 ohms [20 °C {68 °F}]
Suction control valve resistance (specification)	Approx. 2.1 ohms [20 °C {68 °F}] If not within the specification, place a repair order with the DENSO, Ltd. services network.
Fuel warmer resistance (specification)	Approx. 0.85—1.28 ohms
HP-EGR control valve resistance (specification)	0.3—100 ohms [20 °C {68 °F}]
LP-EGR control valve resistance (specification)	0.3—100 ohms [20 °C {68 °F}]
Intake shutter valve resistance (specification)	0.3—100 ohms [20 °C {68 °F}]
Exhaust shutter valve resistance (specification)	0.3—100 ohms [20 °C {68 °F}]
Battery parasitic draw (When the ignition is off, all doors and the bonnet are closed.)	**Vehicles without i-ELOOP** 40—65 mA **Vehicles with i-ELOOP** 25—45 mA
Generator standard voltage [IG-ON] (Without i-ELOOP)	Terminal 1A: B+ Terminal 2A: Approx. 1 V or less Terminal 2B: Approx. 0 V
Generator standard voltage [Idle, 20 °C {68 °F}] (Without i-ELOOP)	Terminal 1A: 13—15 V Terminal 2A: Approx. 3—8 V Terminal 2B: Turn the electrical loads (such as headlights, blower motor, rear window defogger, brake lights) on and verify that the voltage reading increases.
Generator rotor resistance (between slip rings) [20 °C {68 °F}] (Without i-ELOOP)	1.8—2.1 ohms [20 °C {68 °F}]
Generator brush length (Without i-ELOOP)	Standard: 22.5 mm {0.886 in} Minimum: 5.0 mm {0.20 in}
Generator brush spring force (Without i-ELOOP)	Standard: 4.1—5.3 N {0.42—0.54 kgf, 1.0—1.1 lbf} Minimum: 1.7 N {0.17 kgf, 0.38 lbf}
Starter no-load test voltage	11 V
Starter no-load test current	105 A or less

Compression pressure



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Standard: Refer to the graph.
Minimum: Refer to the graph.
Maximum difference between cylinders: 150 kPa {1.53 kgf/cm^2, 21.8 psi}[200 rpm]

Engine oil [European (L.H.D. U.K.) and Taiwan specs.]

Recommended engine oil		Alternative engine oil
Mazda Original Oil Supra DPF 0W-30	Mazda Original Oil Ultra DPF 5W-30	ACEA C3 0W-30, 5W-30

Engine oil [Australian specs.]

Recommended engine oil	
ACEA C1 or JASO DL-1 5W-30	ACEA C3 0W-30, 5W-30

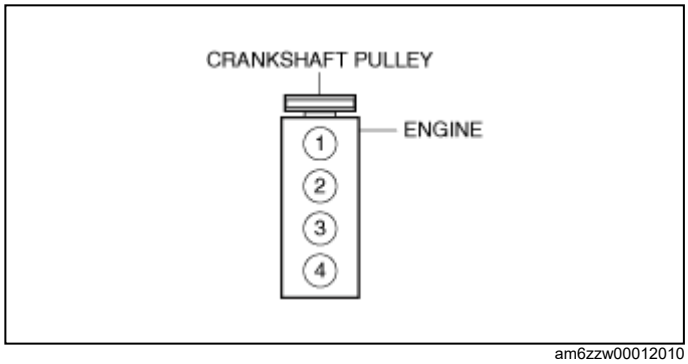
Engine oil [General (L.H.D. R.H.D.) specs.]

Recommended engine oil
ACEA C3 0W-30, 5W-30

Generator generated current (reference value) [Ambient temperature: 20 °C {68 °F}, Engine hot] (With i-stop, without i-ELOOP)

Engine speed (rpm)	Terminal 1A voltage (V)	Generator output current (A)
1,000	13	125
1,000	15	128
2,000	13	148
2,000	15	163

Cylinder number



am6zzw00012010